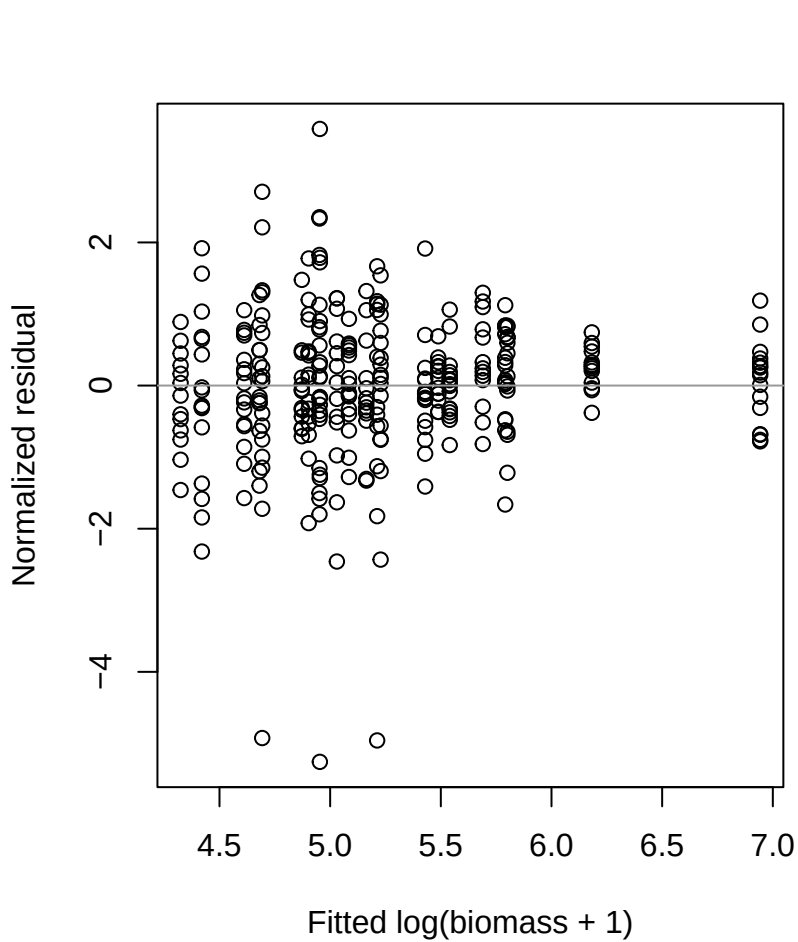
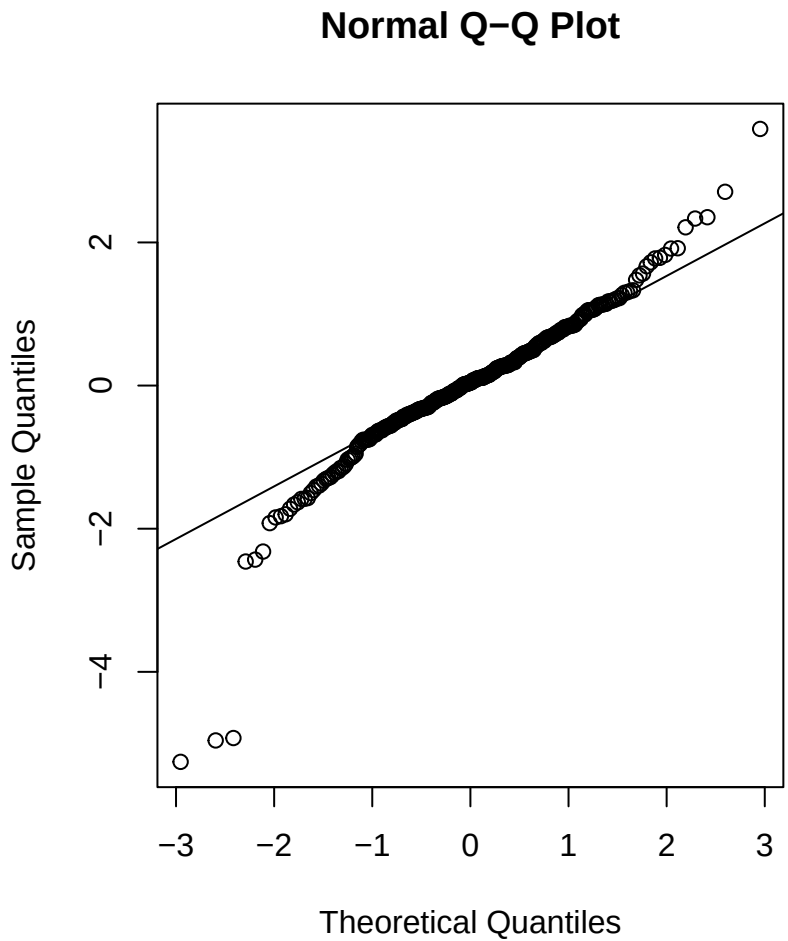
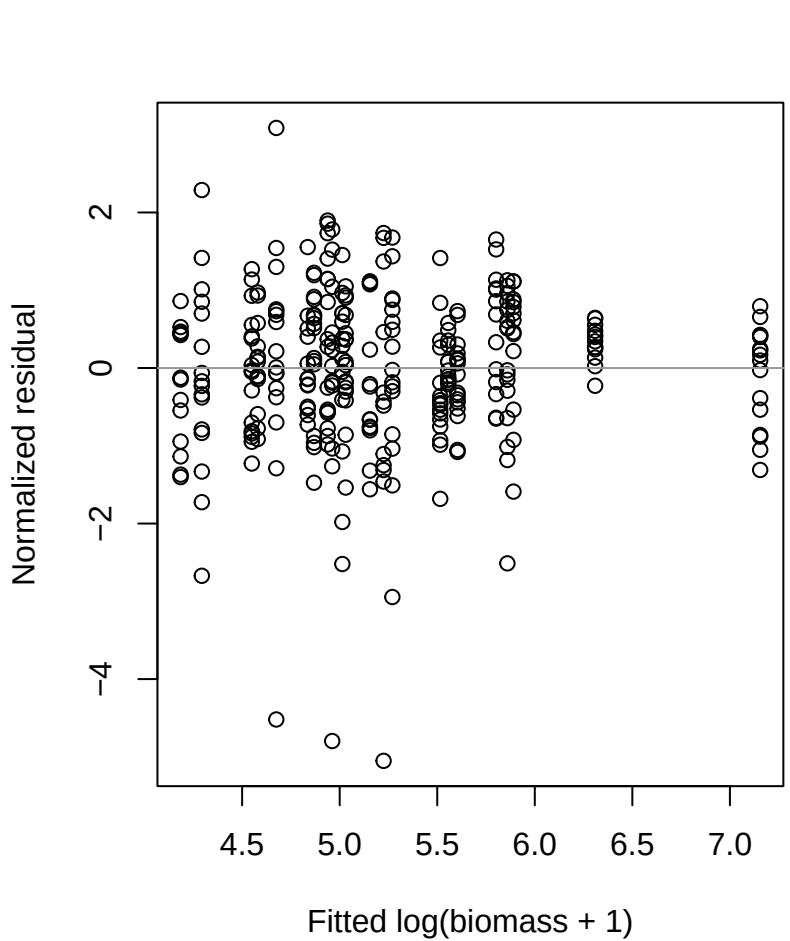
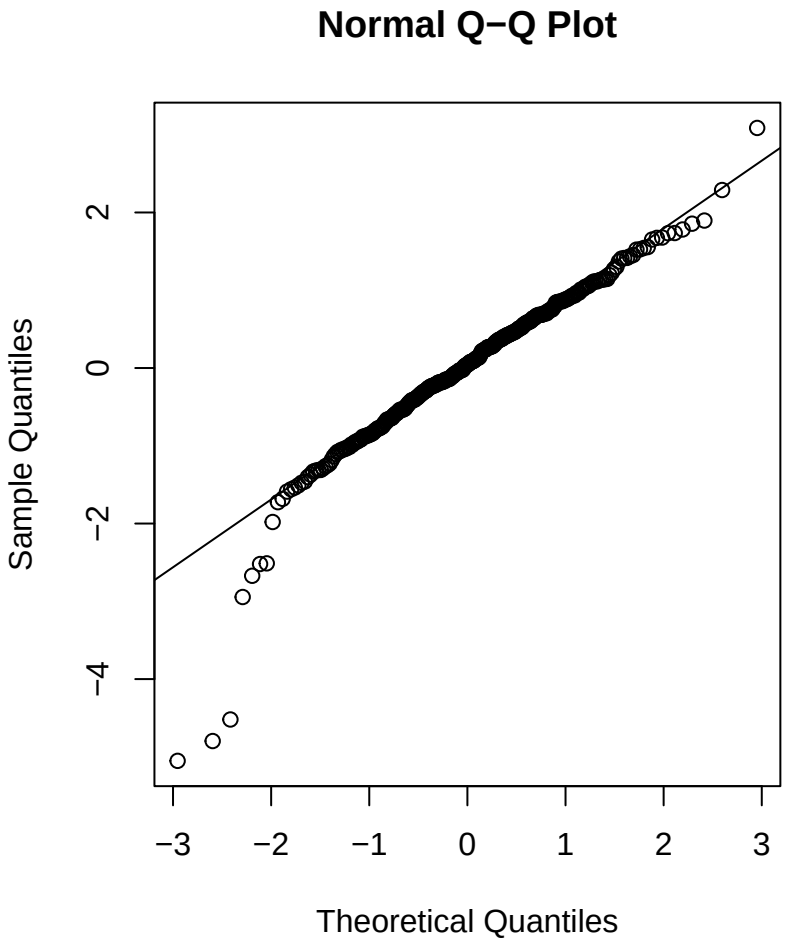
red_biomass__2019__2004__site_calendar_AR1

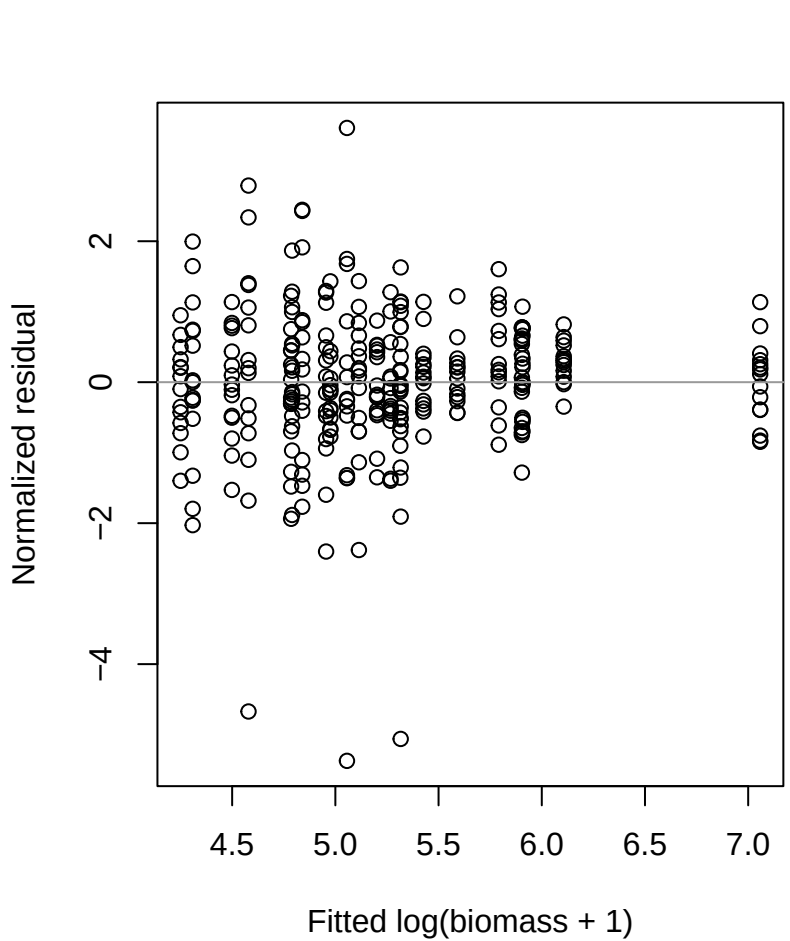


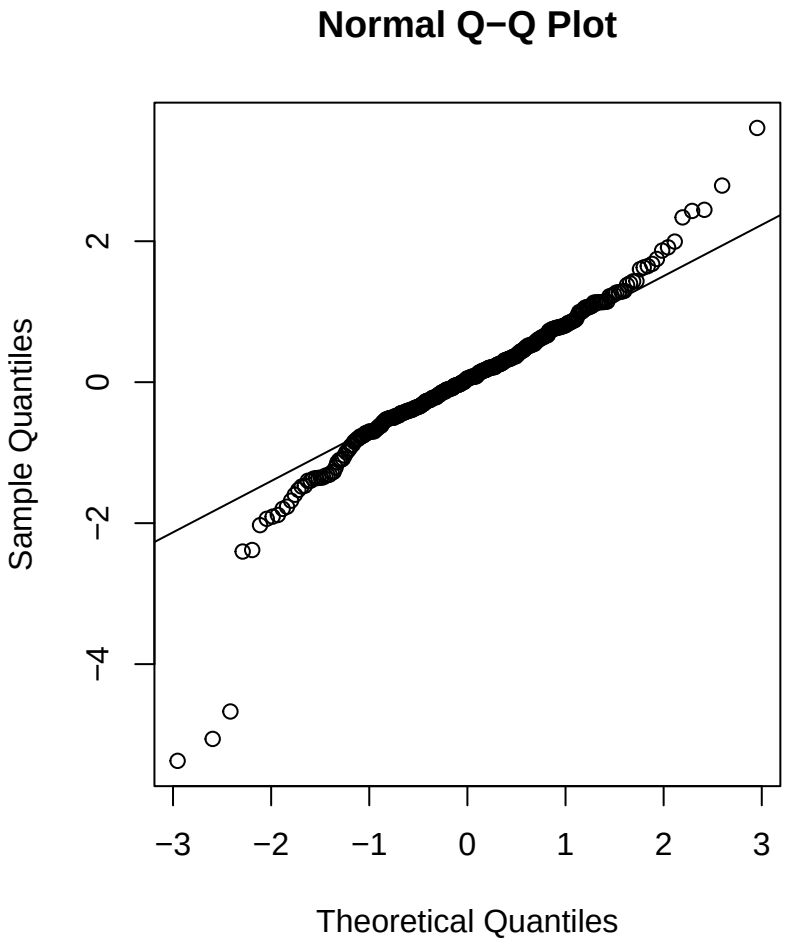



Normalized residual
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
7.0
Fitted log(biomass + 1)

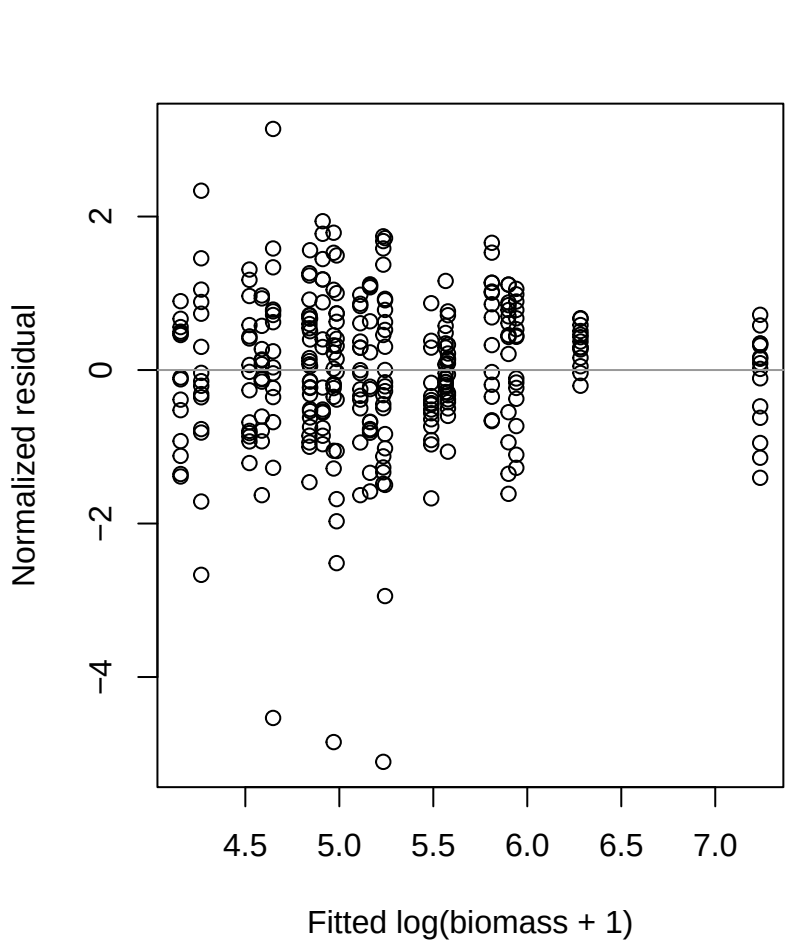

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

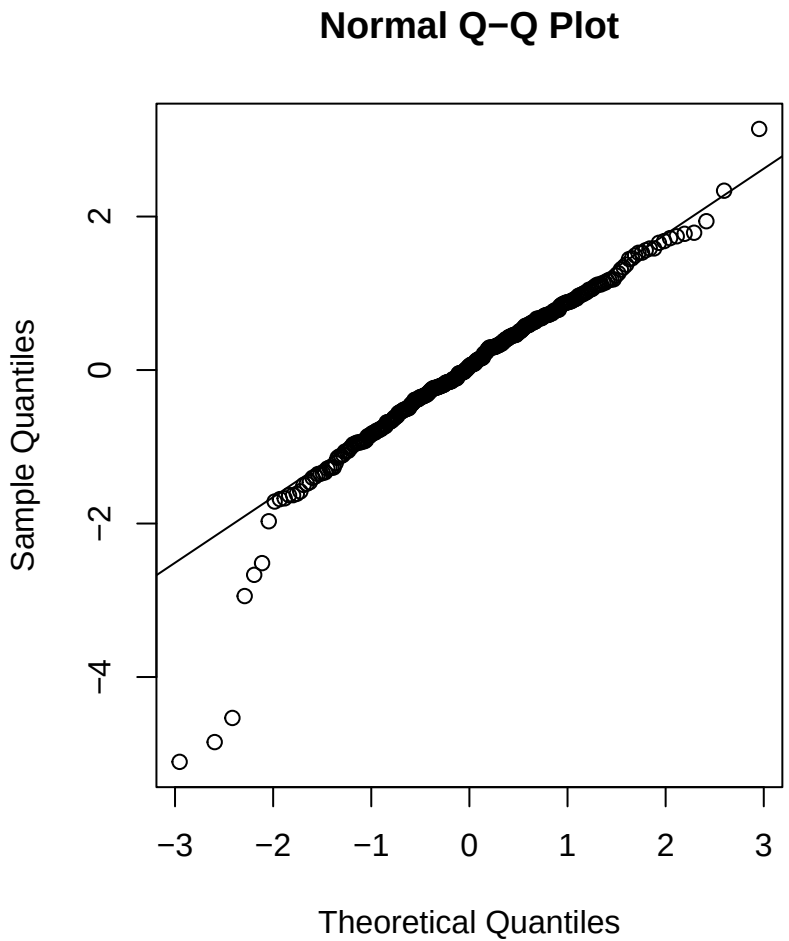
red_biomass__2019__2005__site_calendar_AR1



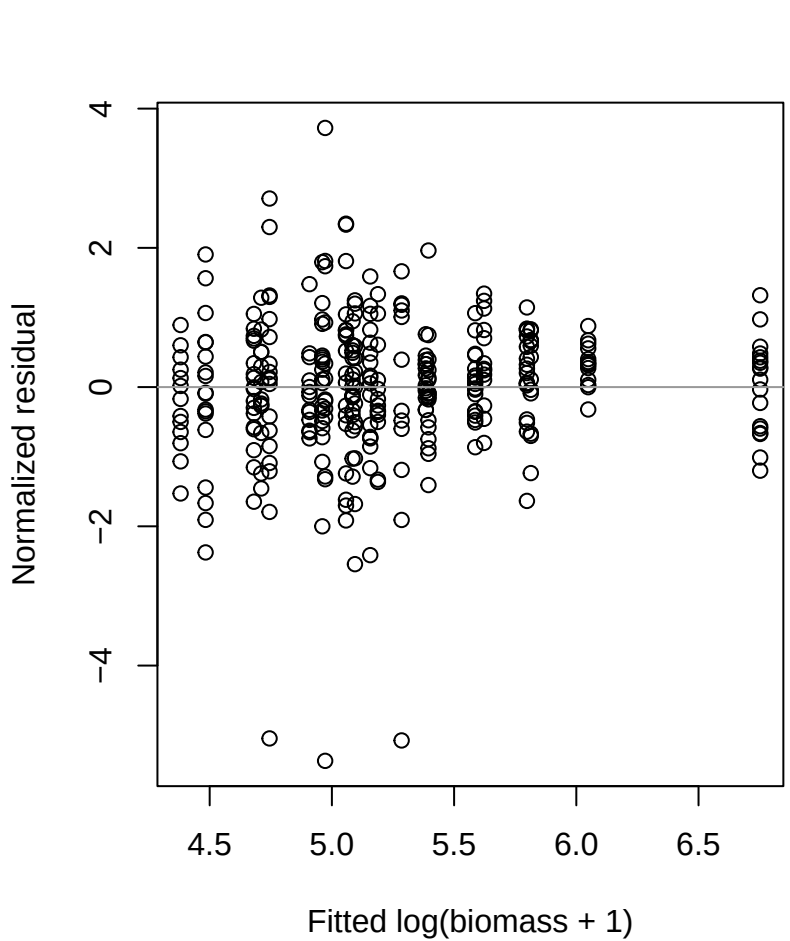


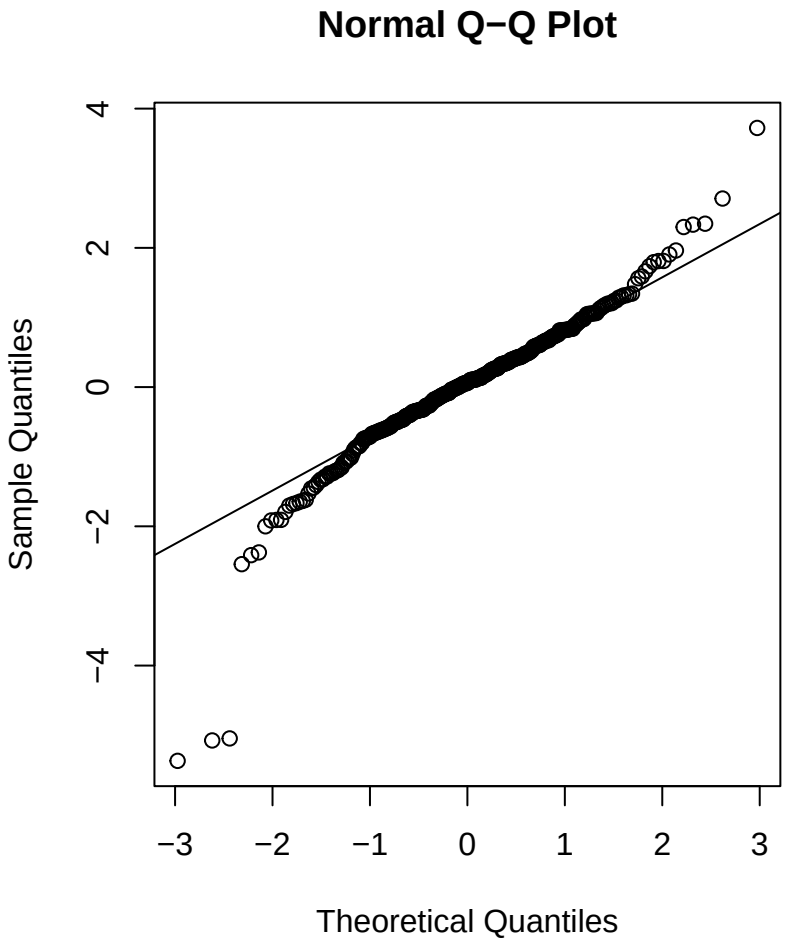
red_biomass__2019__2005__site_independent

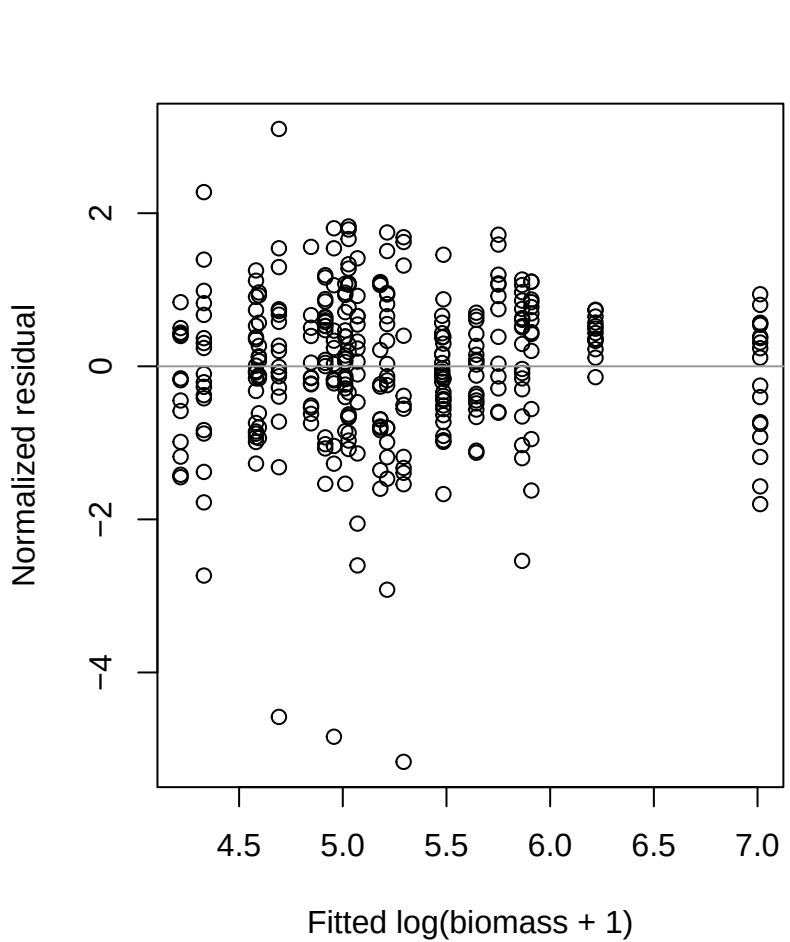


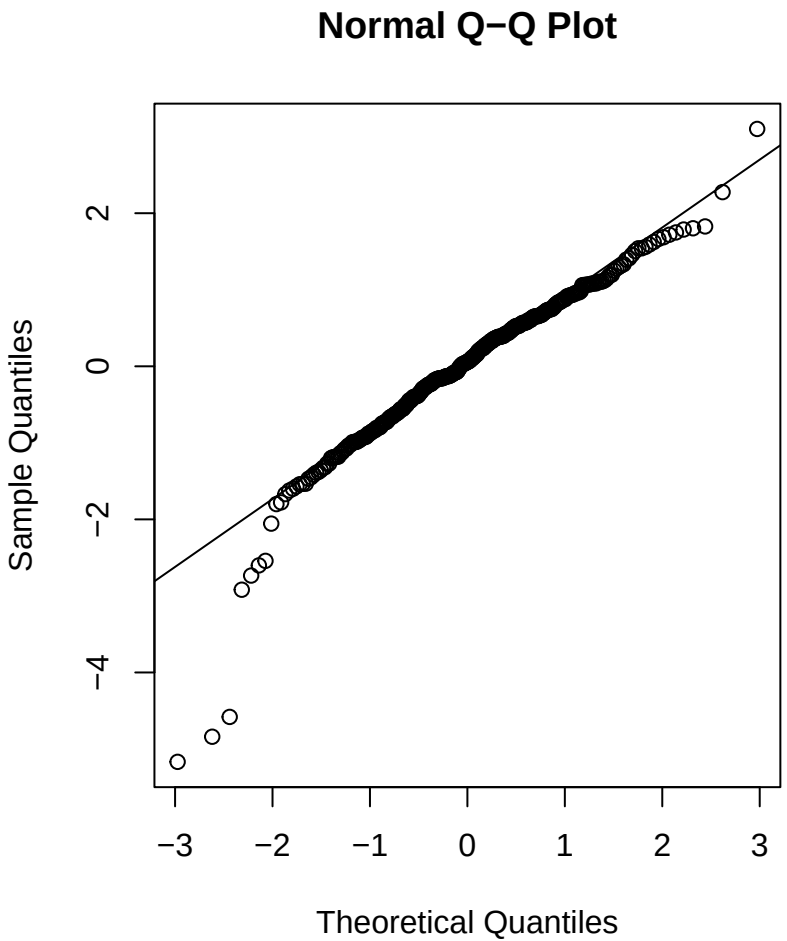


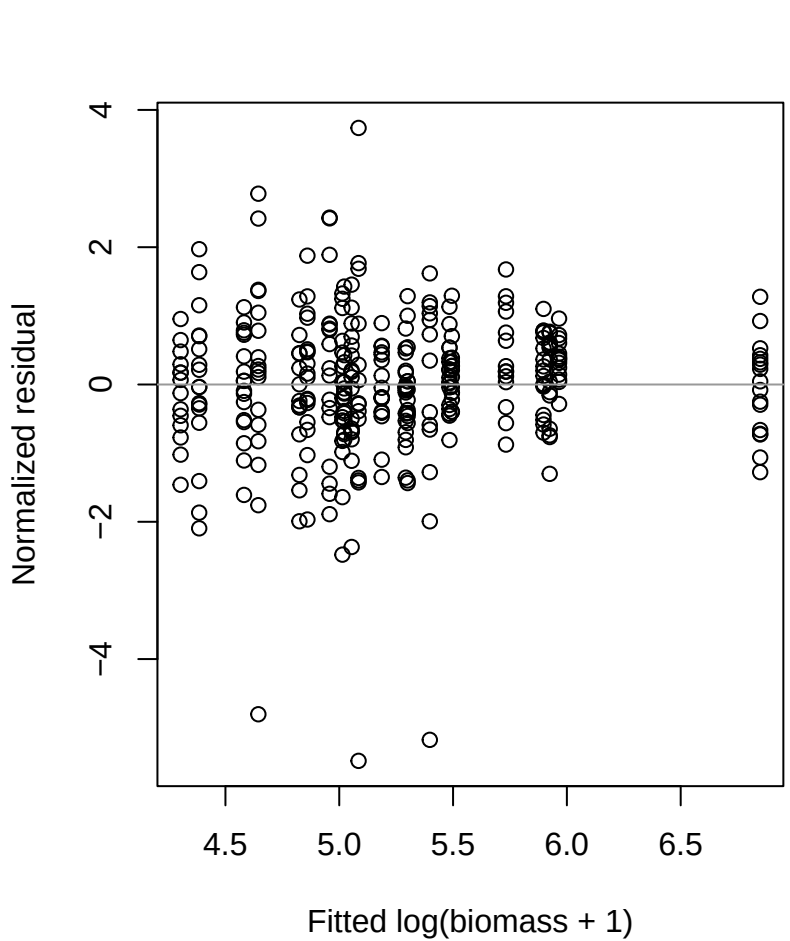
red_biomass__2022__2004__site_calendar_AR1

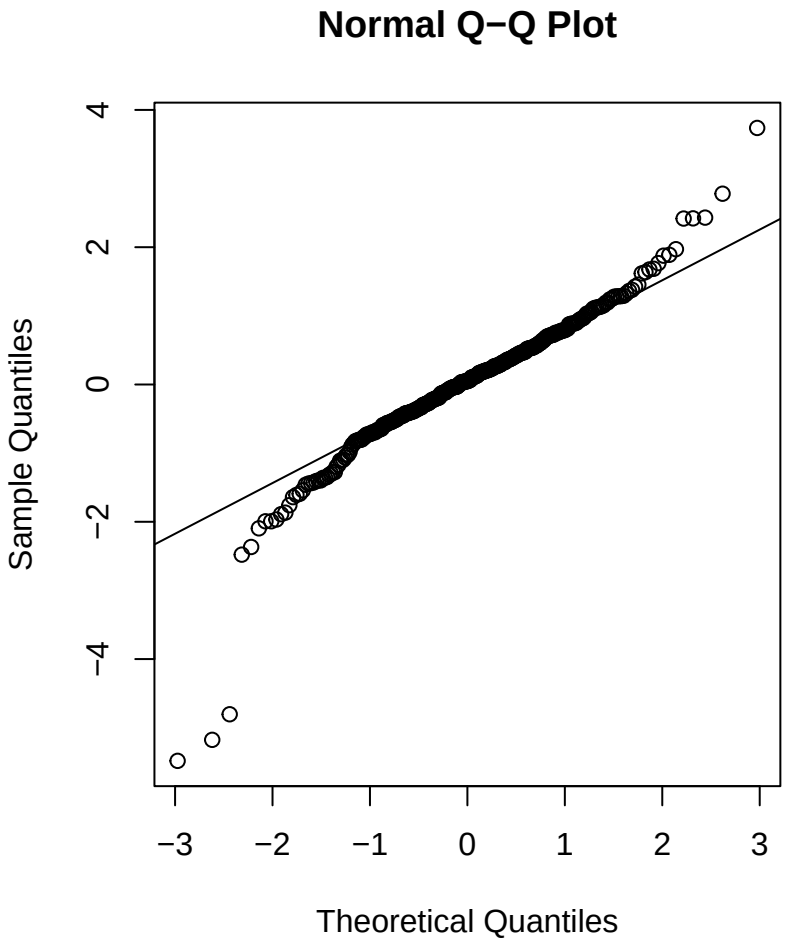




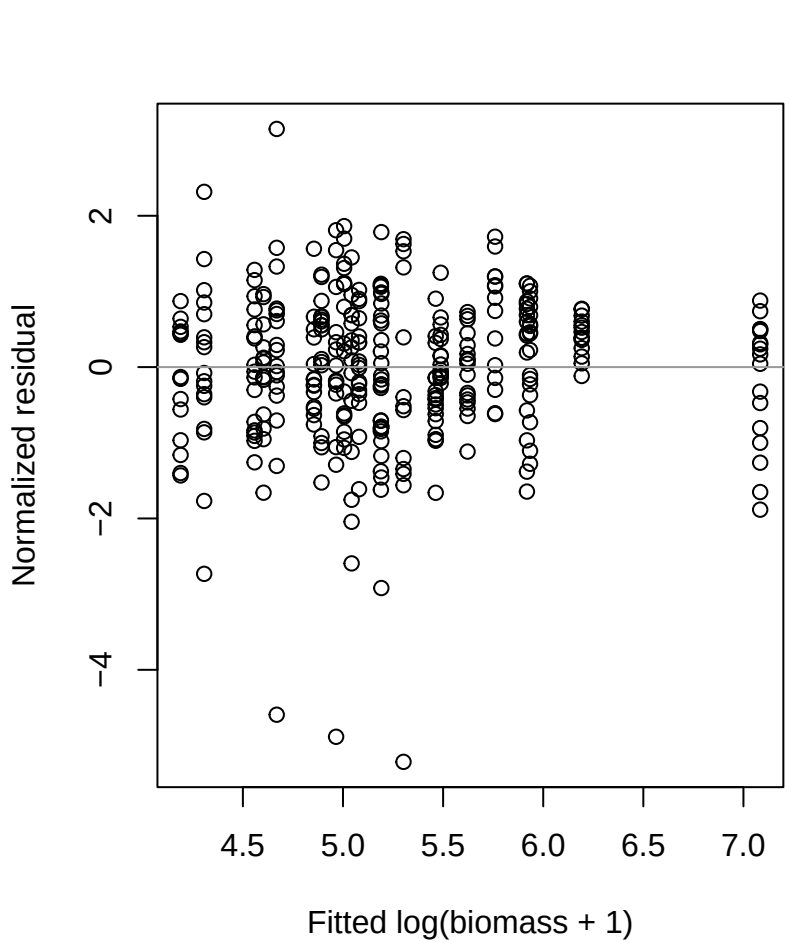

Normalized residual
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
7.0
Fitted log(biomass + 1)

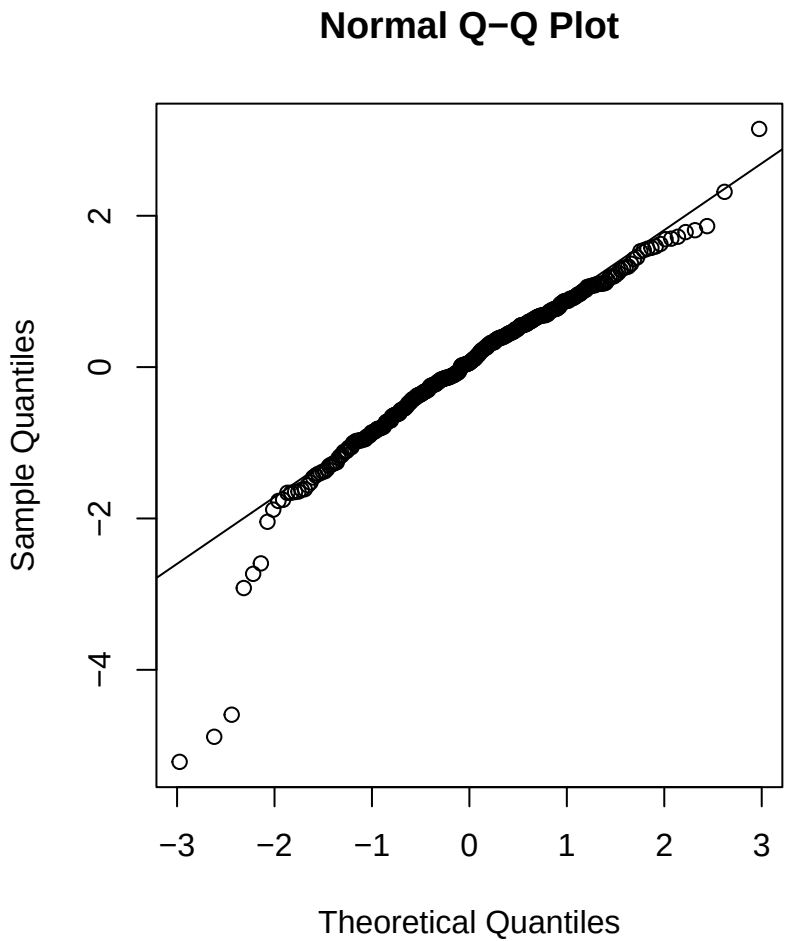

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

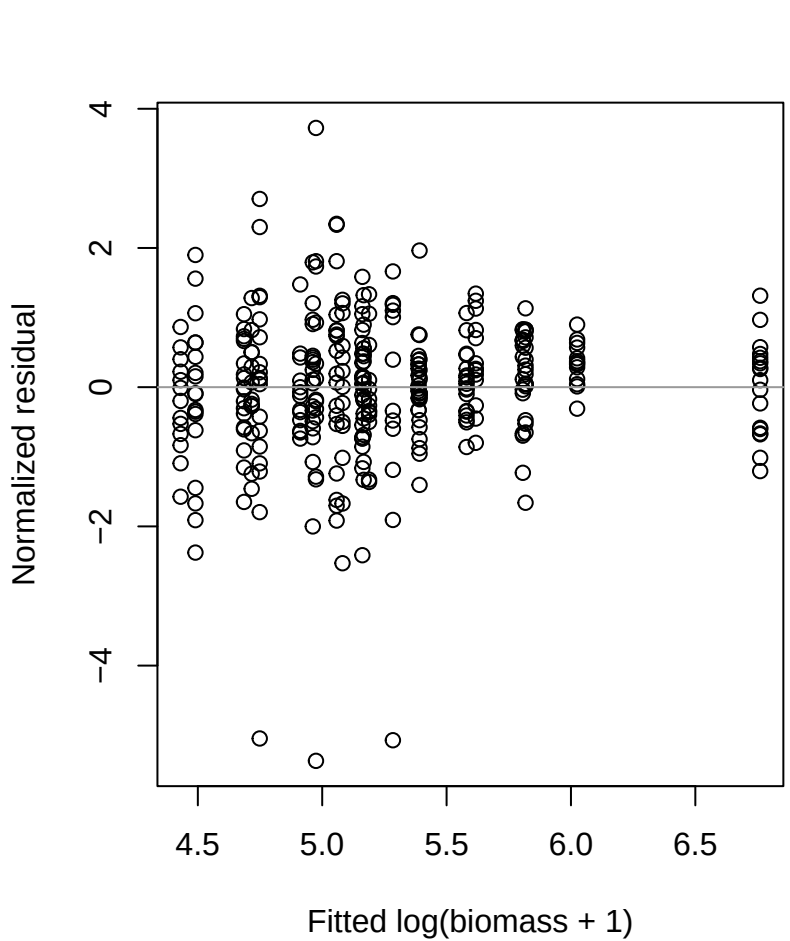

Normalized residual
4
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
Fitted log(biomass + 1)

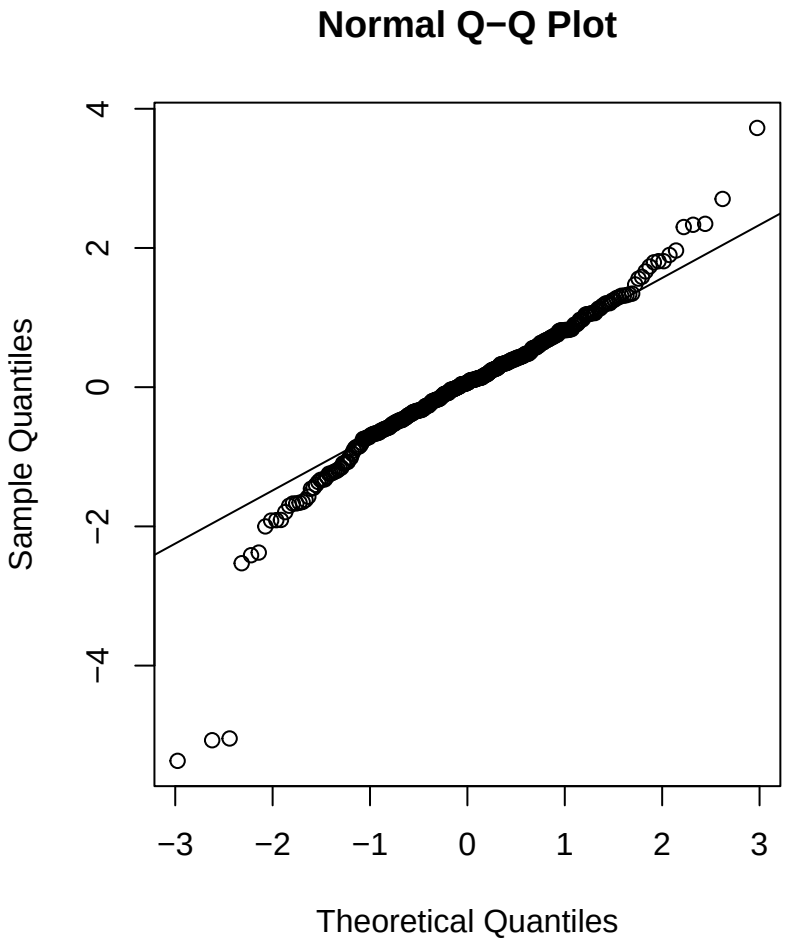

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

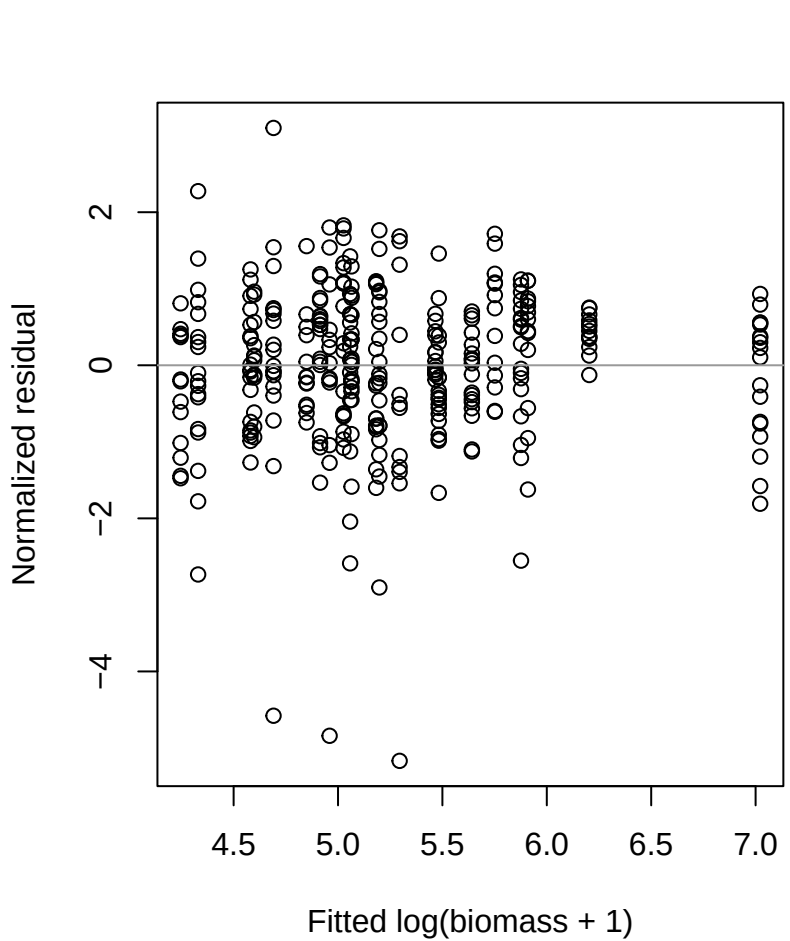
red_biomass__2022__2005__site_independent

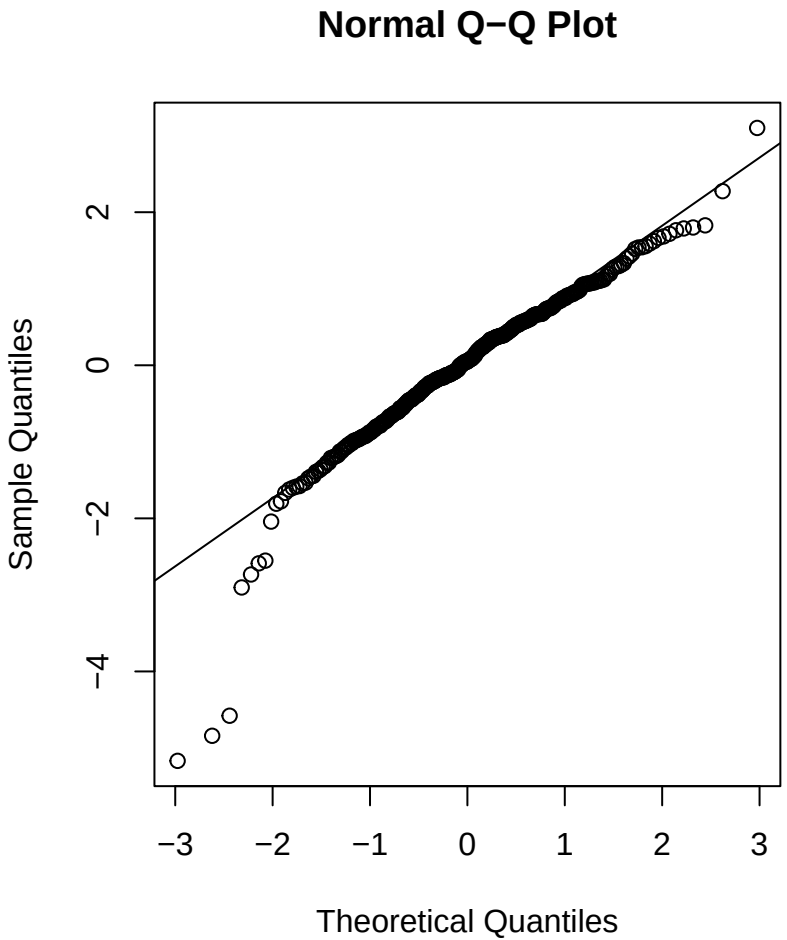


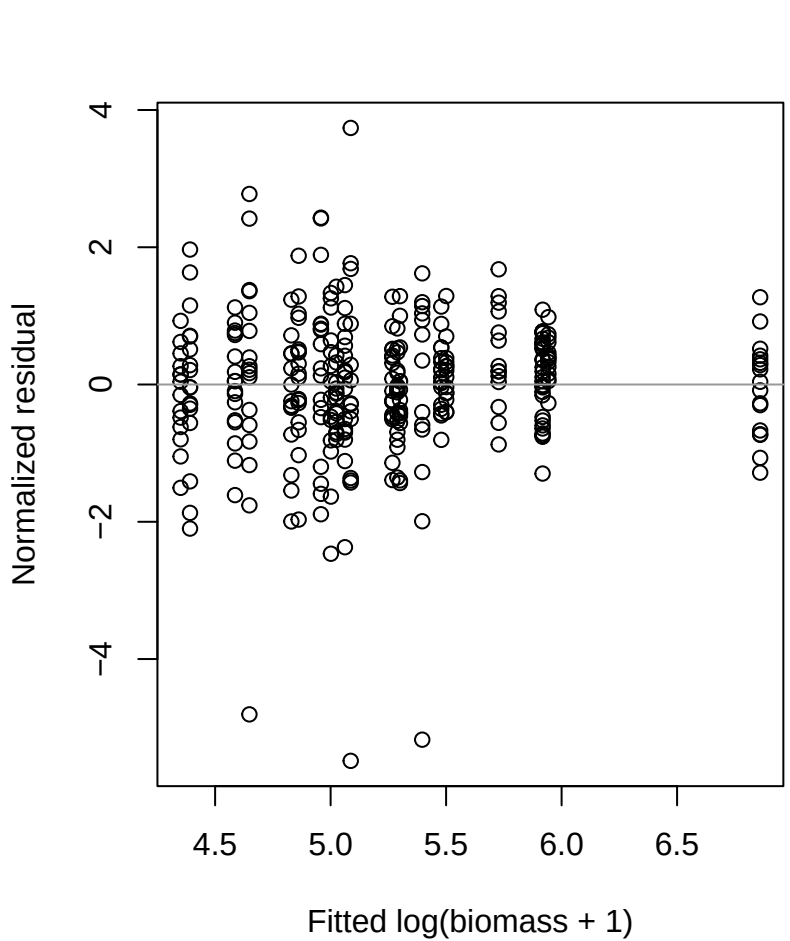


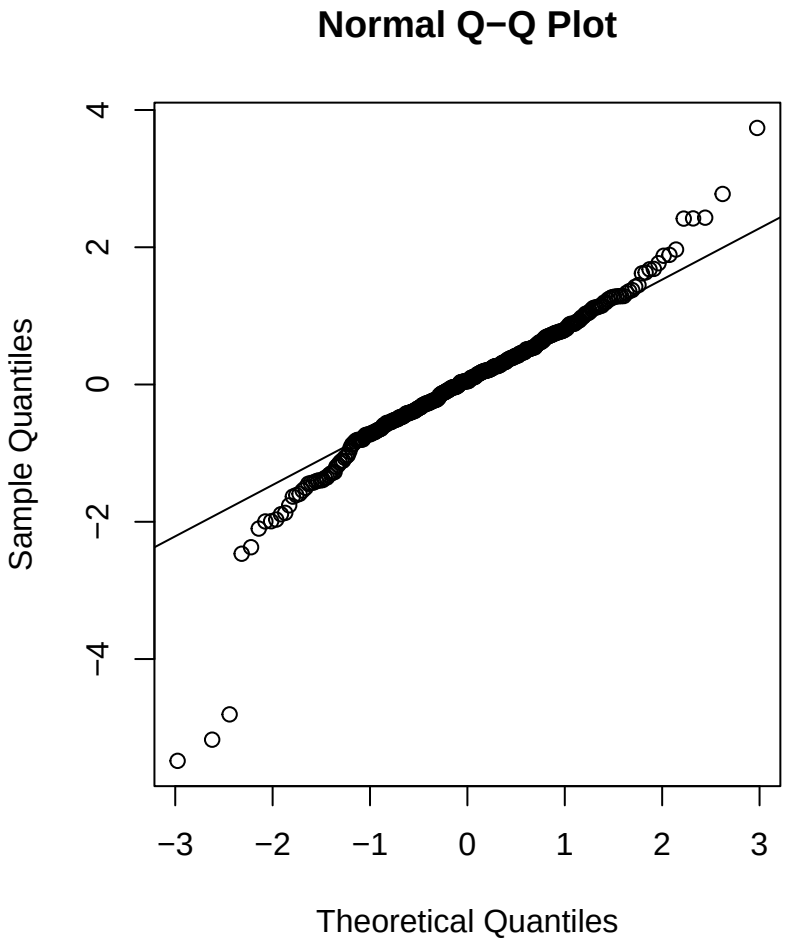

Normalized residual
4
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
Fitted log(biomass + 1)

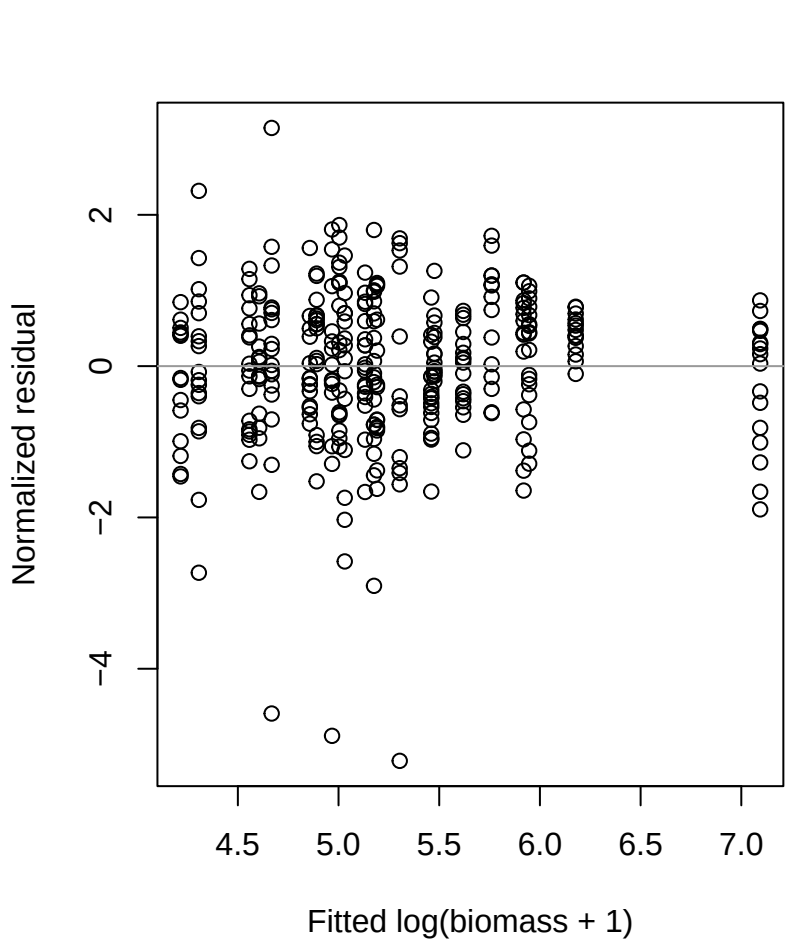

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

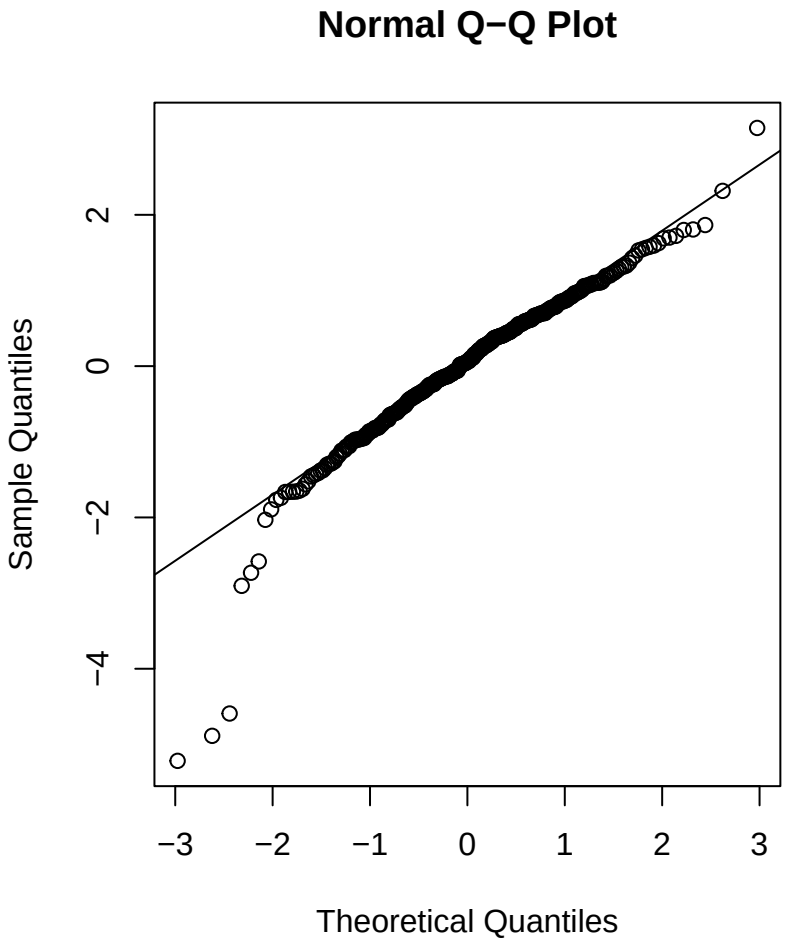

Normalized residual
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
7.0
Fitted log(biomass + 1)


Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

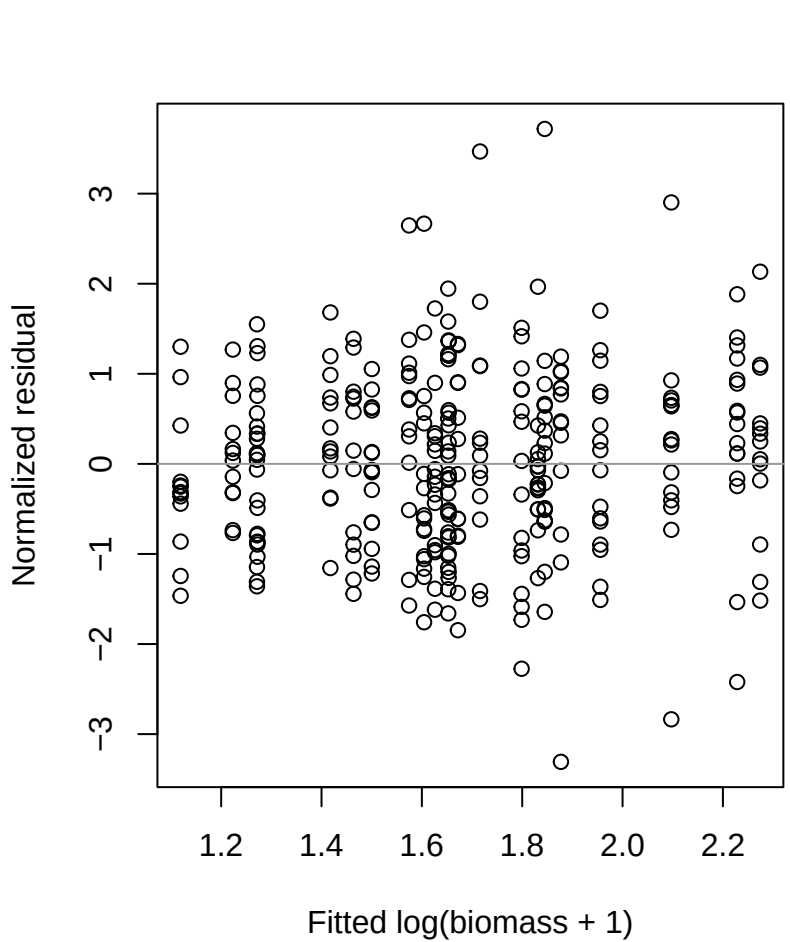

Normalized residual
4
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
Fitted log(biomass + 1)

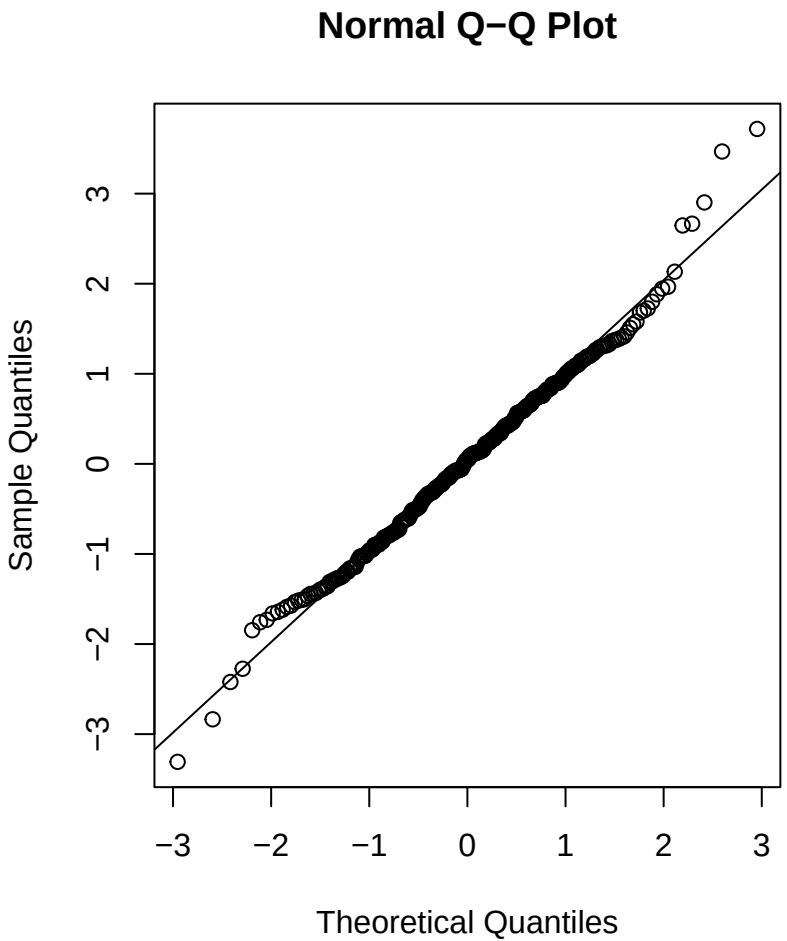

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Normalized residual
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
7.0
Fitted log(biomass + 1)

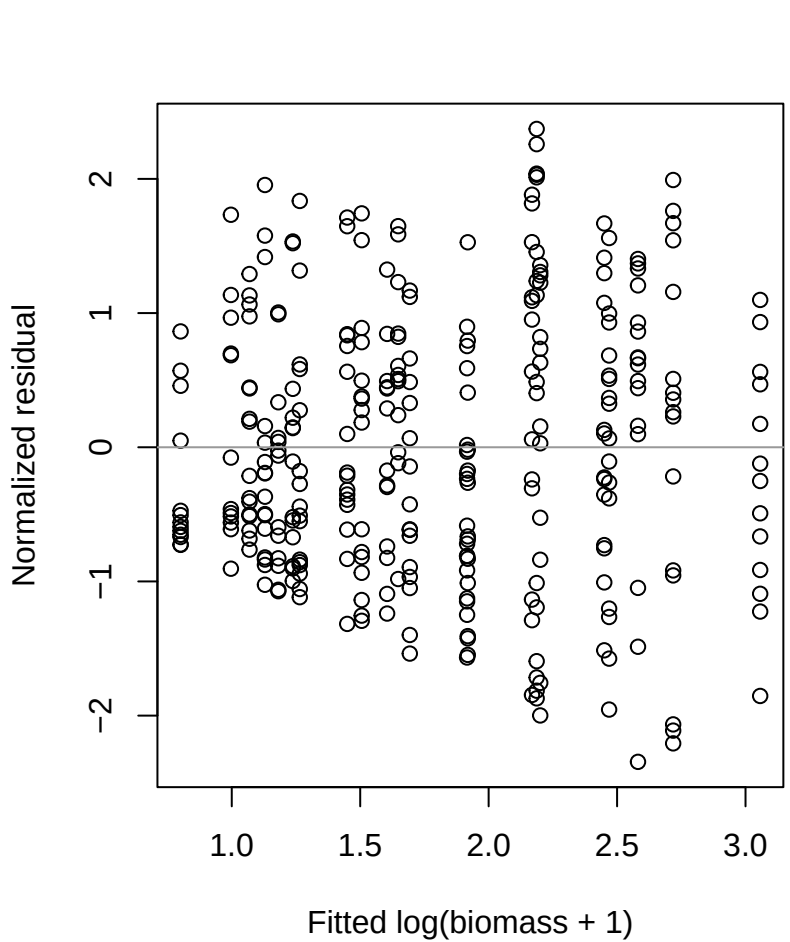

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

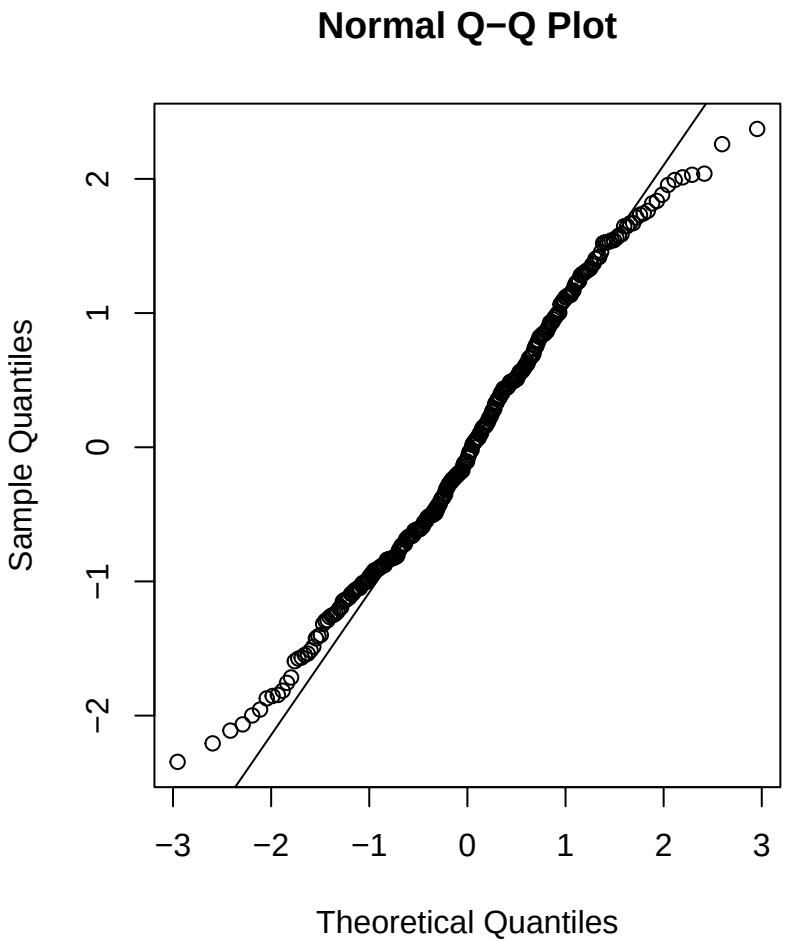
red_small_mass__2019__2004__site_calendar_AR1



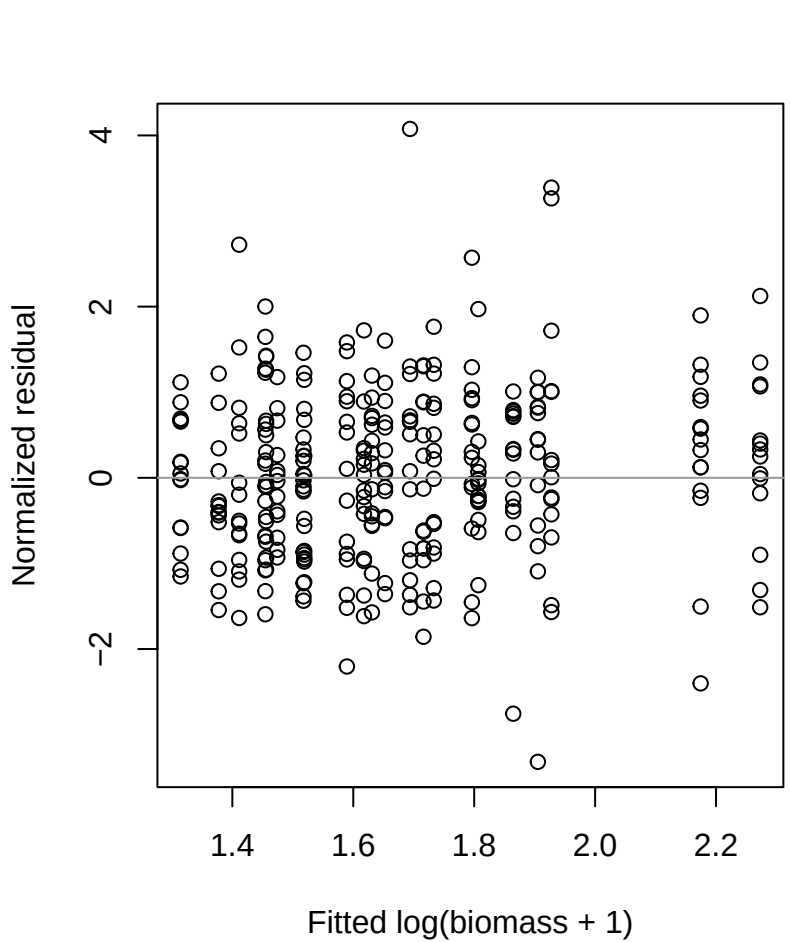


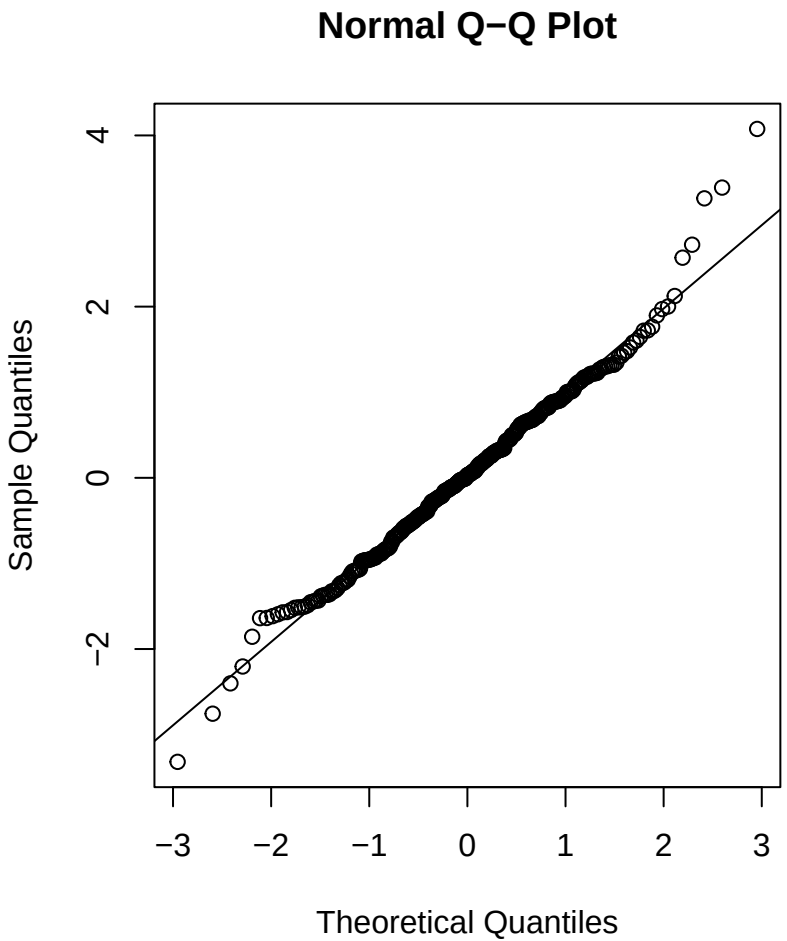
red_small_mass__2019__2004__site_independent



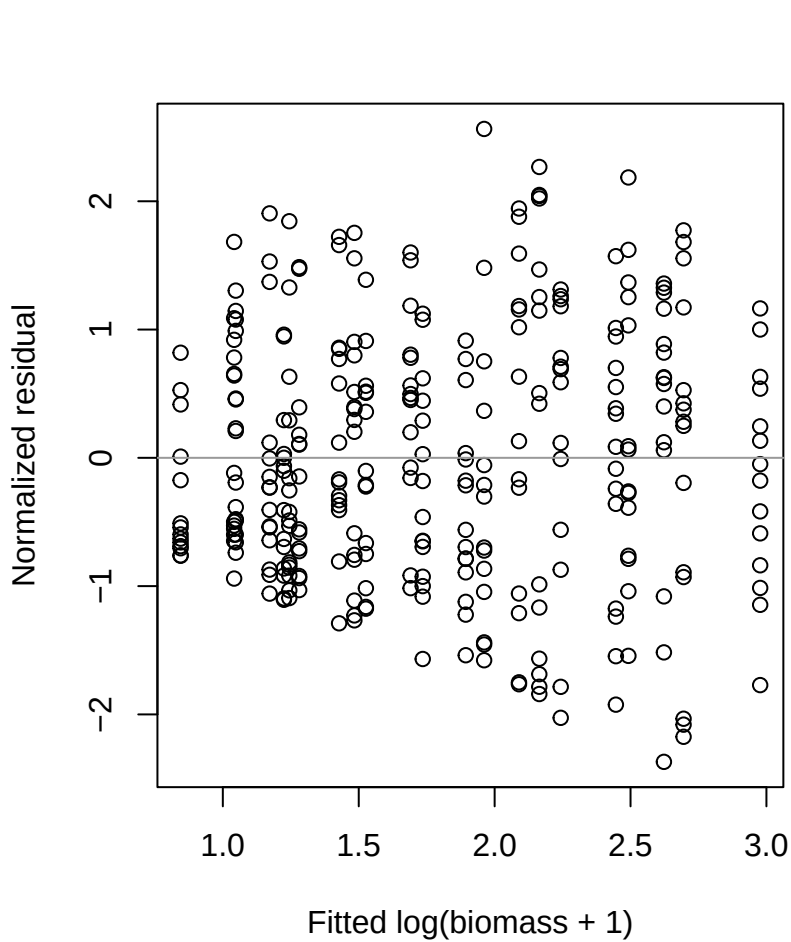


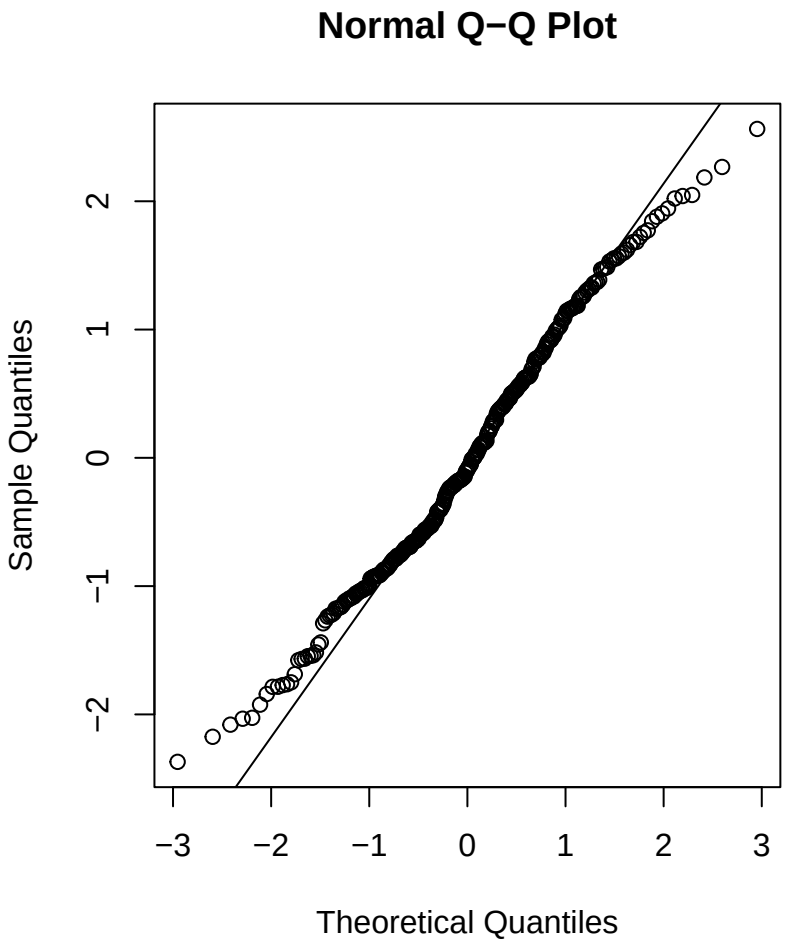
red_small_mass__2019__2005__site_calendar_AR1

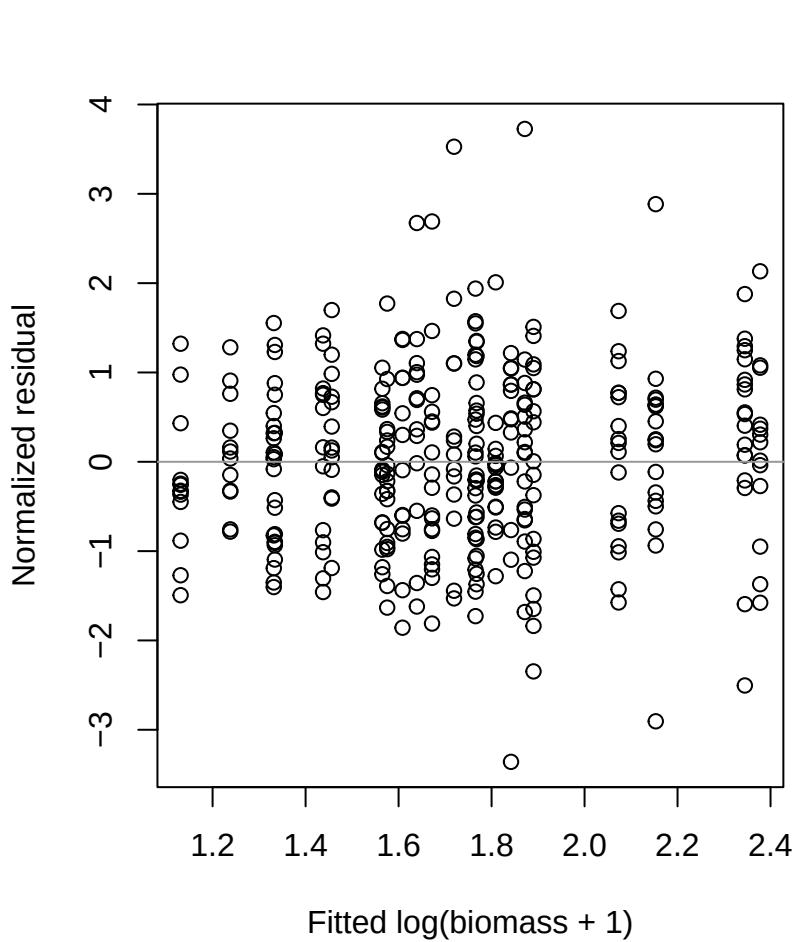


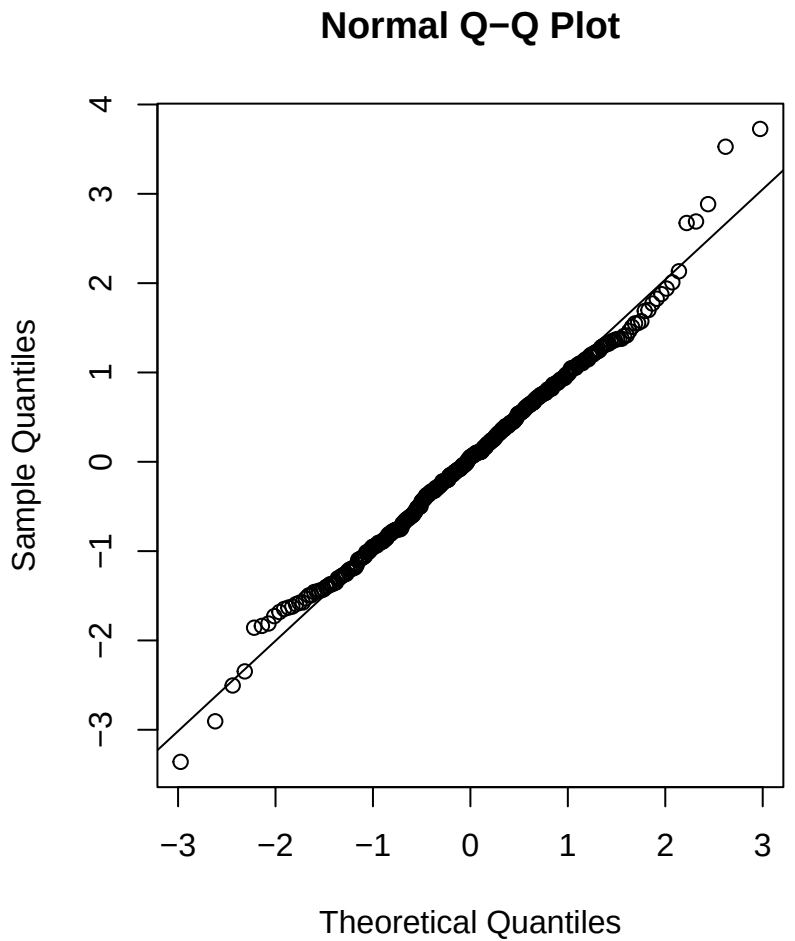


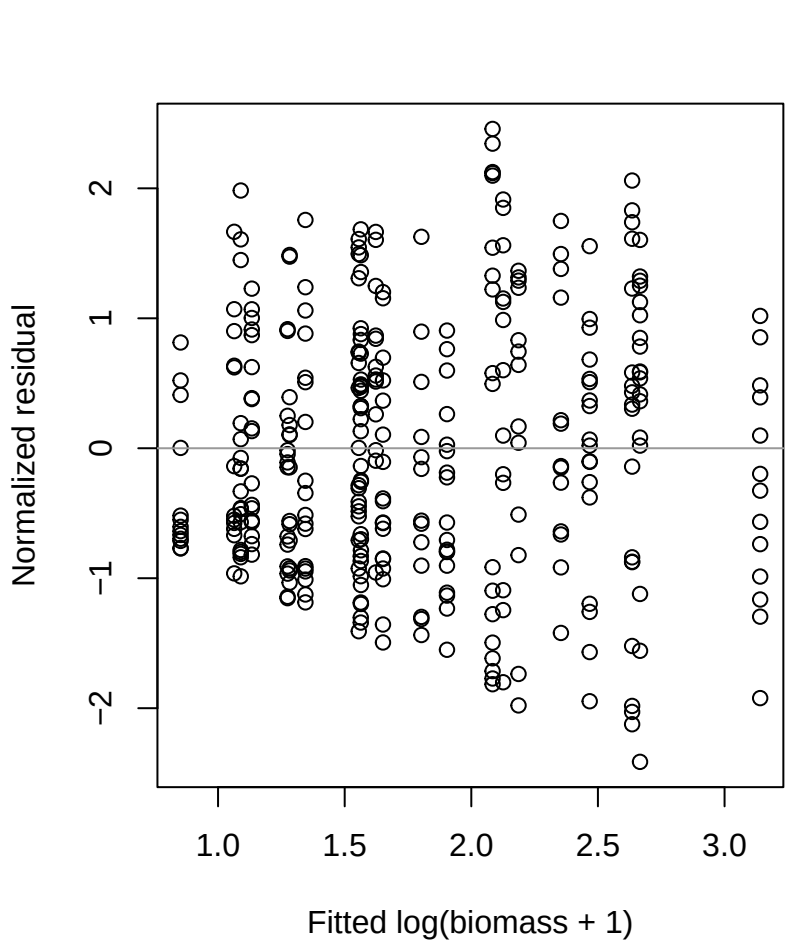
red_small_mass__2019__2005__site_independent

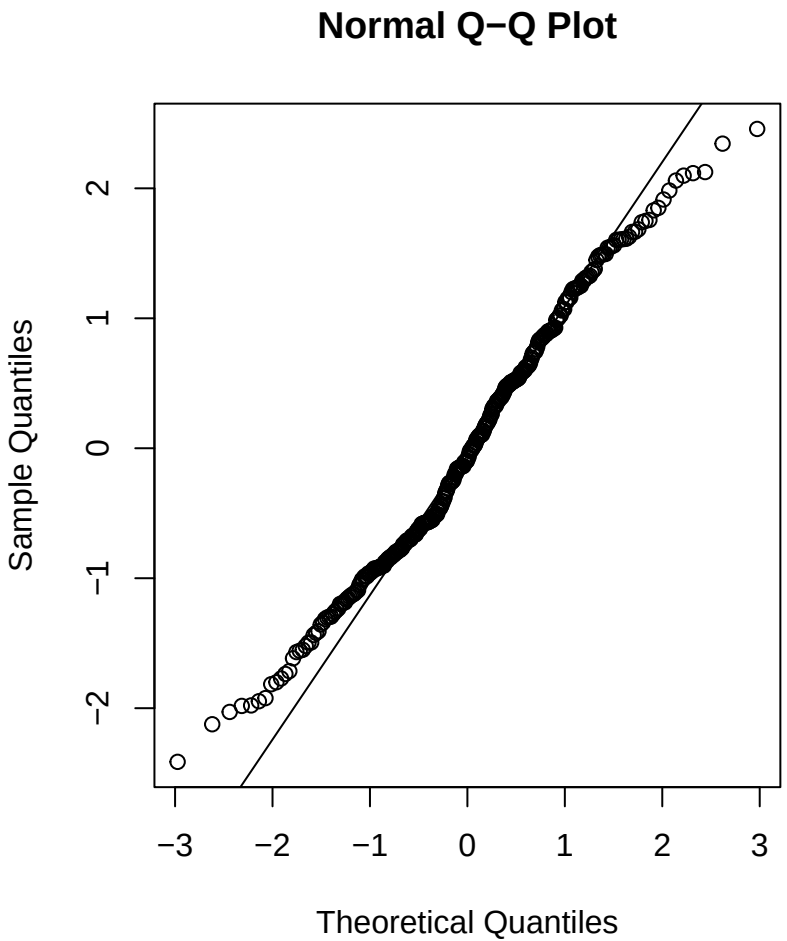




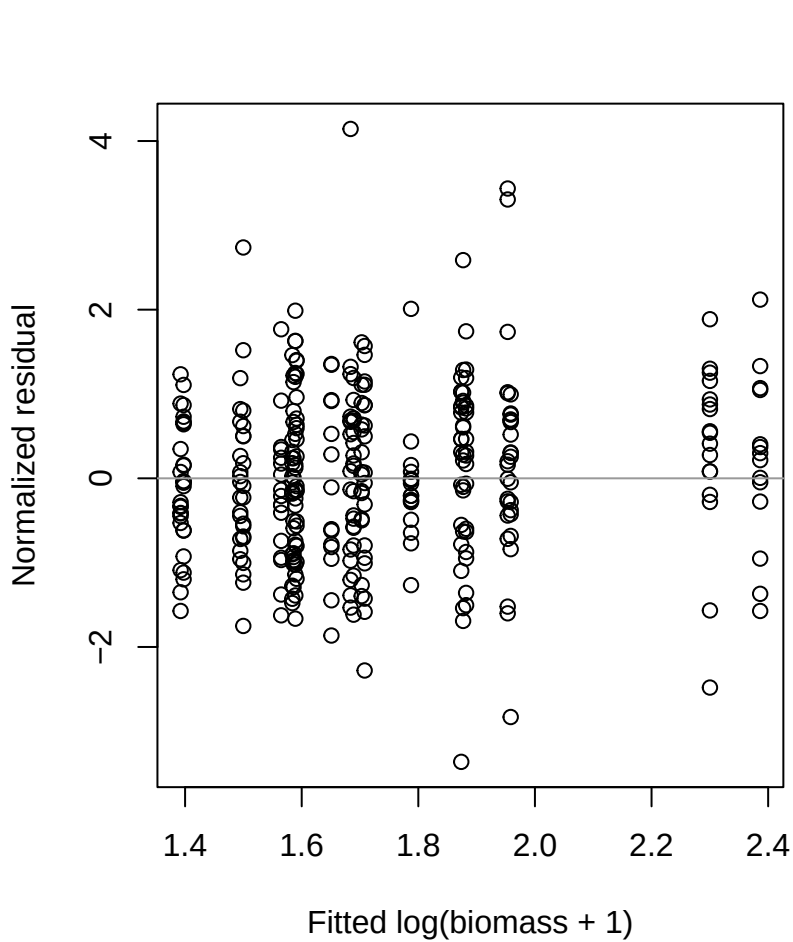

Normalized residual
Fitted log(biomass + 1)
−3
−2
−1
0
1
2
3
4
1.2
1.4
1.6
1.8
2.0
2.2
2.4

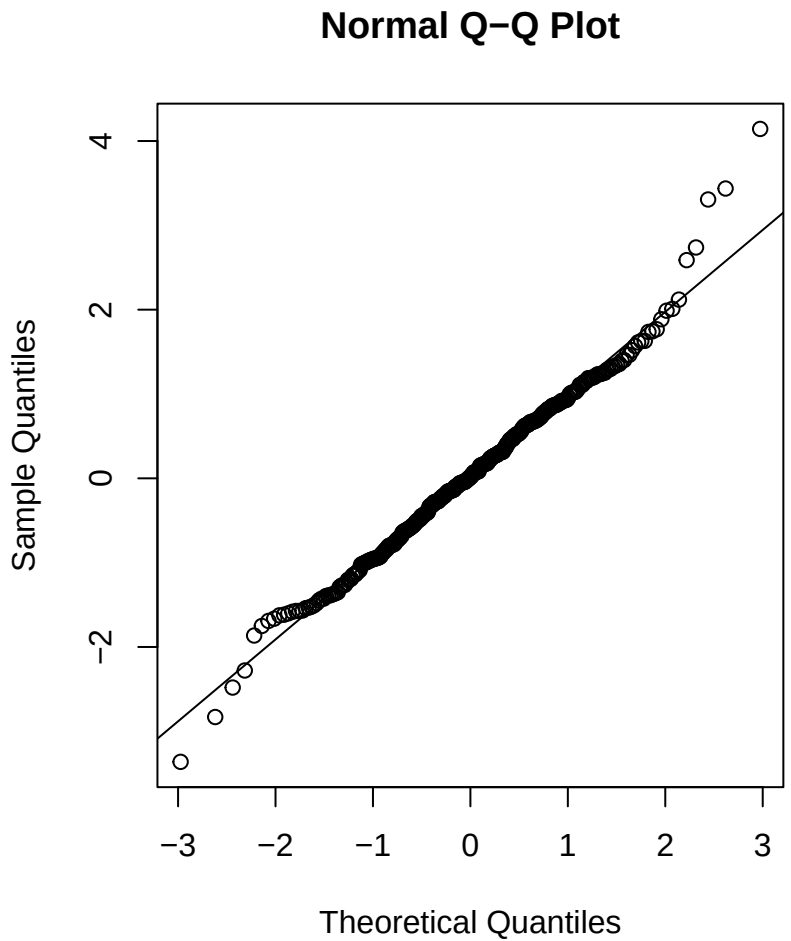

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
4

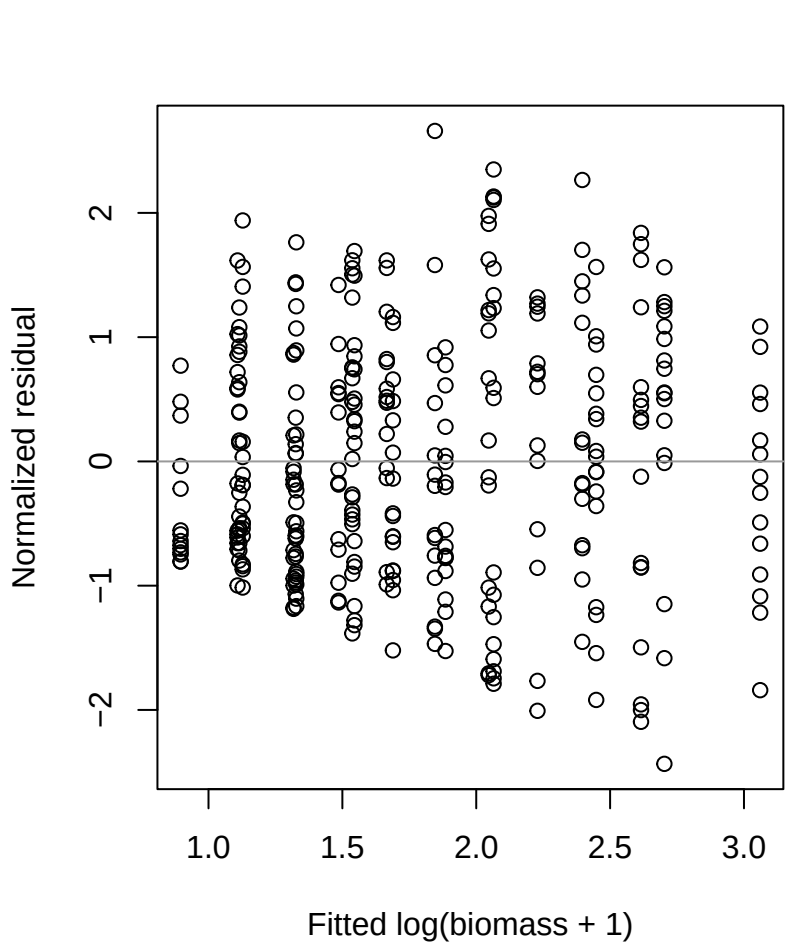

Normalized residual
2
1
0
−1
−2
1.0
1.5
2.0
2.5
3.0
Fitted log(biomass + 1)

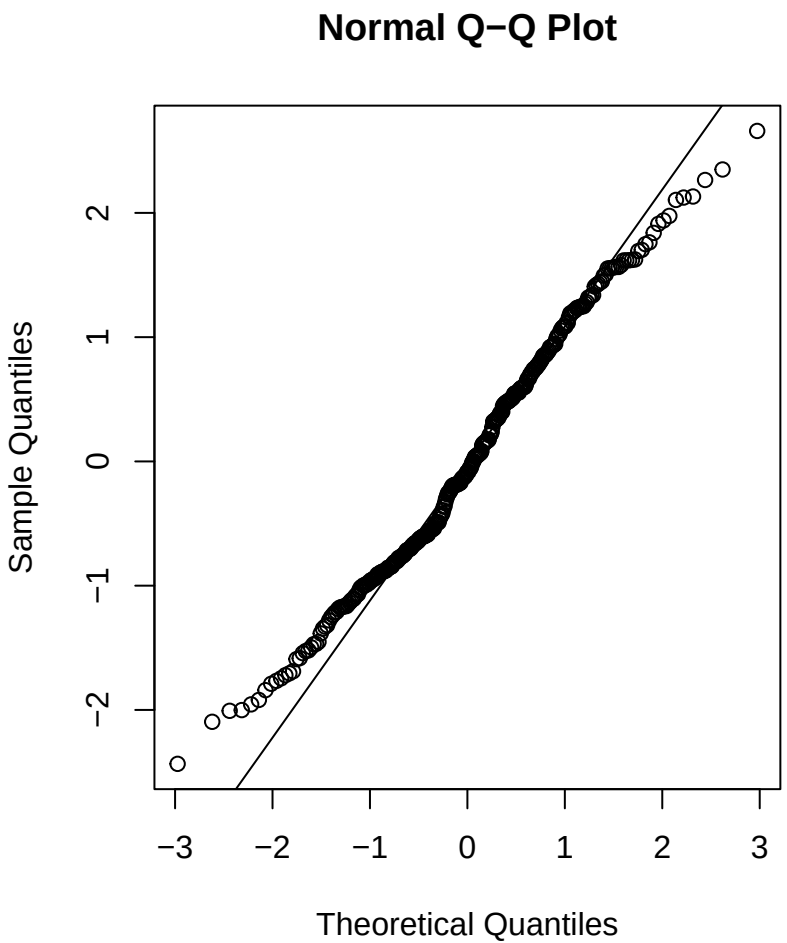

Normal Q−Q Plot
Sample Quantiles
2
1
0
−1
−2
−3
−2
−1
0
1
2
3
Theoretical Quantiles

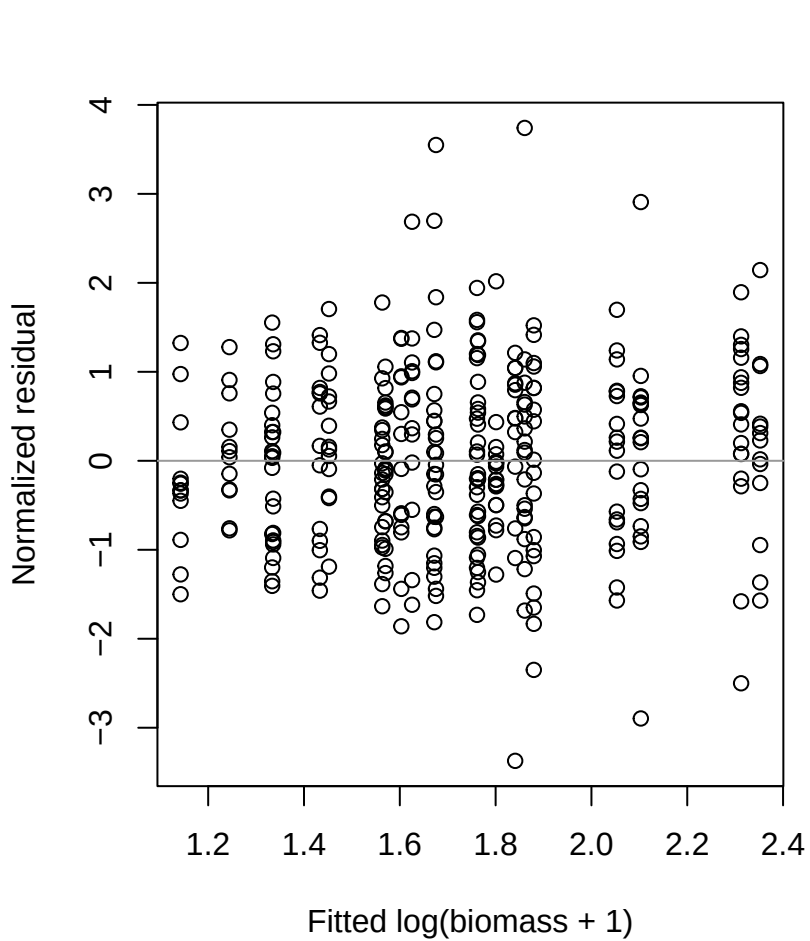
red_small_mass__2022__2005__site_calendar_AR1

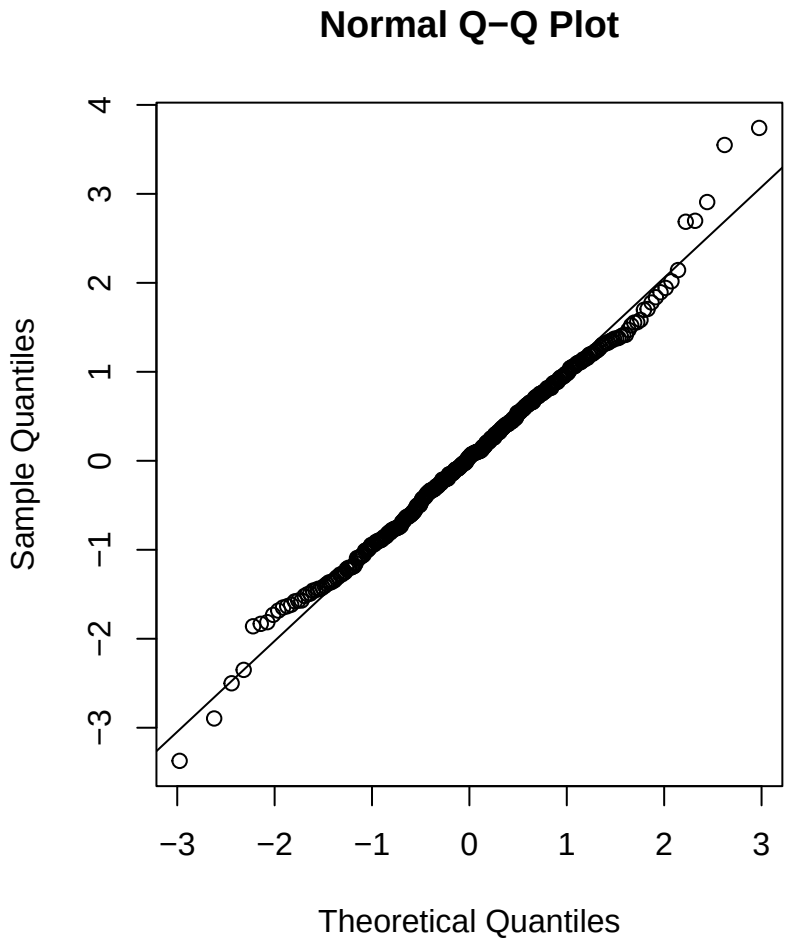




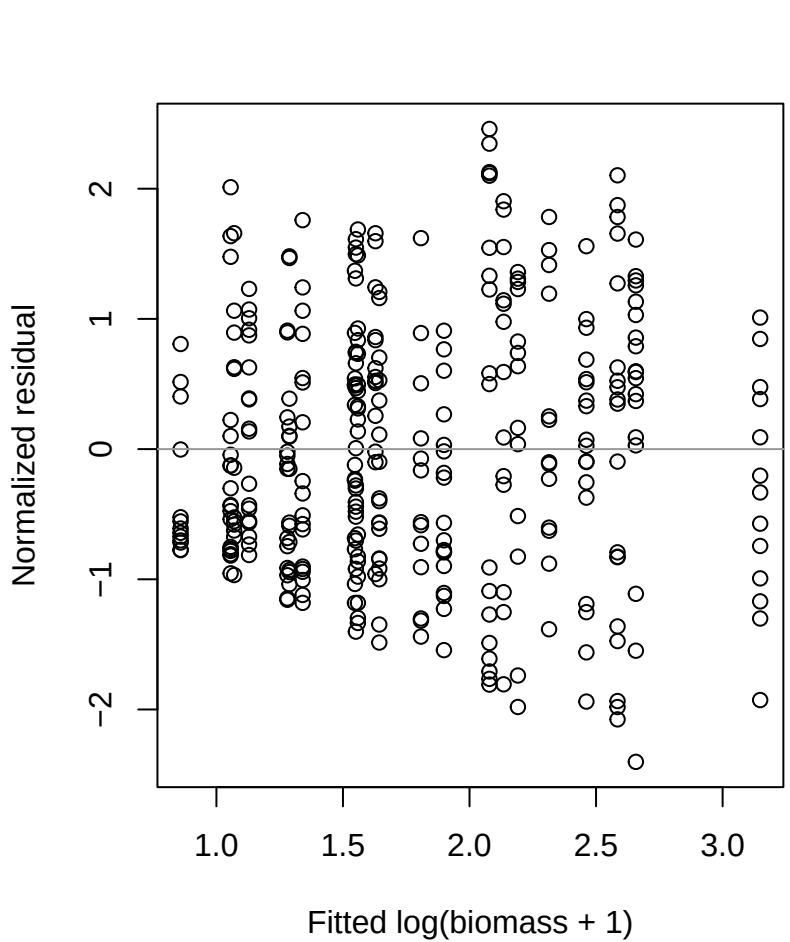

Normalized residual
Fitted log(biomass + 1)
1.0
1.5
2.0
2.5
3.0
−2
−1
0
1
2

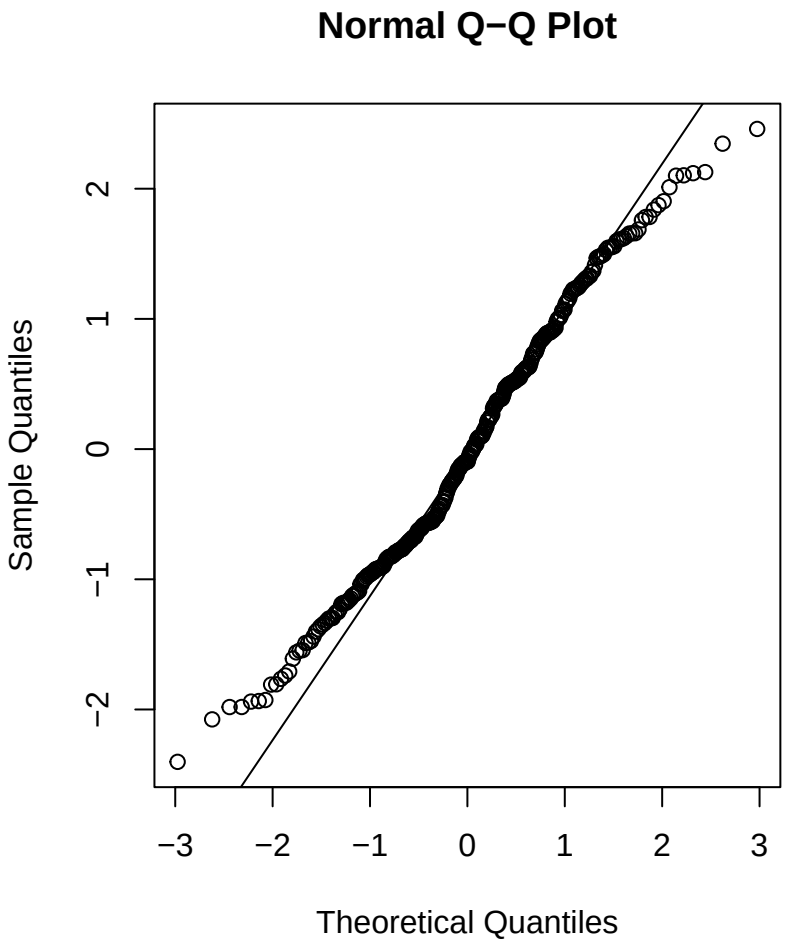

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3


Normalized residual
Fitted log(biomass + 1)
−3
−2
−1
0
1
2
3
4
1.2
1.4
1.6
1.8
2.0
2.2
2.4

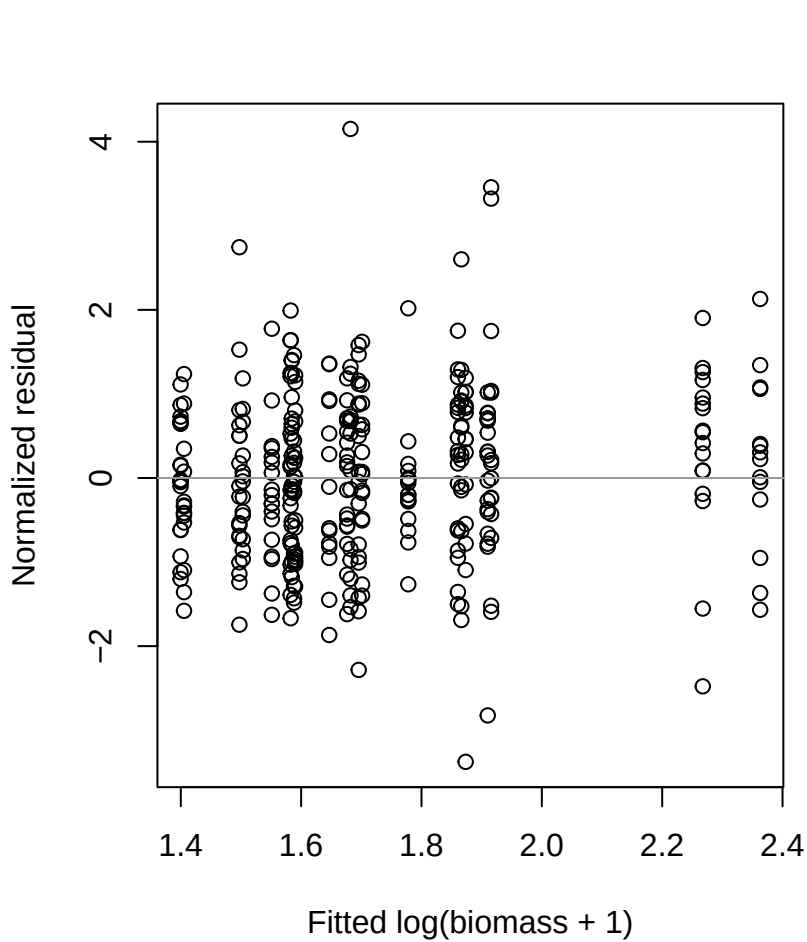

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
4

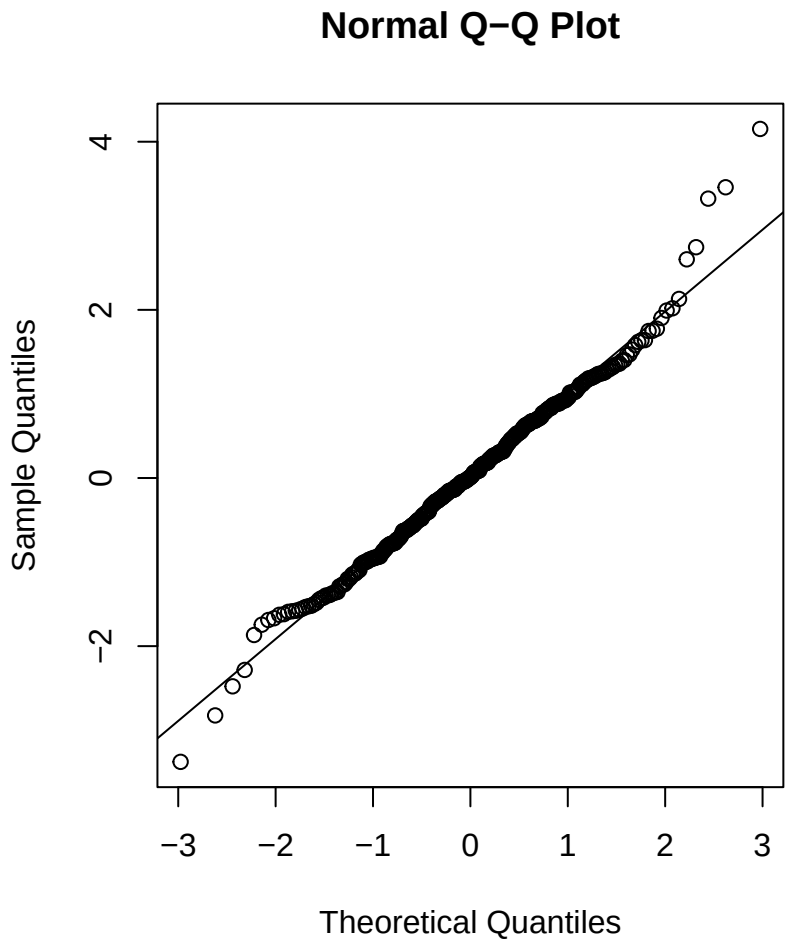
red_small_mass__2023__2004__site_independent

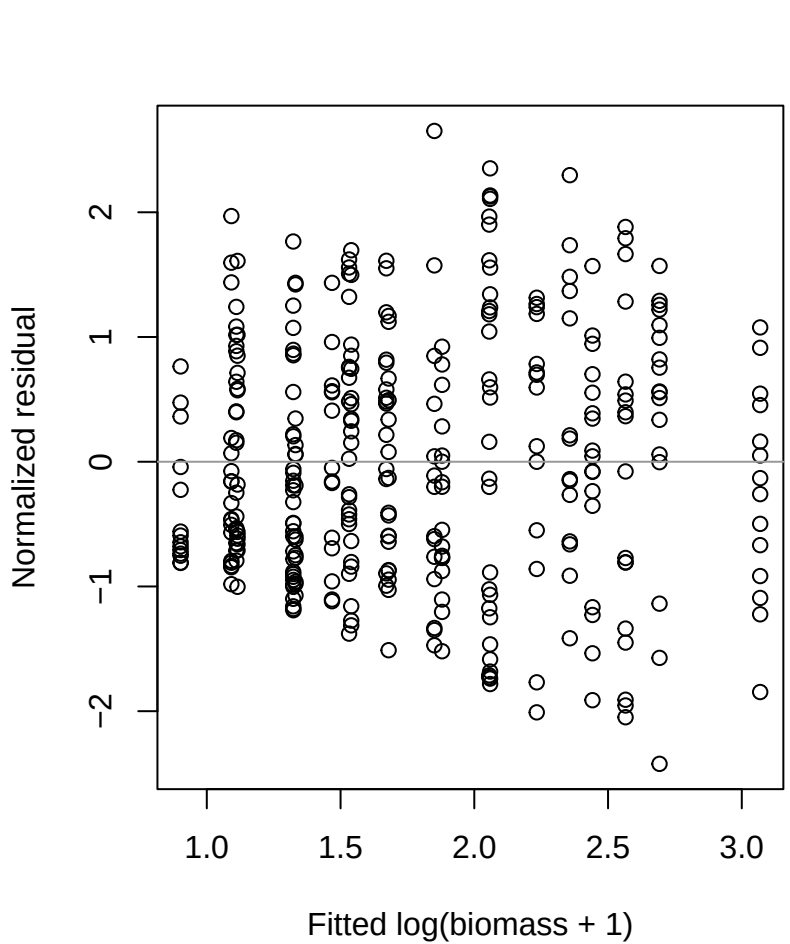


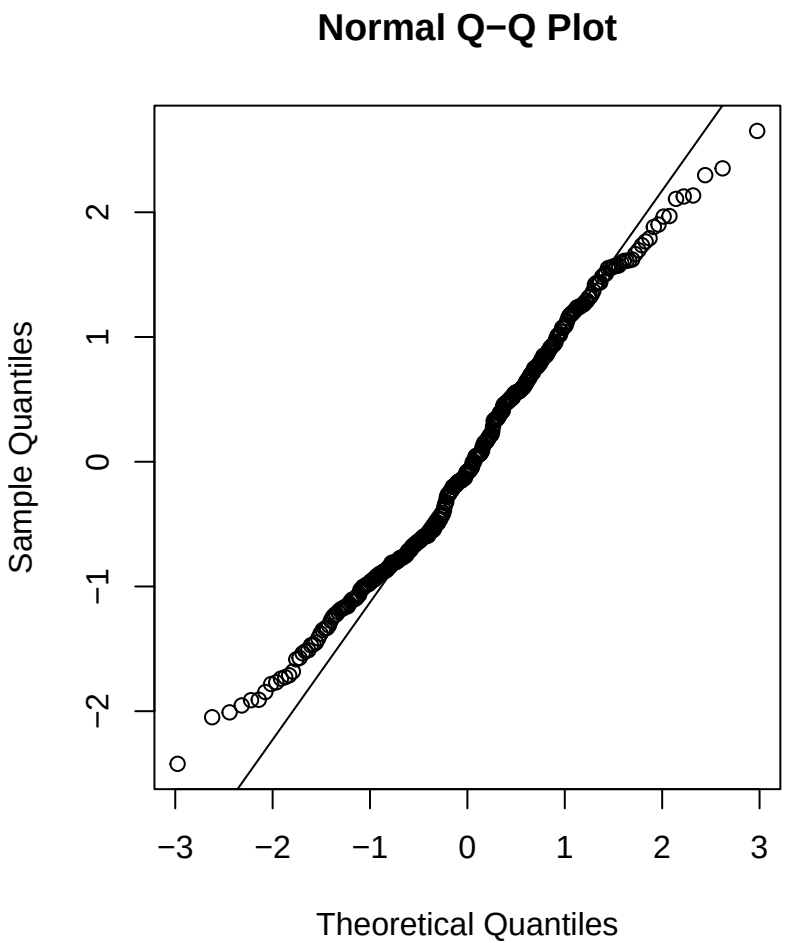


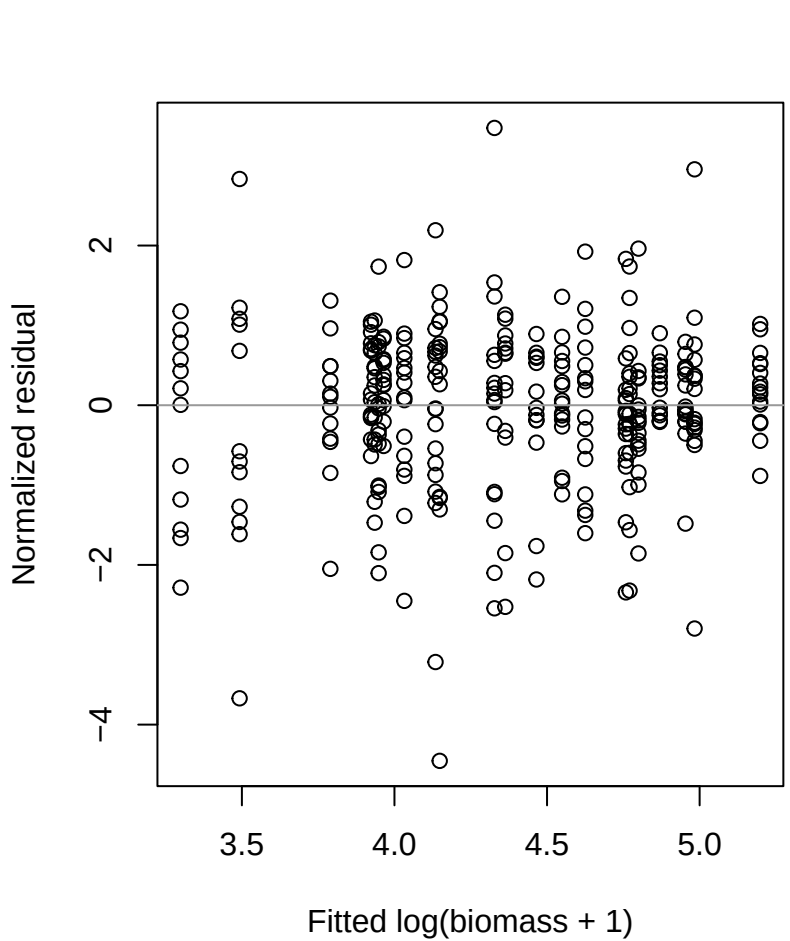
red_small_mass__2023__2005__site_calendar_AR1

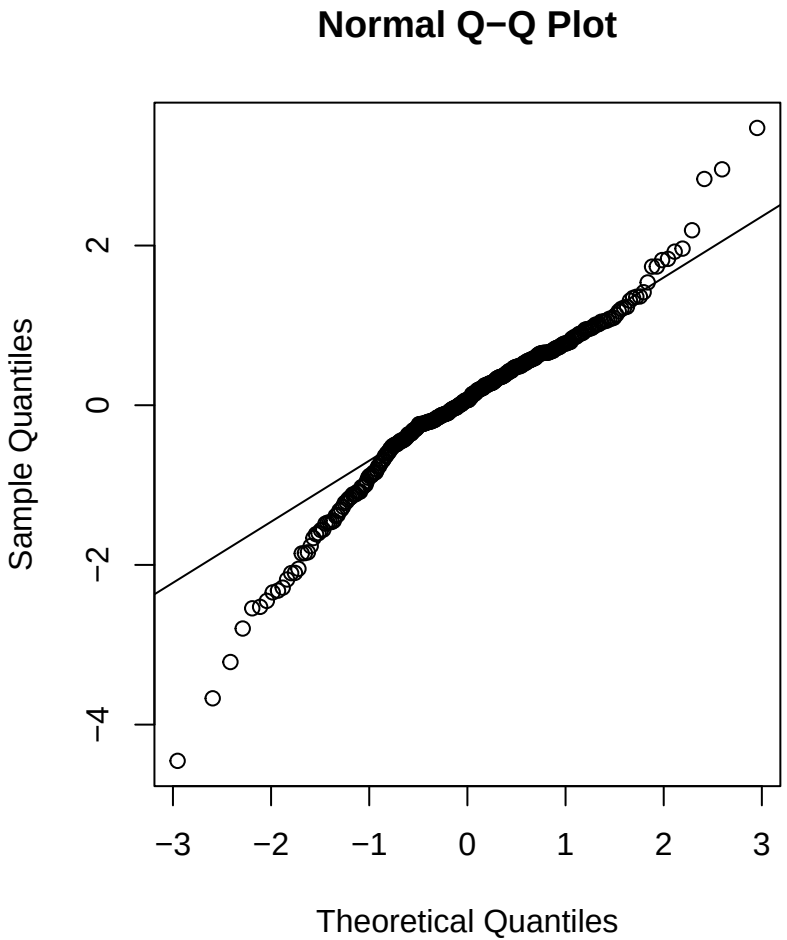




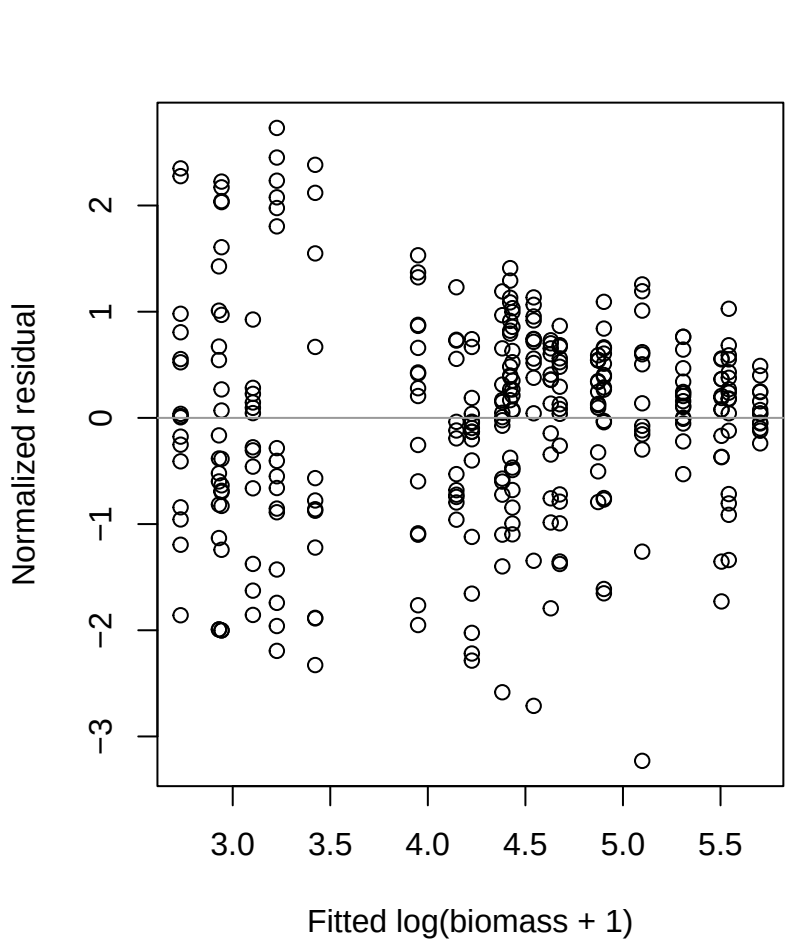

Normalized residual
Fitted log(biomass + 1)
−2
−1
0
1
2
1.0
1.5
2.0
2.5
3.0

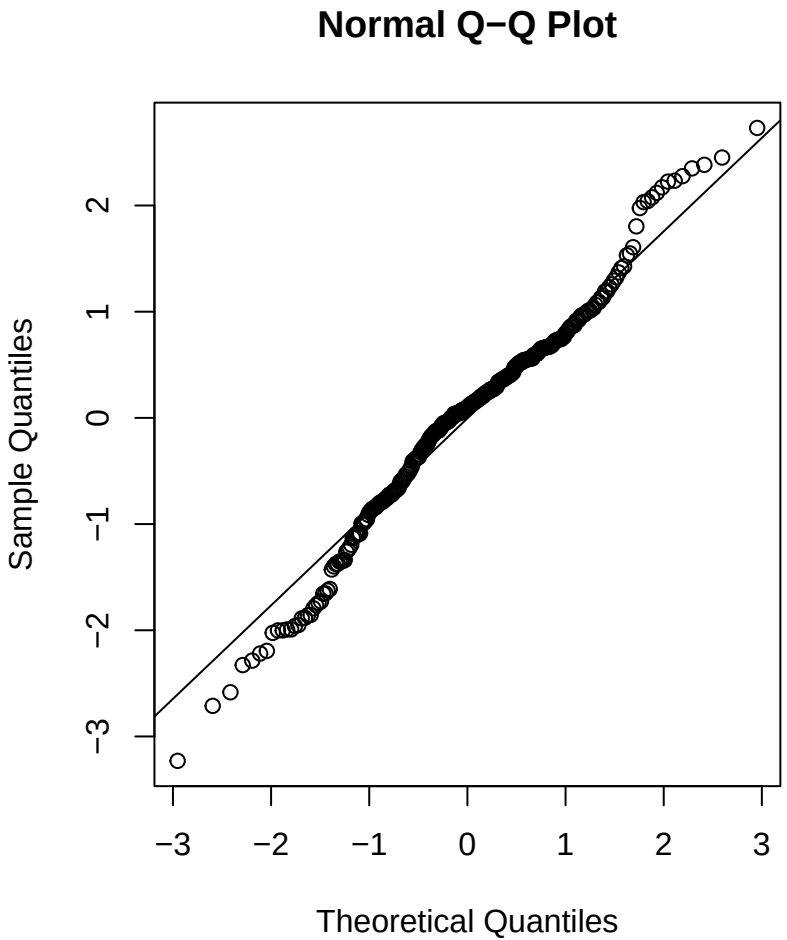

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2
−3
−2
−1
0
1
2
3

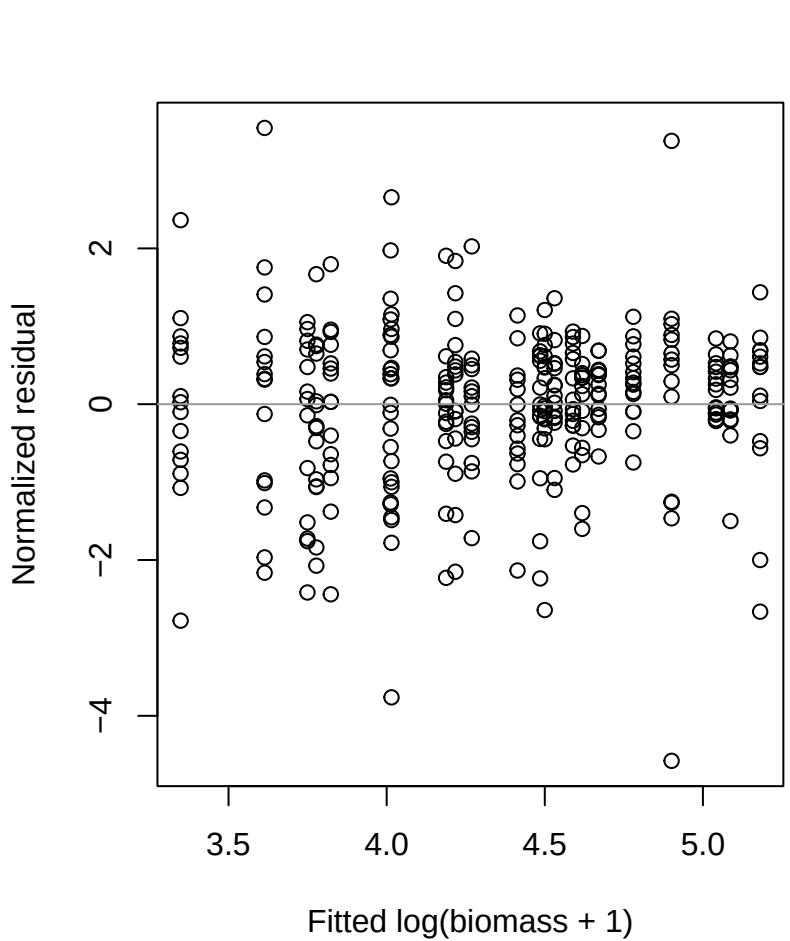

Normalized residual
2
0
−2
−4
3.5
4.0
4.5
5.0
Fitted log(biomass + 1)

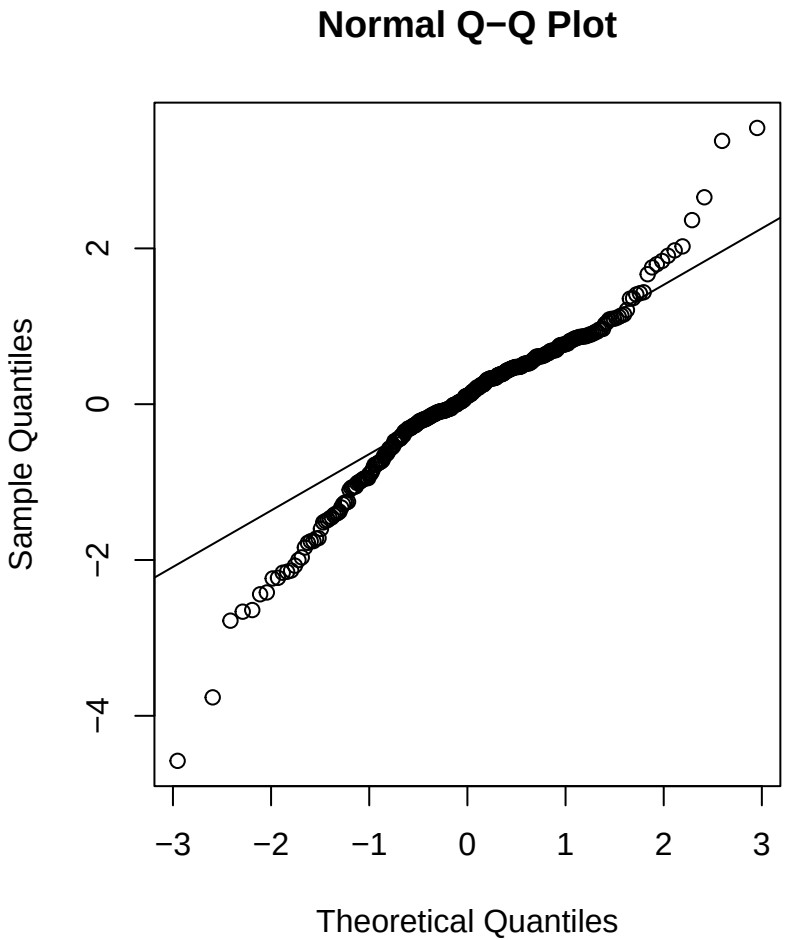

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

purple_biomass__2019__2004__site_independent

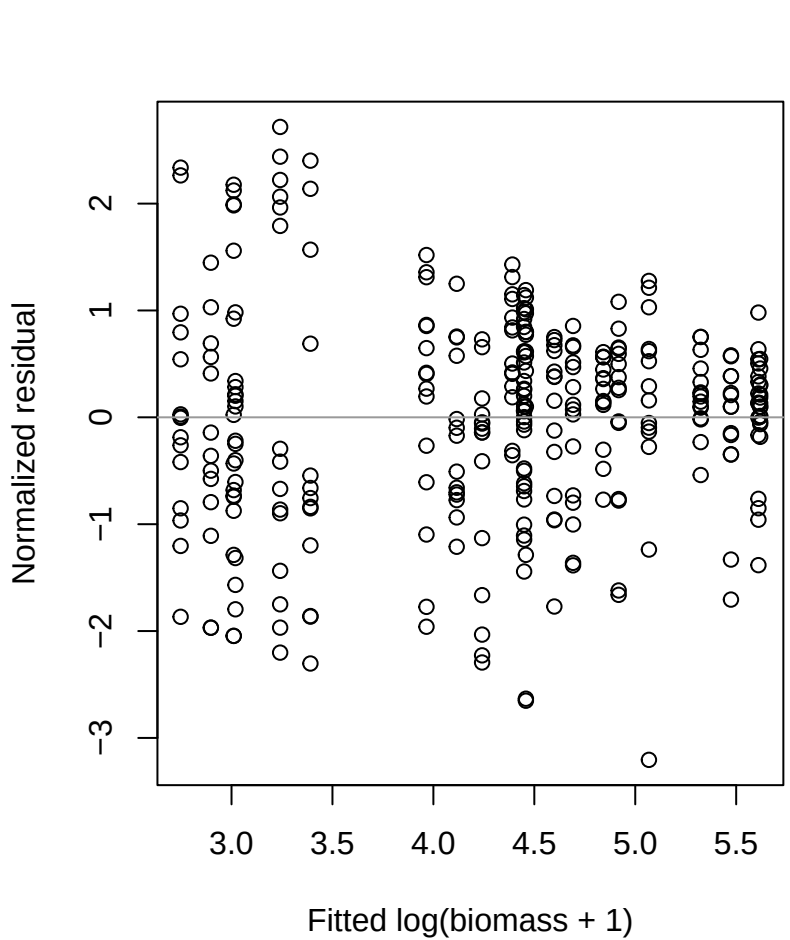


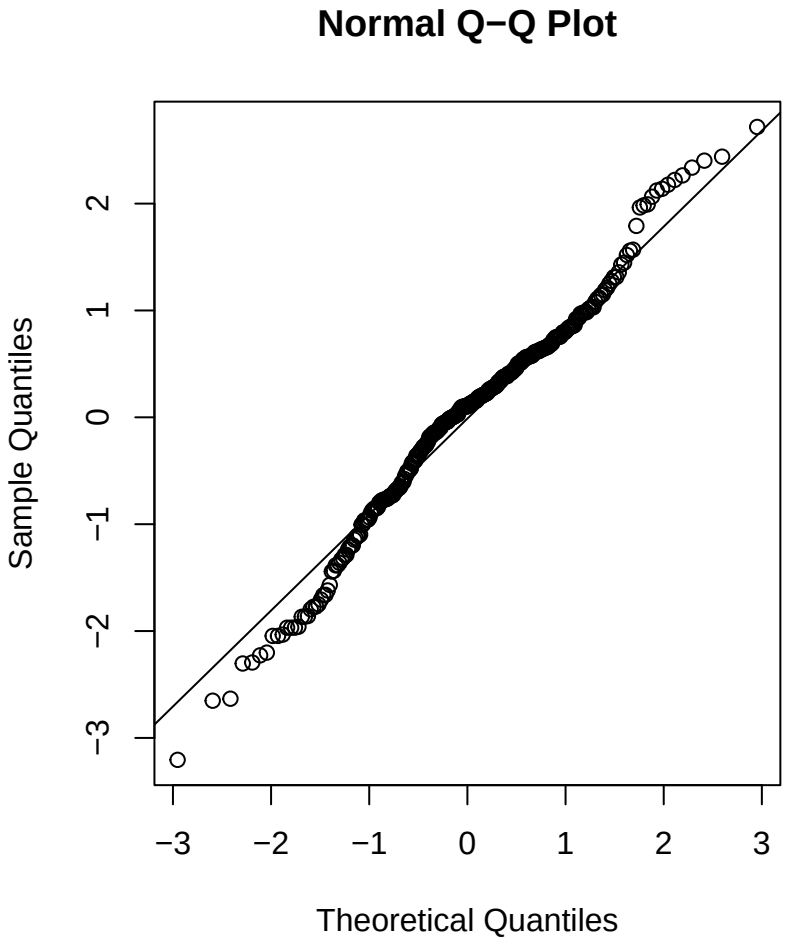


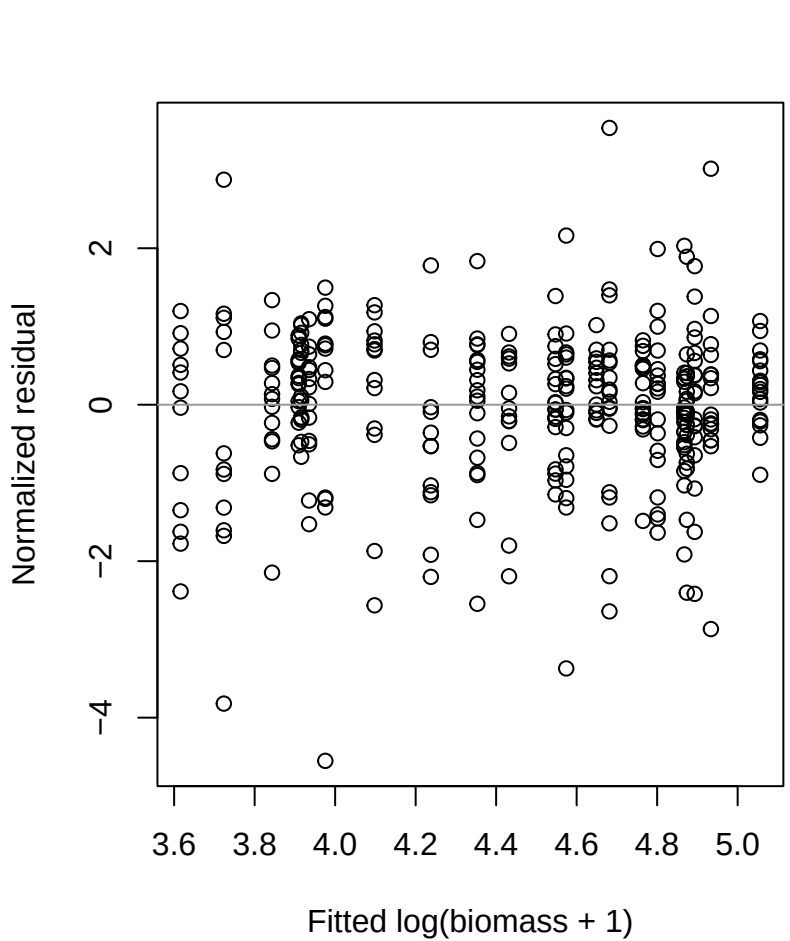

Normalized residual
2
0
−2
−4
3.5
4.0
4.5
5.0
Fitted log(biomass + 1)

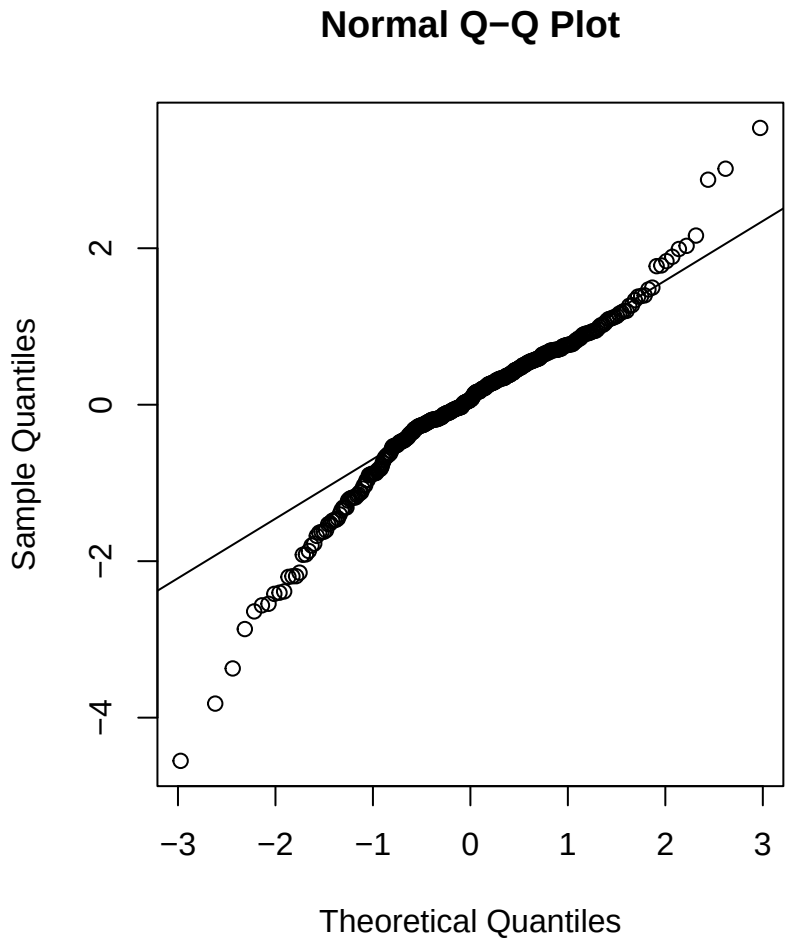

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

purple_biomass__2019__2005__site_independent

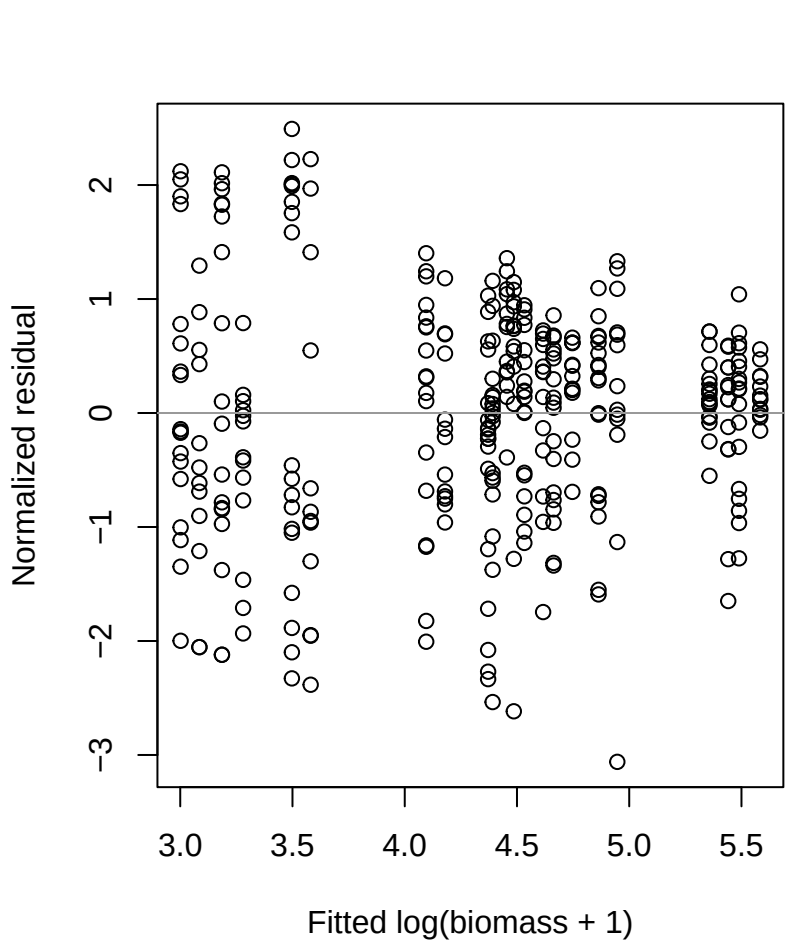


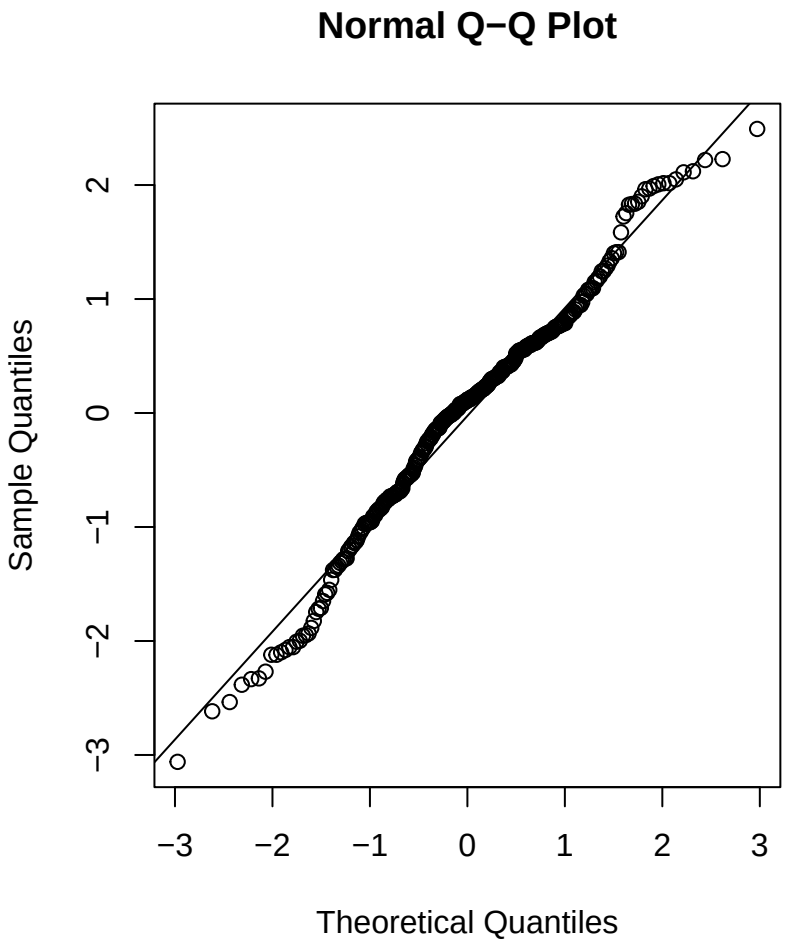


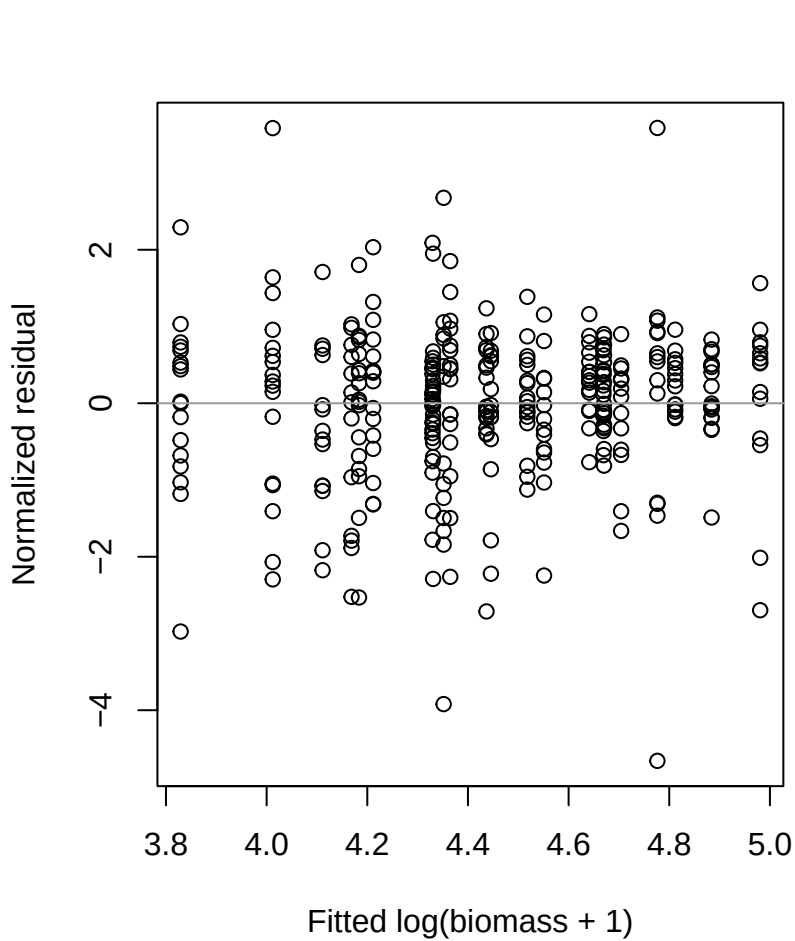

Normalized residual
2
0
−2
−4
3.6
3.8
4.0
4.2
4.4
4.6
4.8
5.0
Fitted log(biomass + 1)

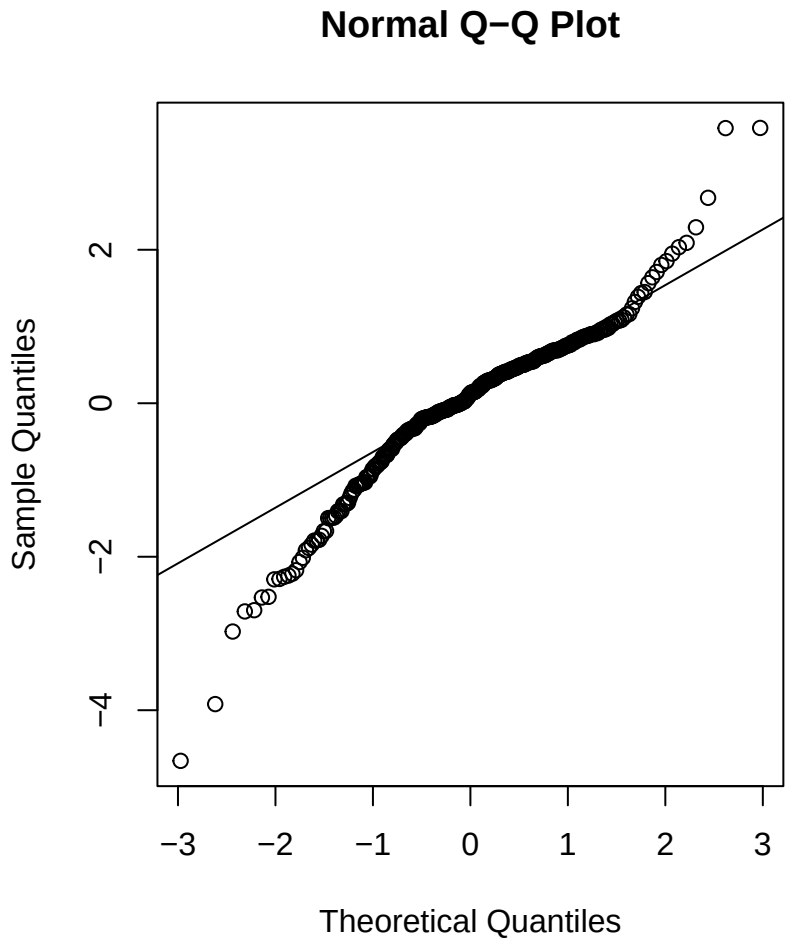

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

purple_biomass__2022__2004__site_independent

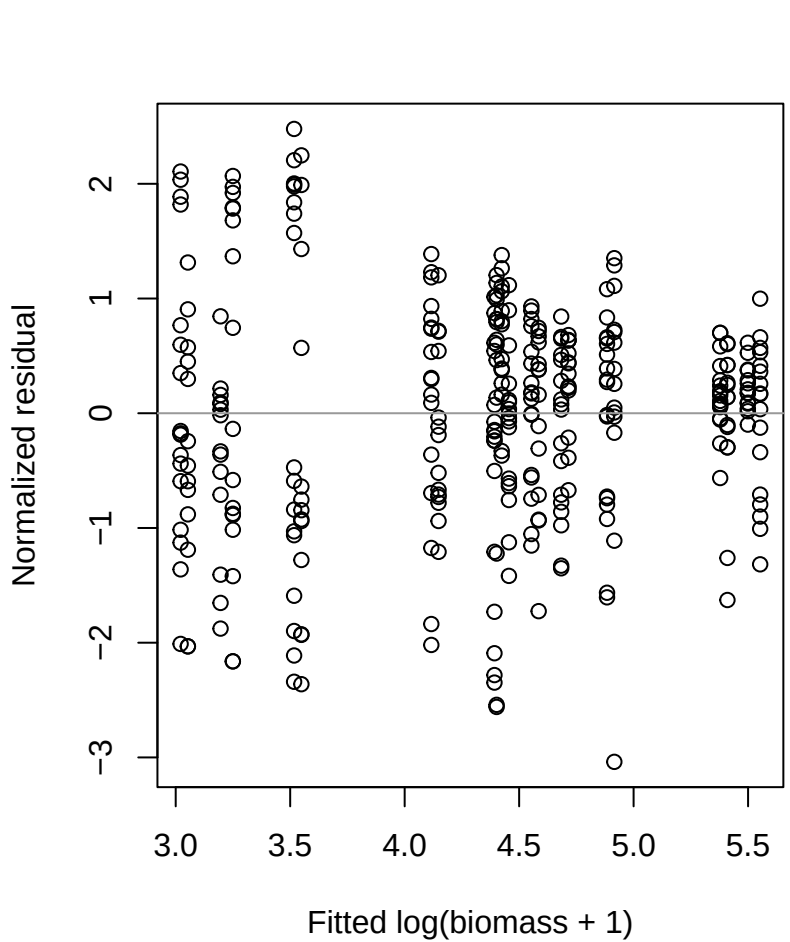


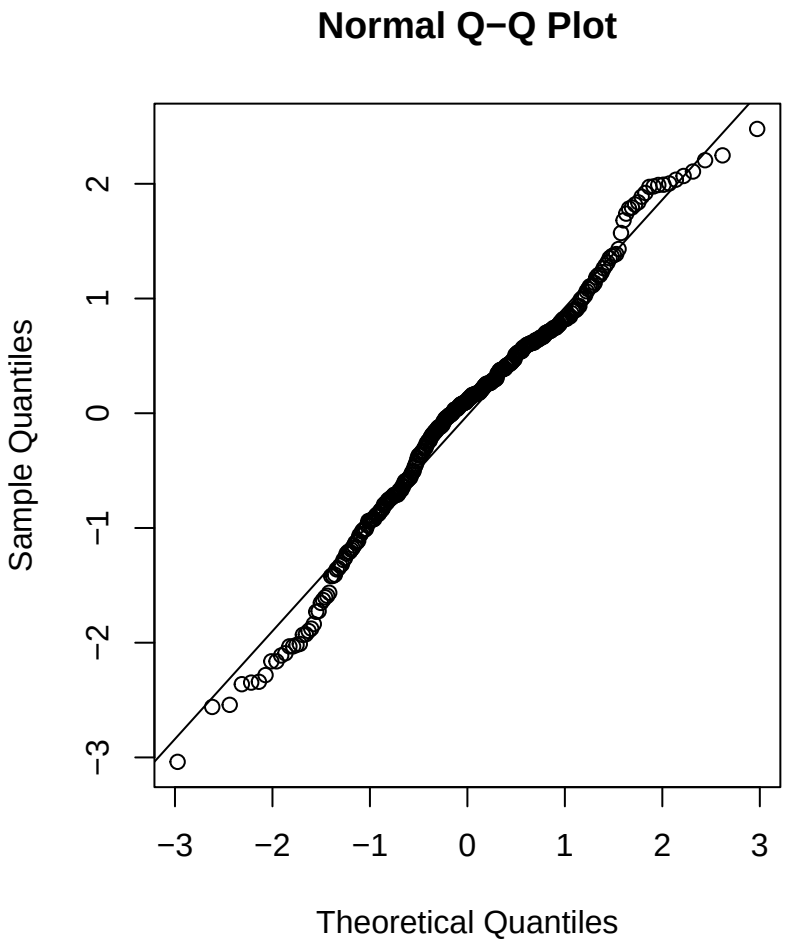


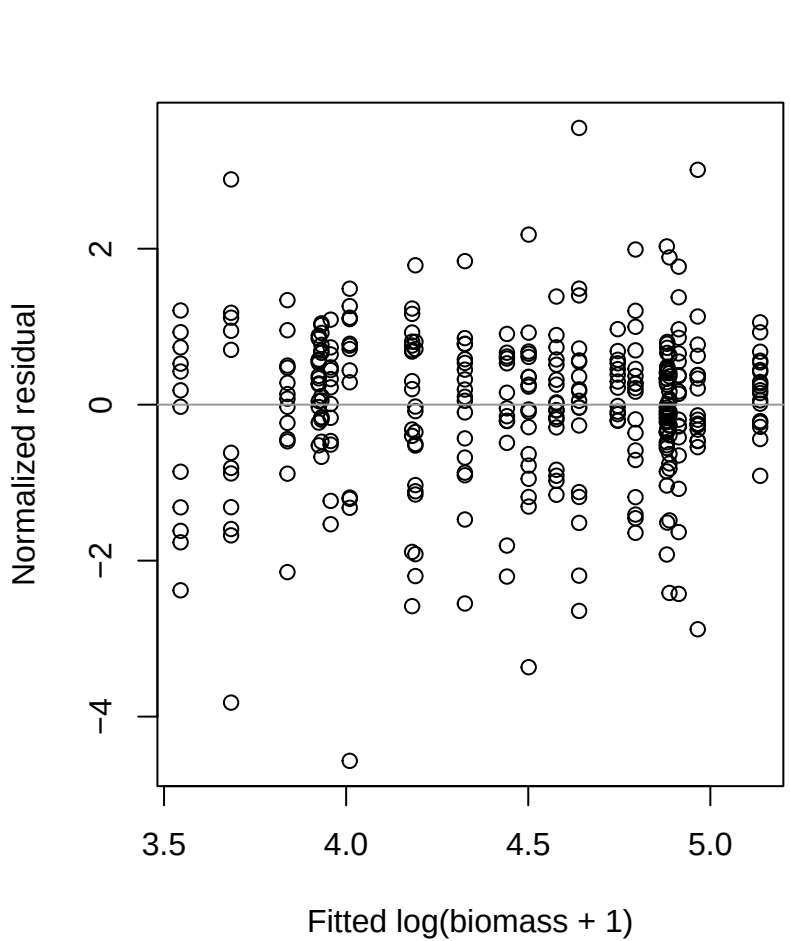

Normalized residual
2
0
−2
−4
3.8
4.0
4.2
4.4
4.6
4.8
5.0
Fitted log(biomass + 1)

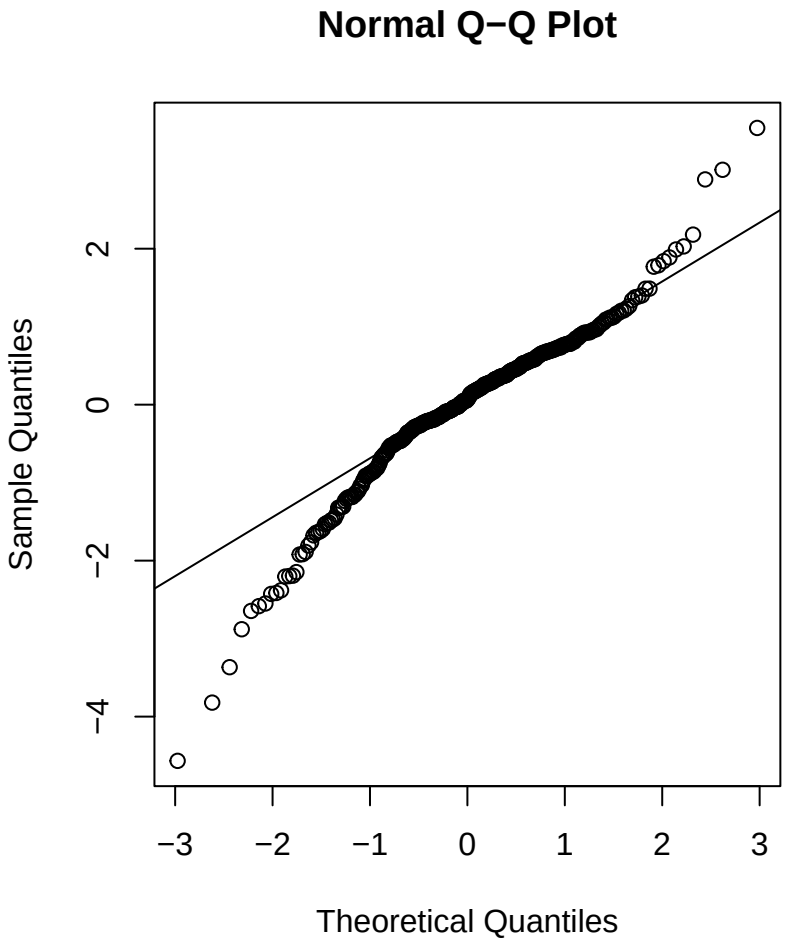

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

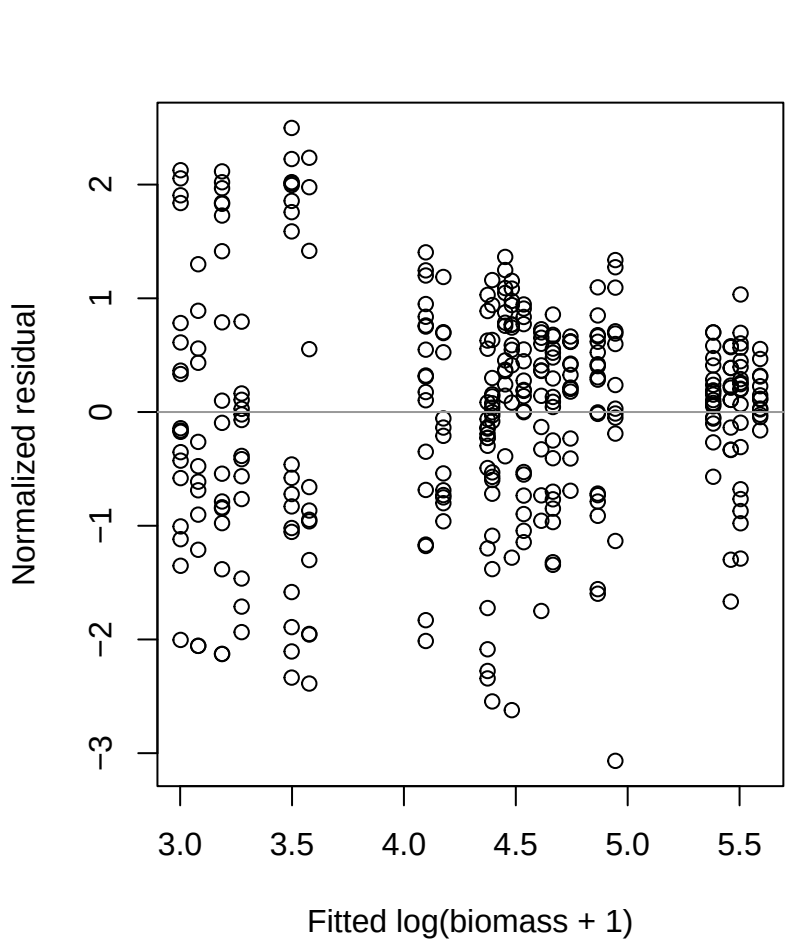
purple_biomass__2022__2005__site_independent

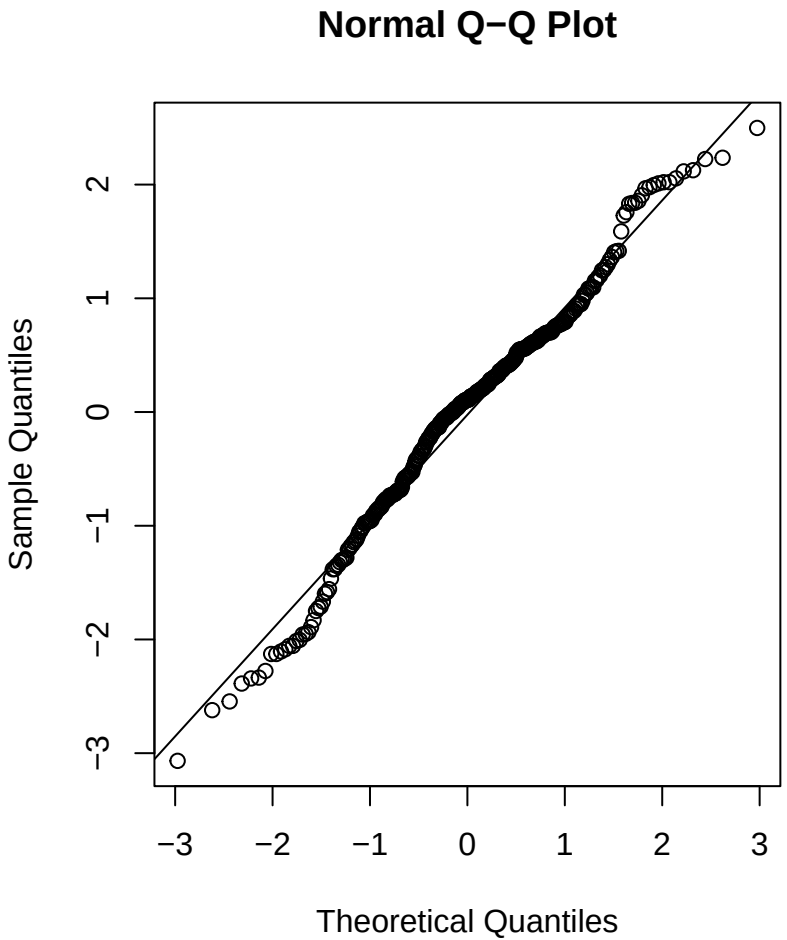


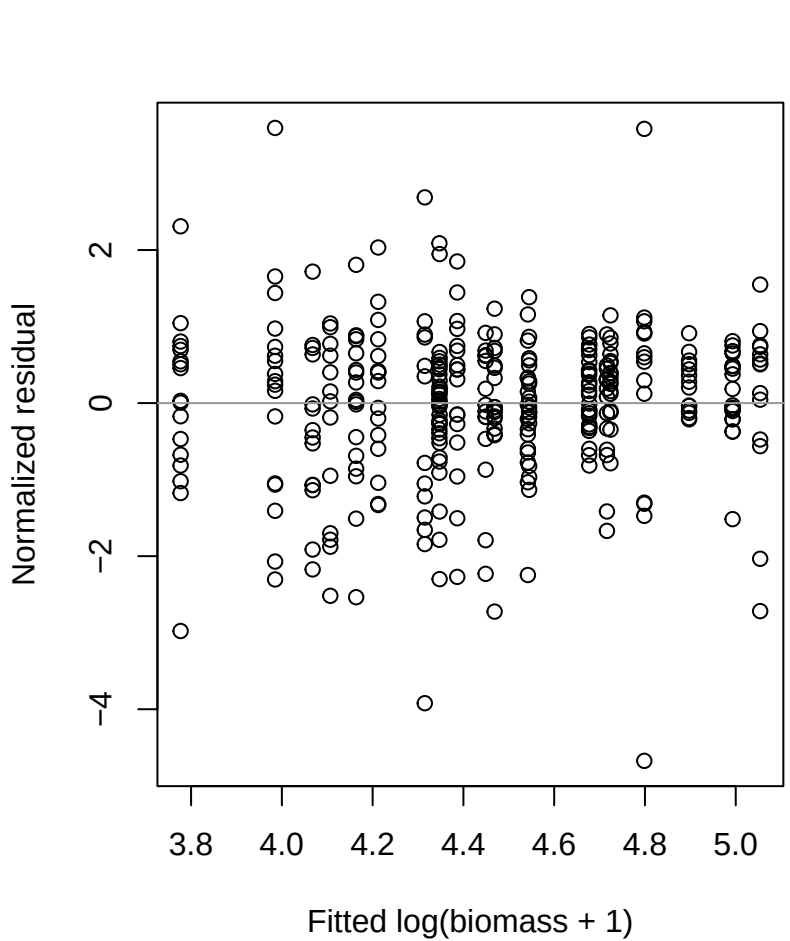


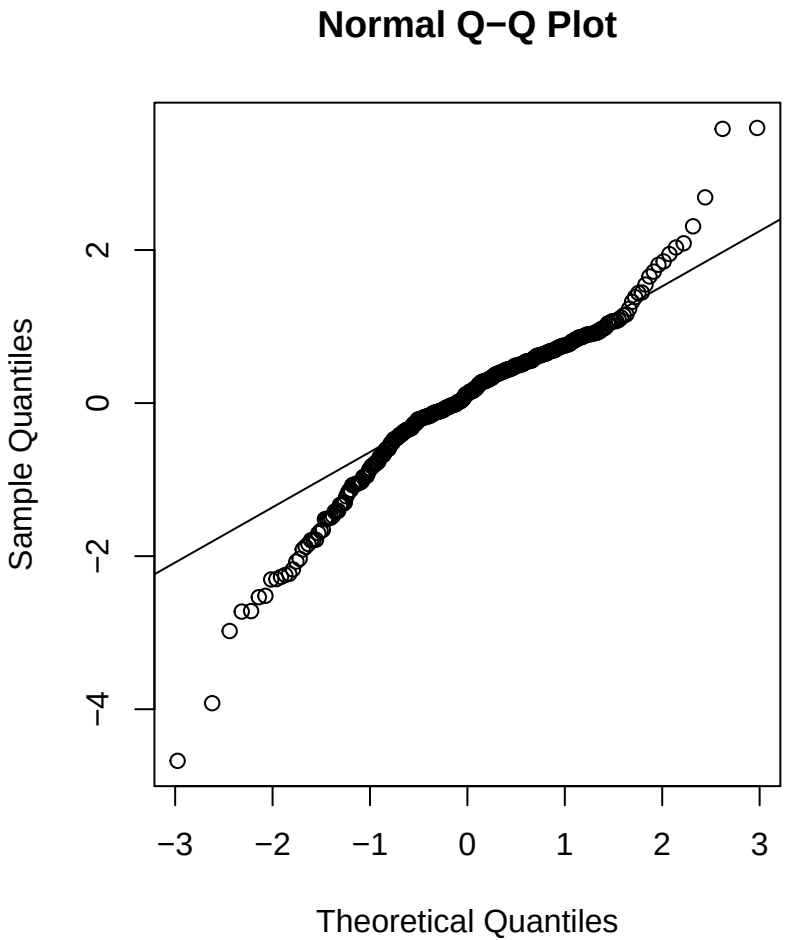

Normalized residual
2
0
−2
−4
3.5
4.0
4.5
5.0
Fitted log(biomass + 1)


Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

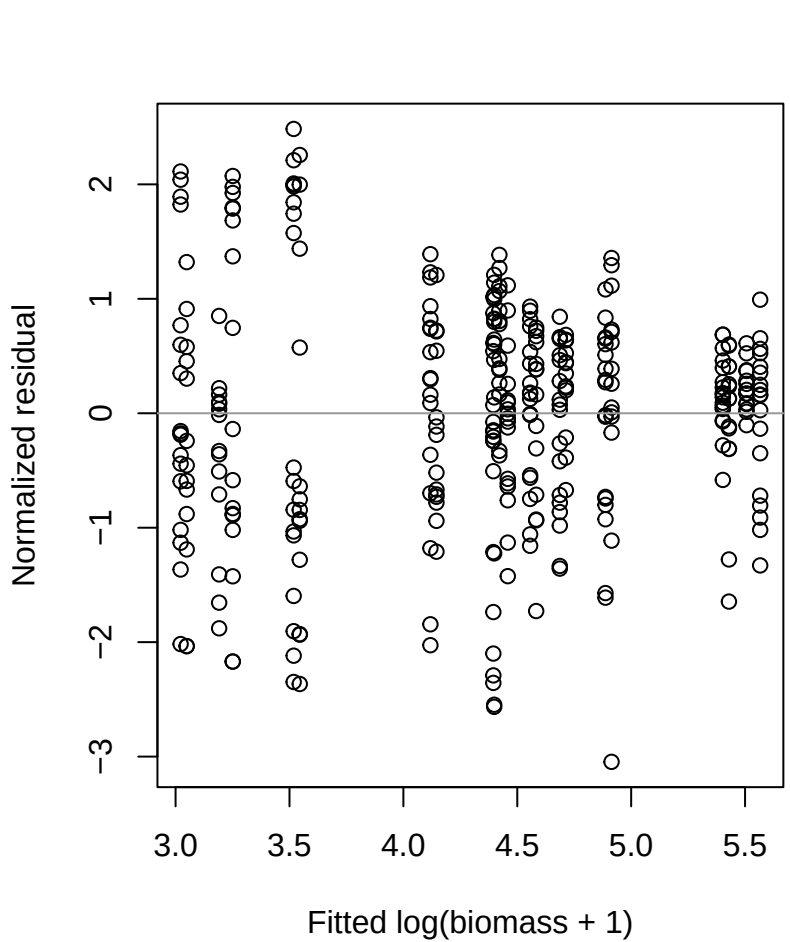

Normalized residual
Fitted log(biomass + 1)
3.0
3.5
4.0
4.5
5.0
5.5
−3
−2
−1
0
1
2

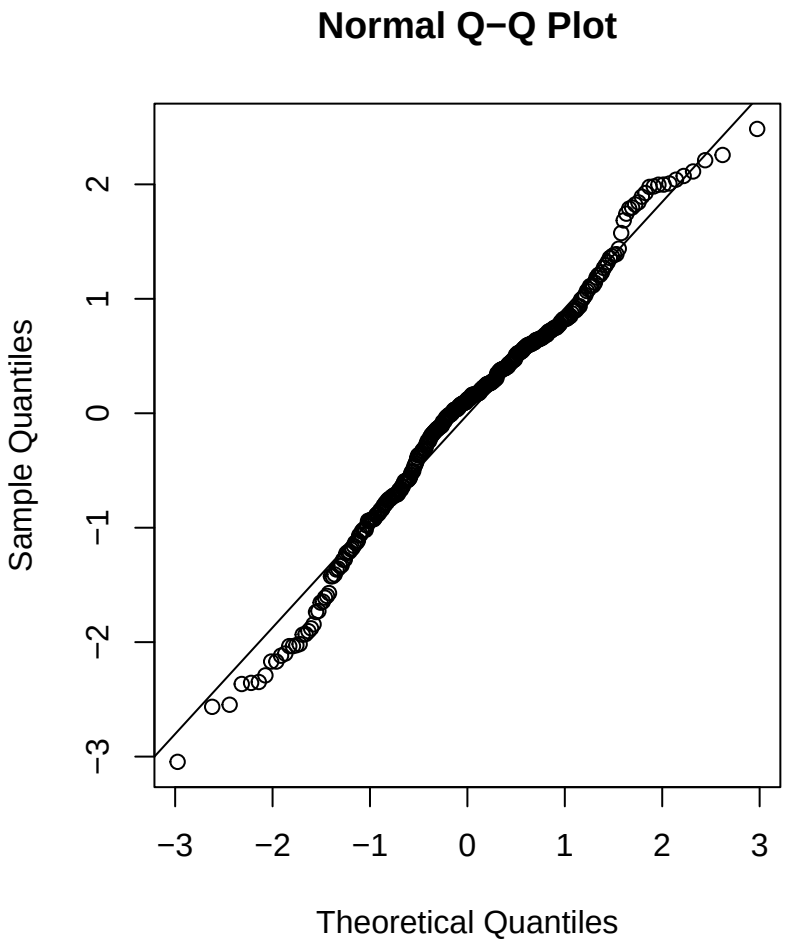

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

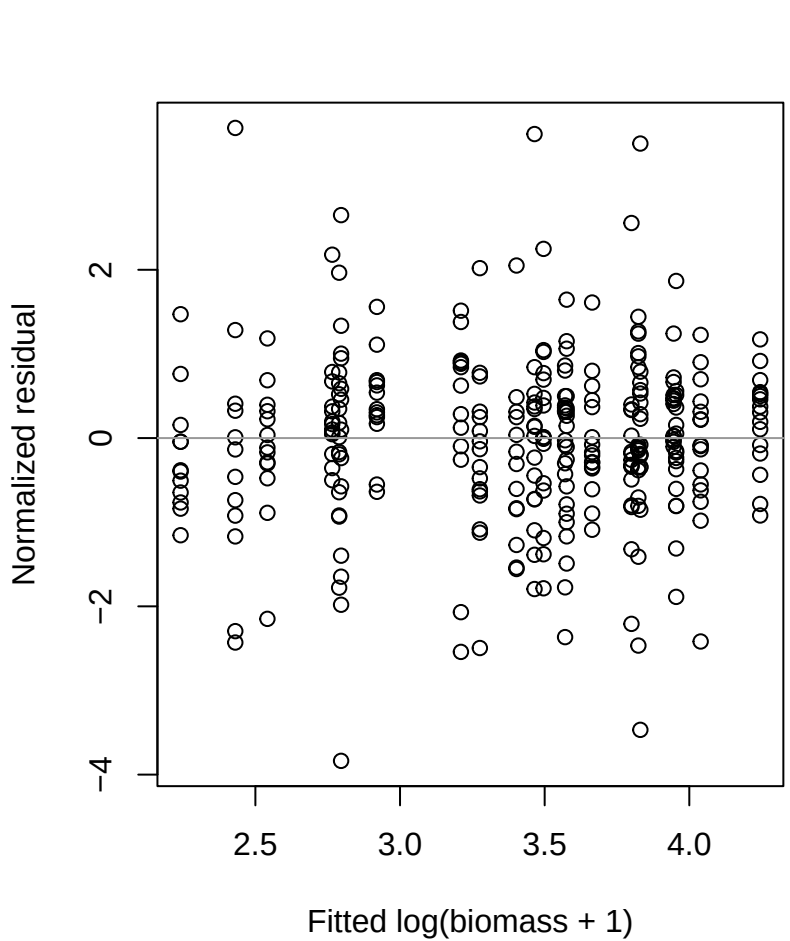

Normalized residual
2
0
−2
−4
3.8
4.0
4.2
4.4
4.6
4.8
5.0
Fitted log(biomass + 1)

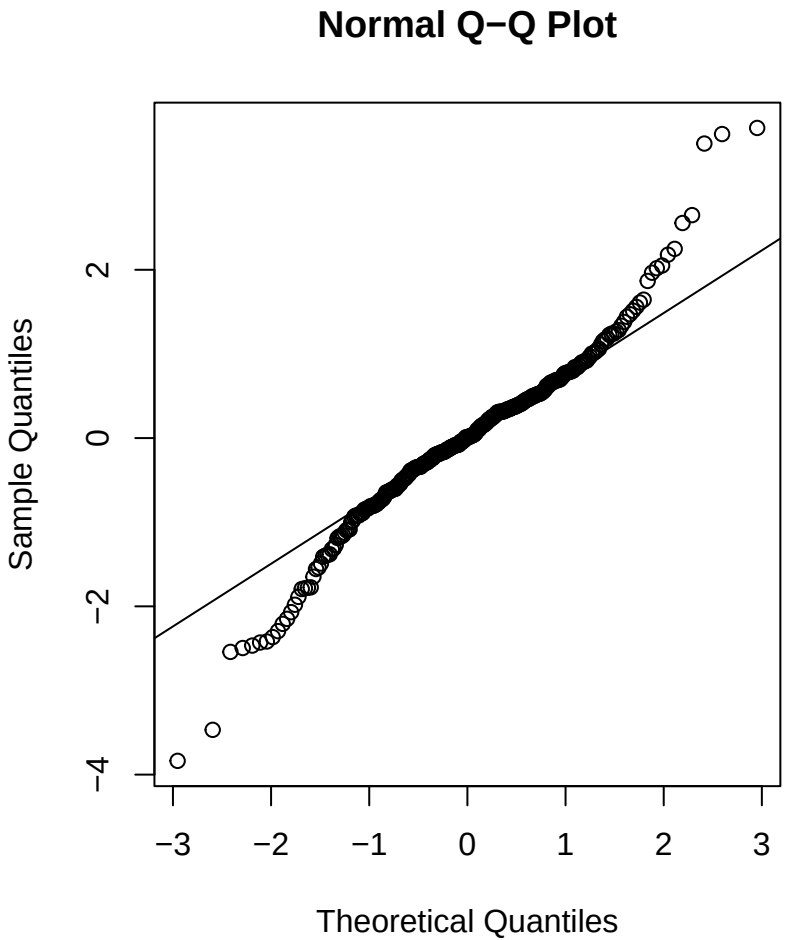

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

purple_biomass__2023__2005__site_independent

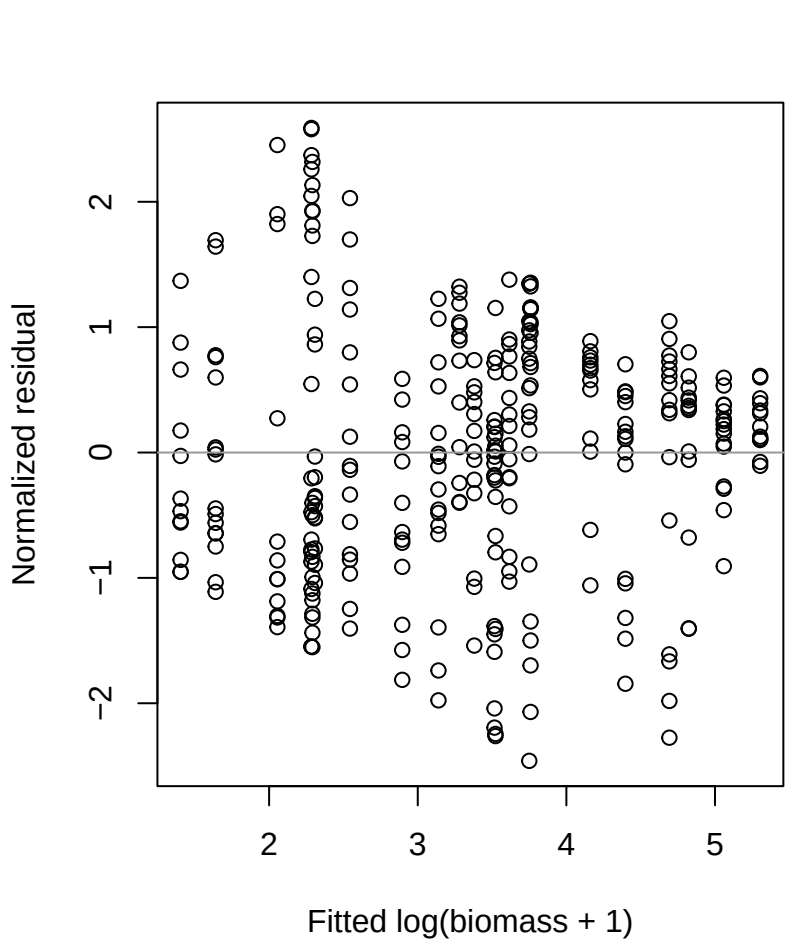


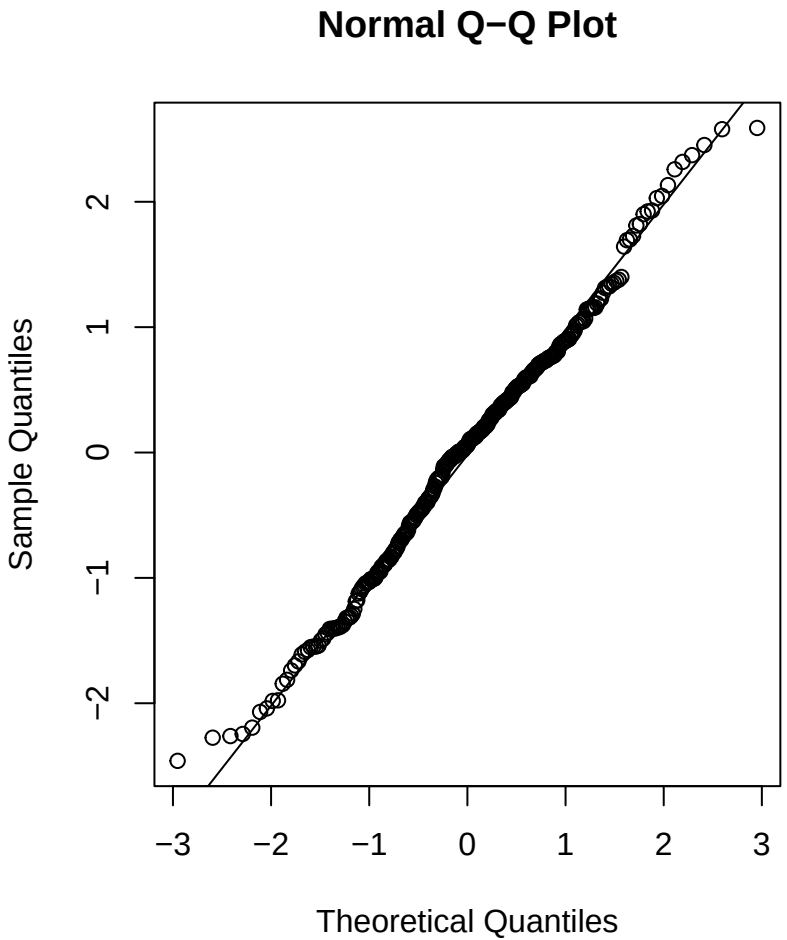



Normalized residual
2
0
−2
−4
2.5
3.0
3.5
4.0
Fitted log(biomass + 1)

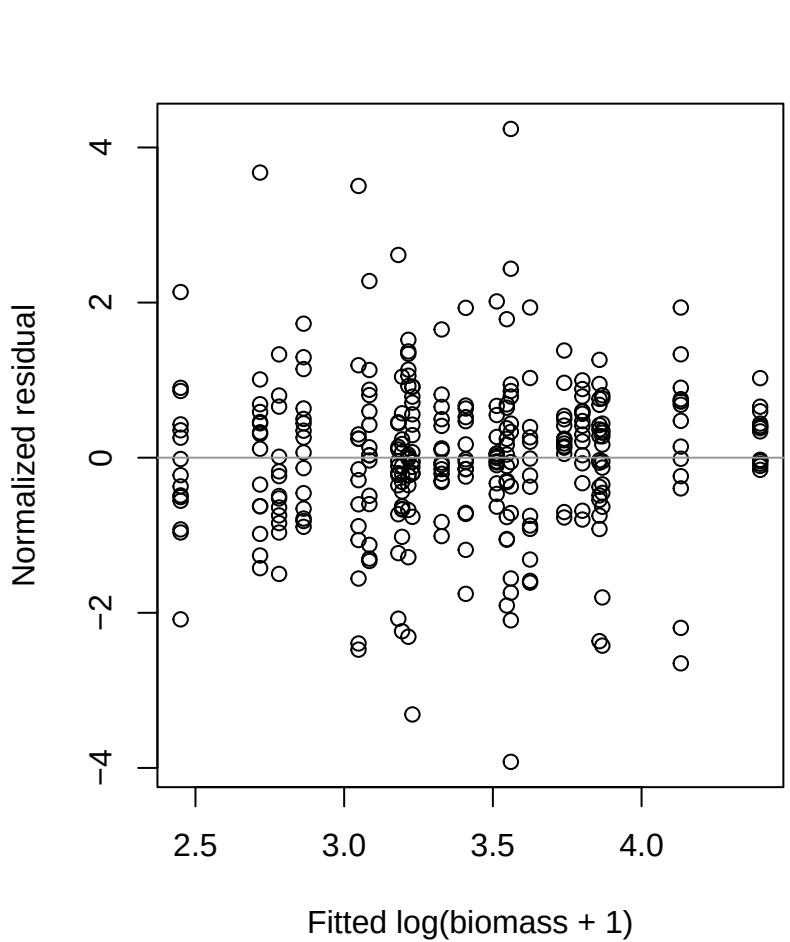

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

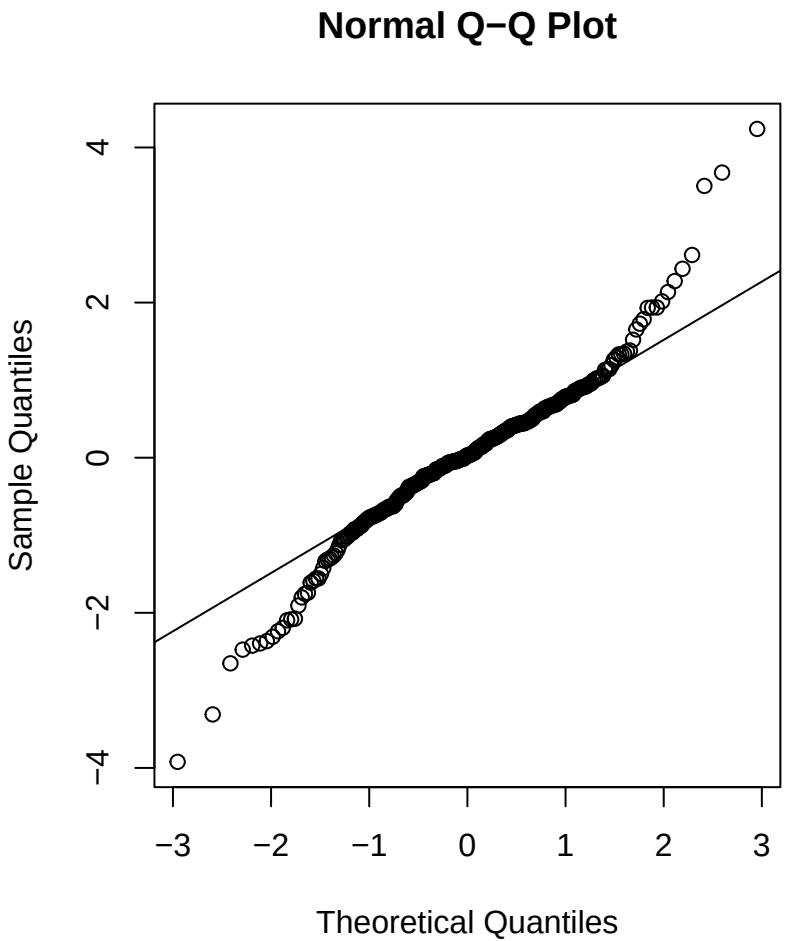
purple_small_mass__2019__2004__site_independent



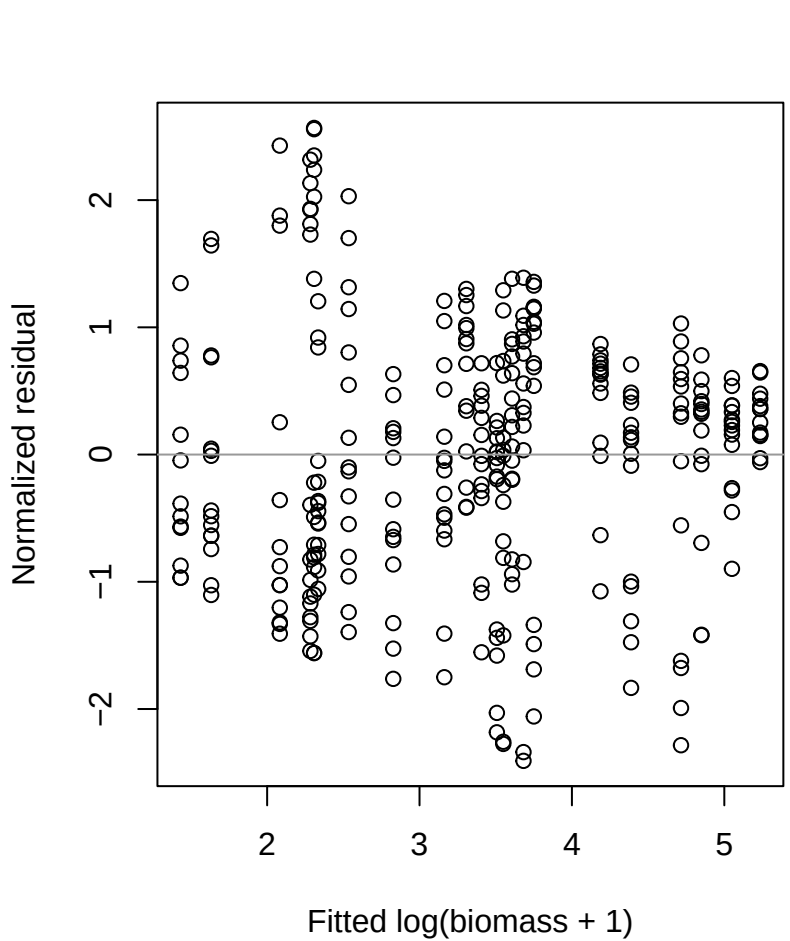


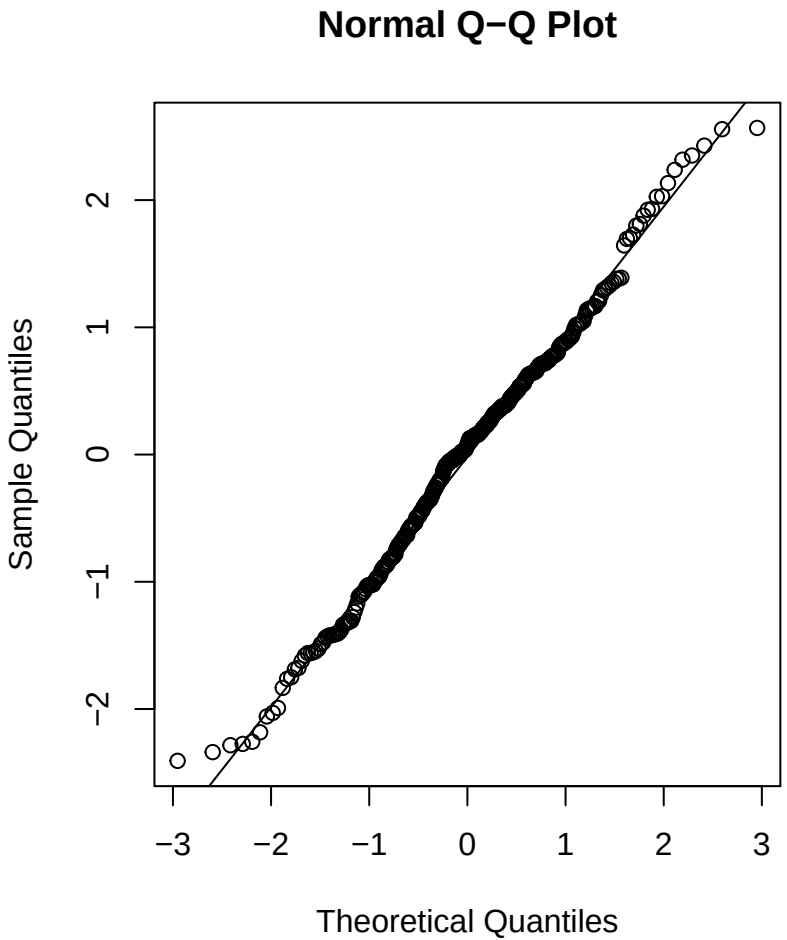
purple_small_mass__2019__2005__site_calendar_AR1

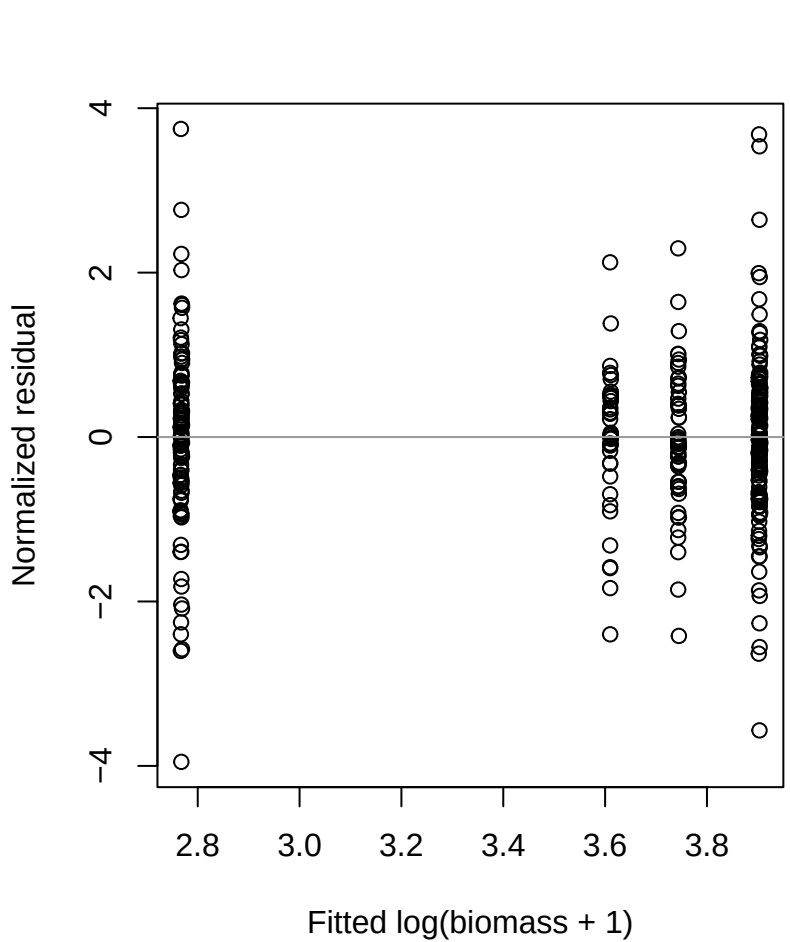


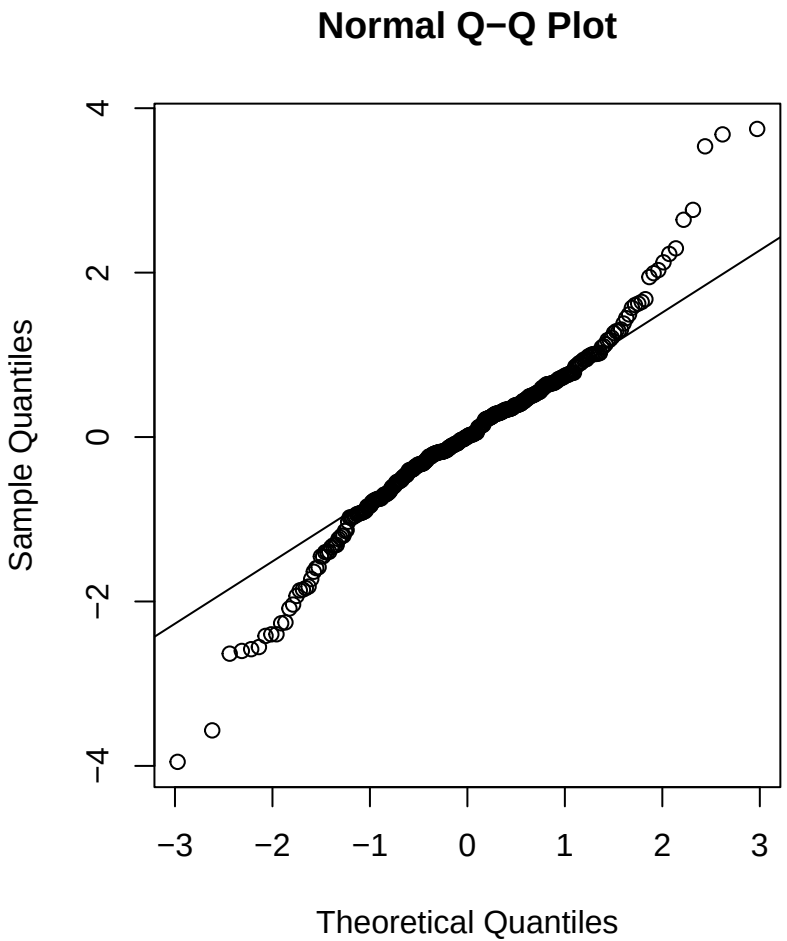


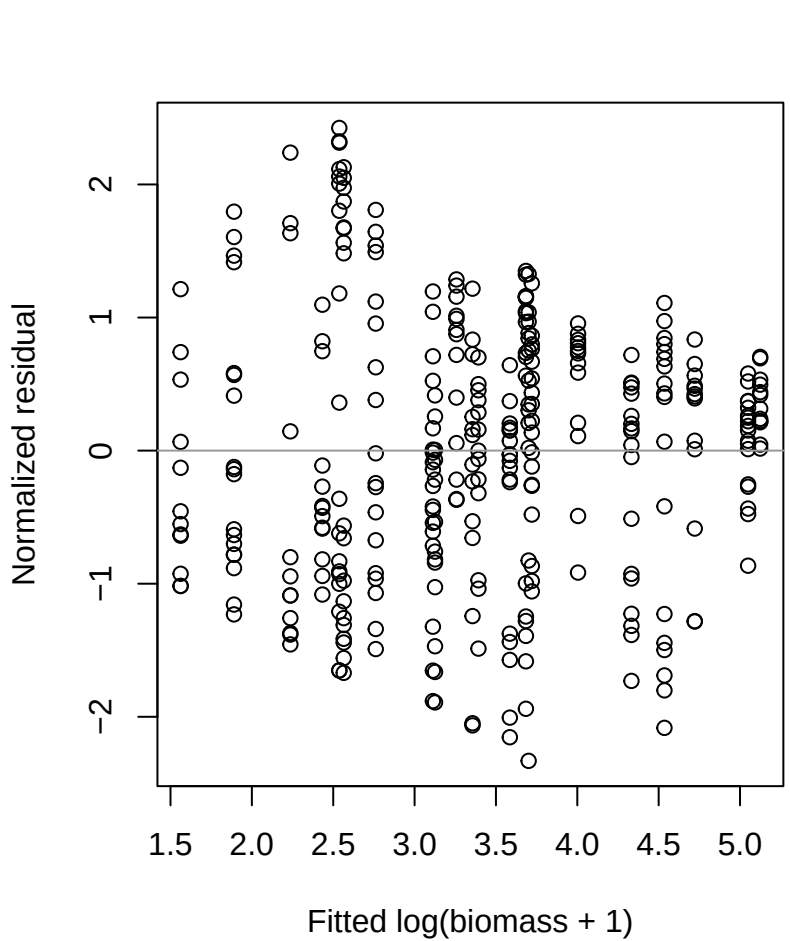
purple_small_mass__2019__2005__site_independent

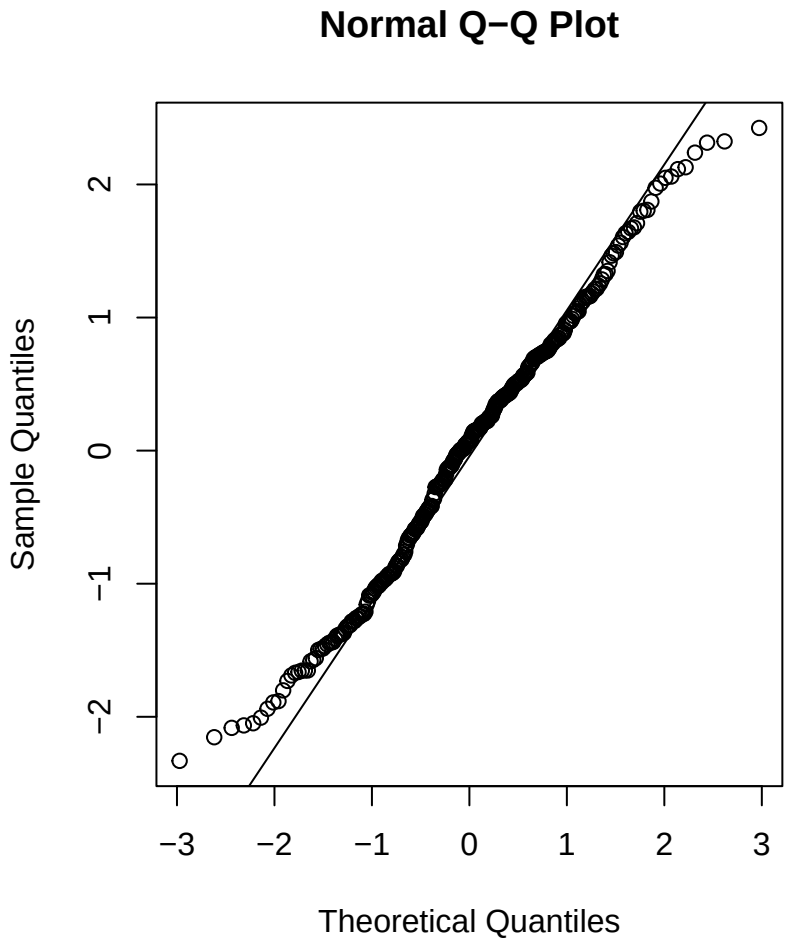




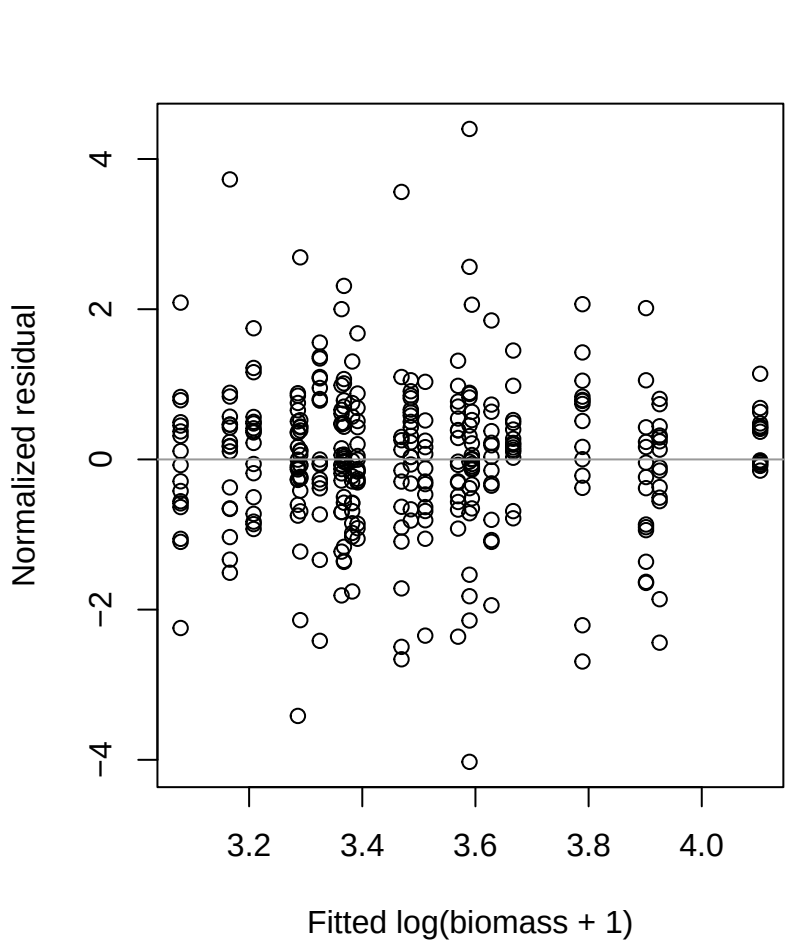

Normalized residual
4
2
0
−2
−4
2.8
3.0
3.2
3.4
3.6
3.8
Fitted log(biomass + 1)

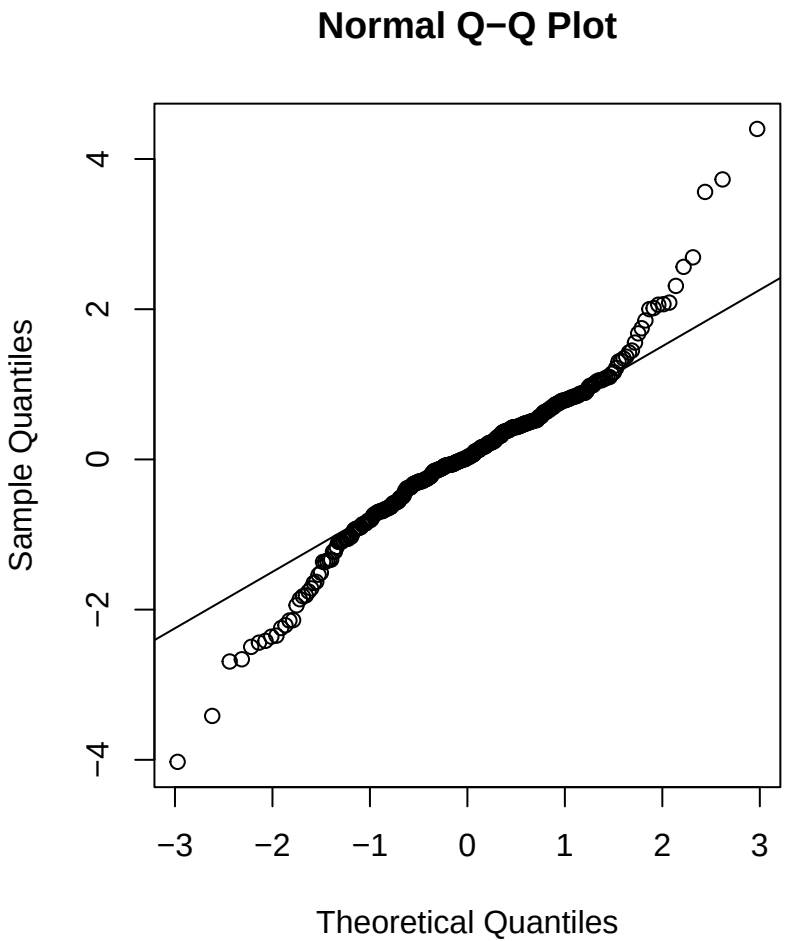

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Normalized residual
−2
−1
0
1
2
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Fitted log(biomass + 1)

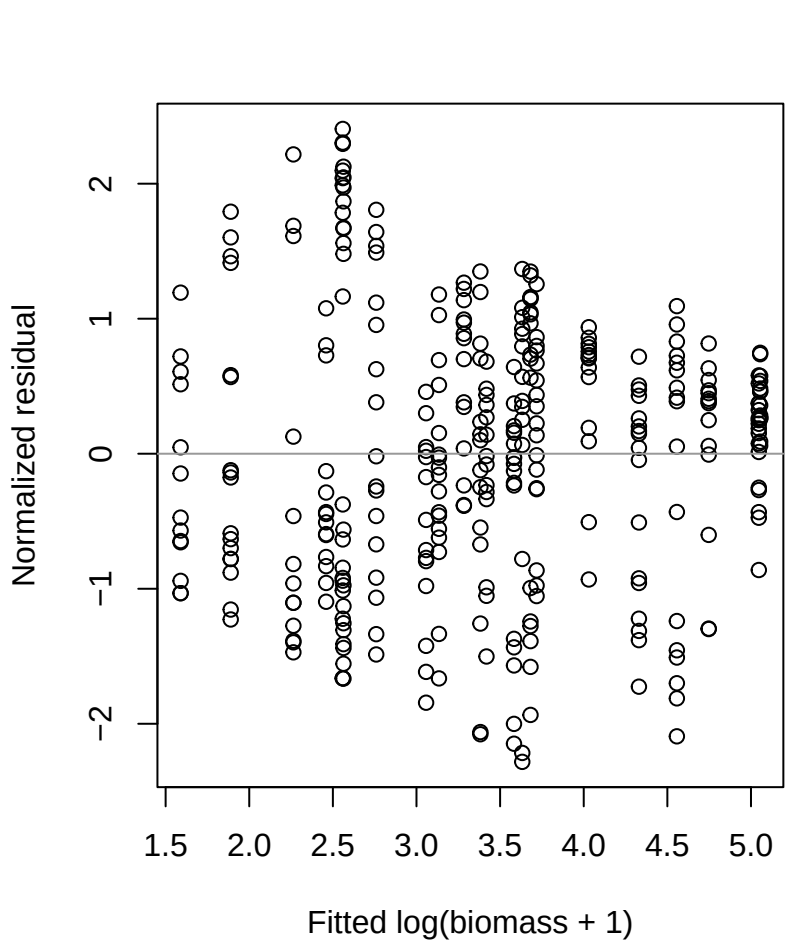

Normal Q−Q Plot
Sample Quantiles
−2
−1
0
1
2
−3
−2
−1
0
1
2
3
Theoretical Quantiles

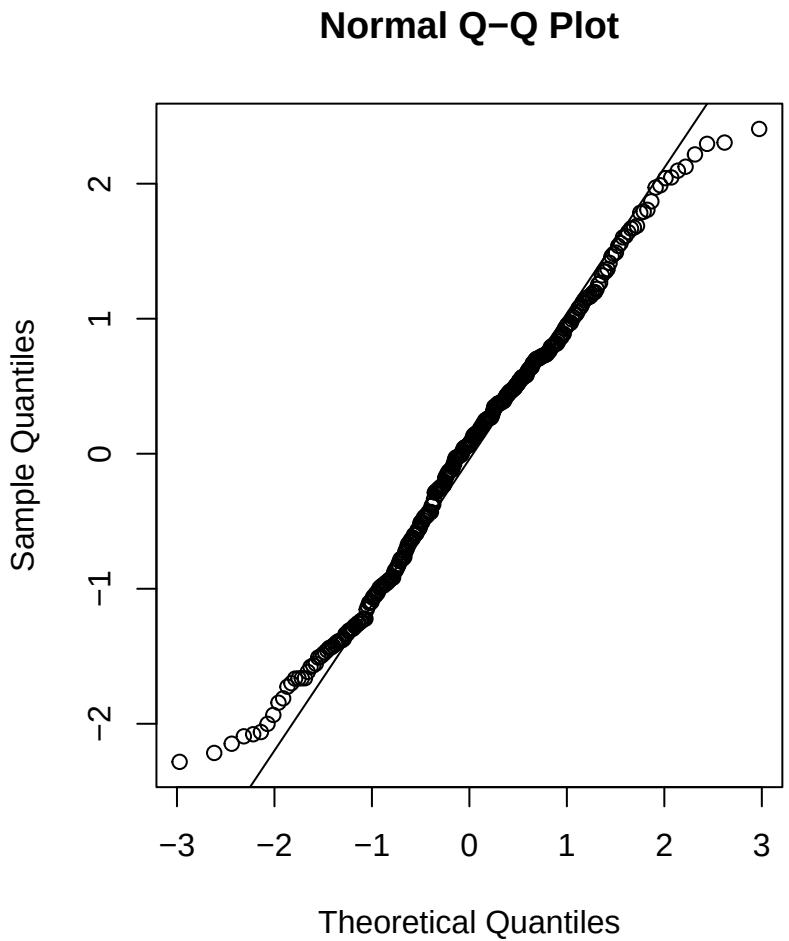
purple_small_mass__2022__2005__site_calendar_AR1



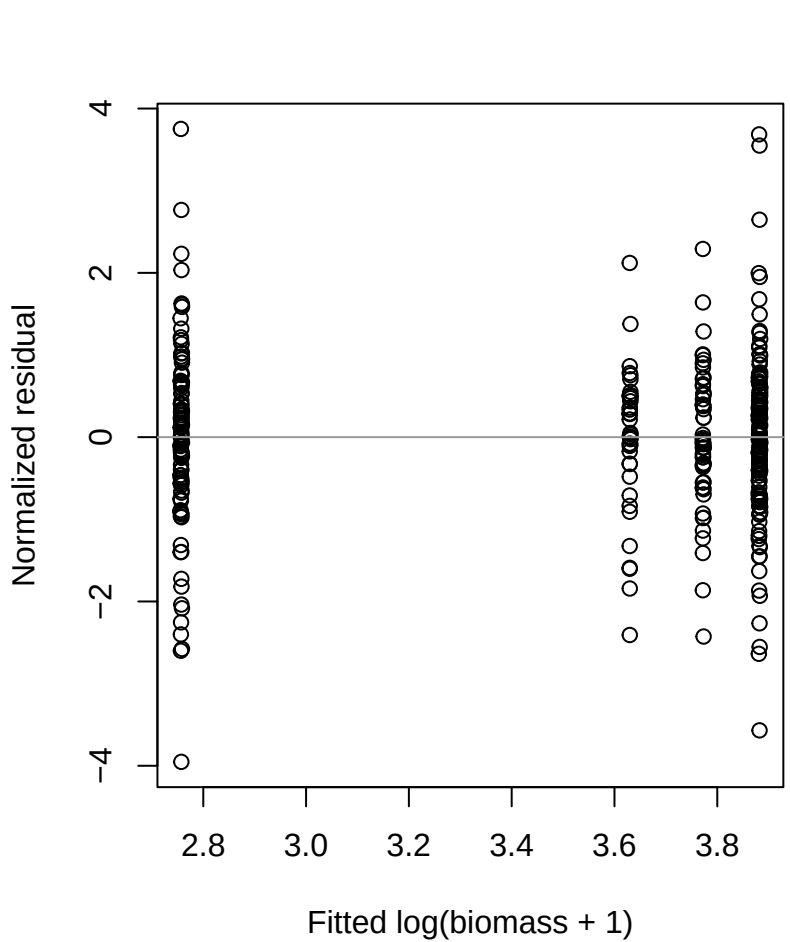


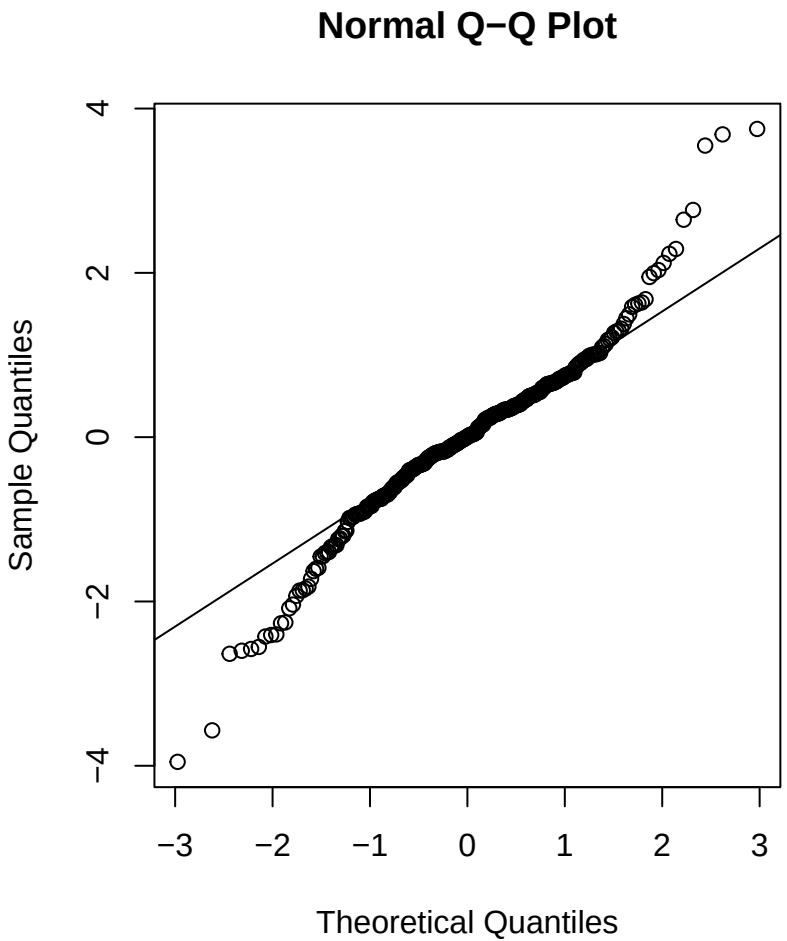
purple_small_mass__2022__2005__site_independent



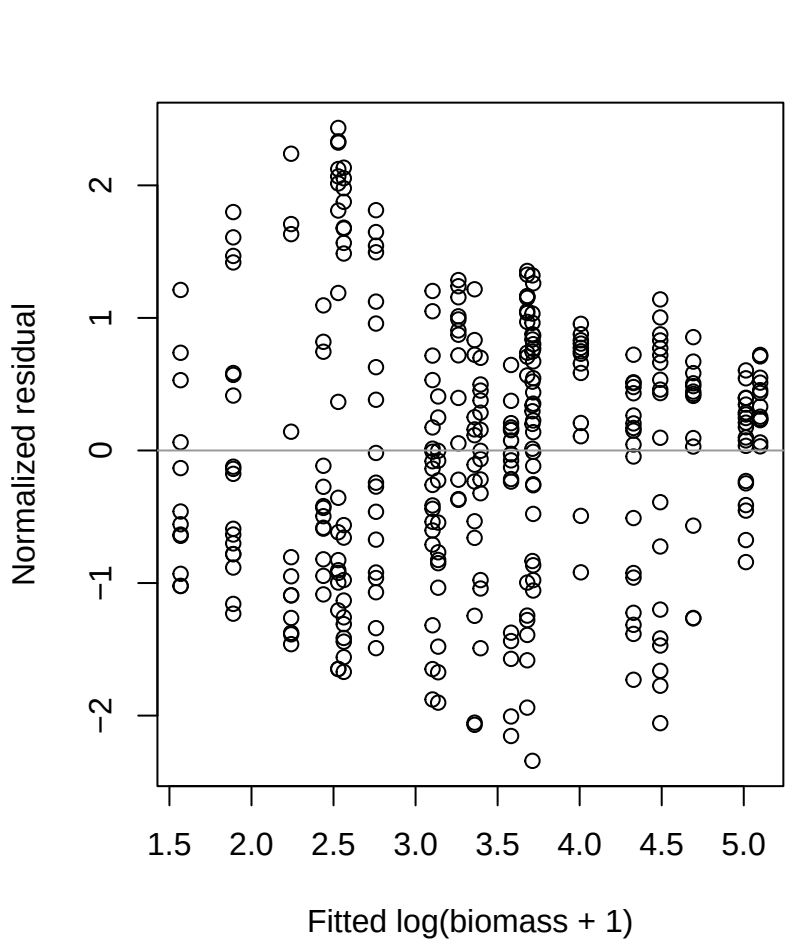


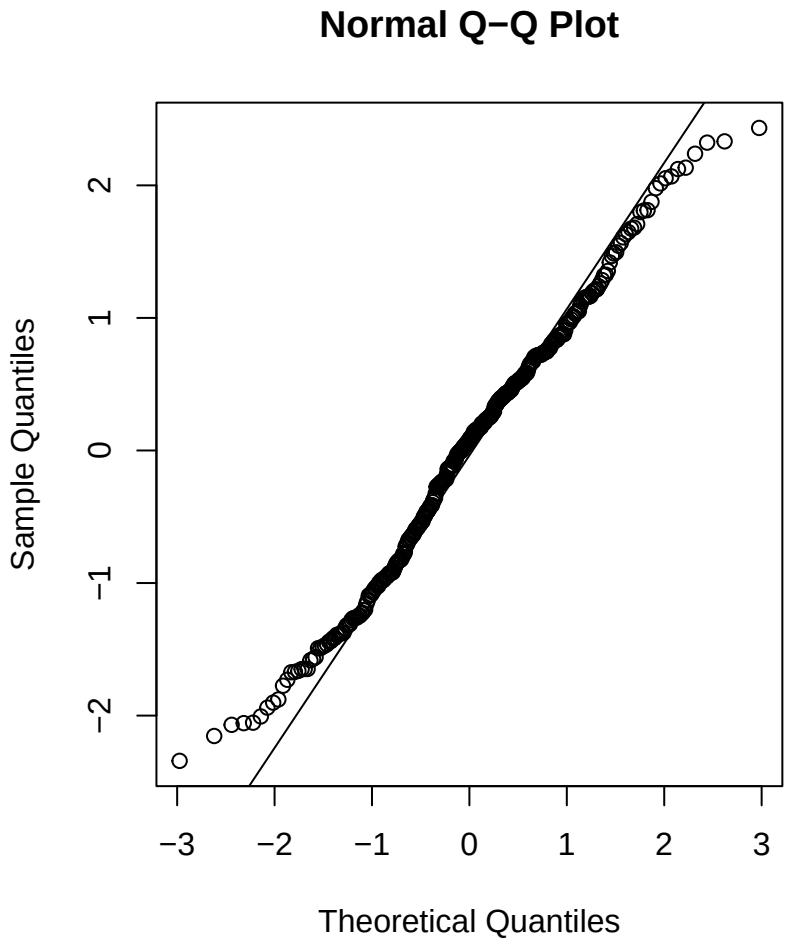
purple_small_mass__2023__2004__site_calendar_AR1

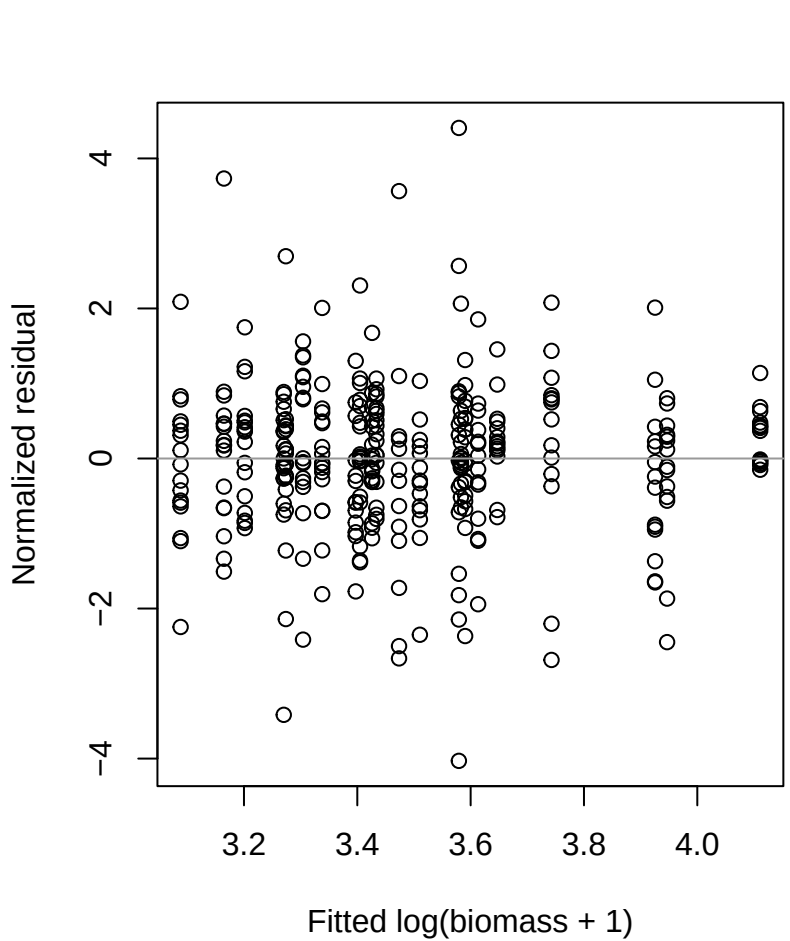


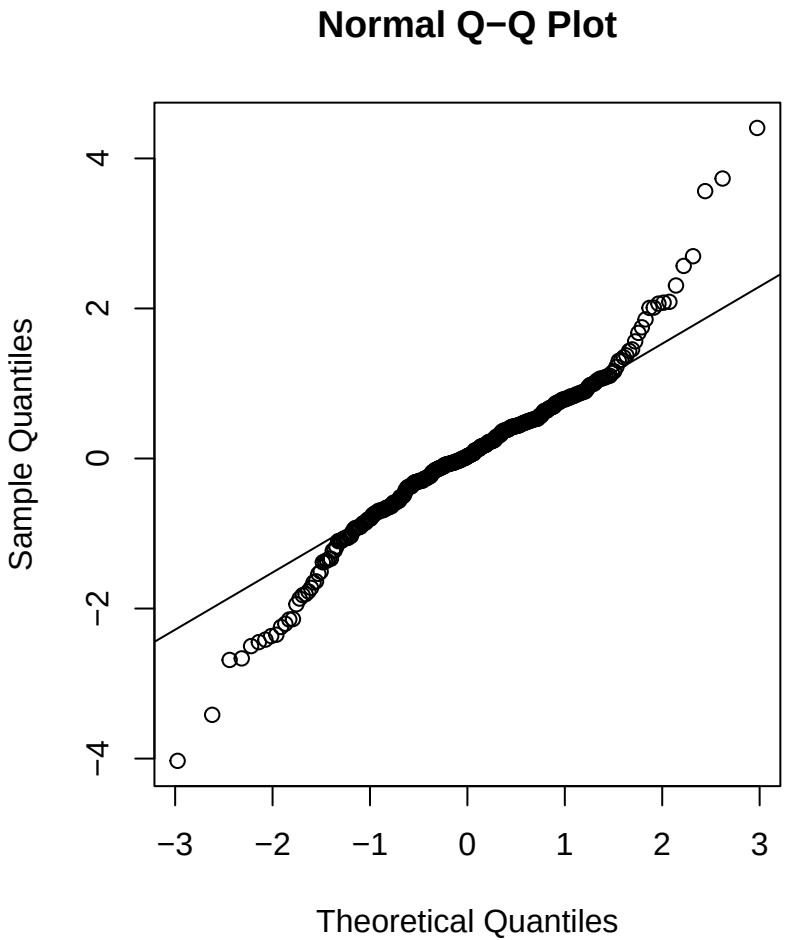


purple_small_mass__2023__2004__site_independent

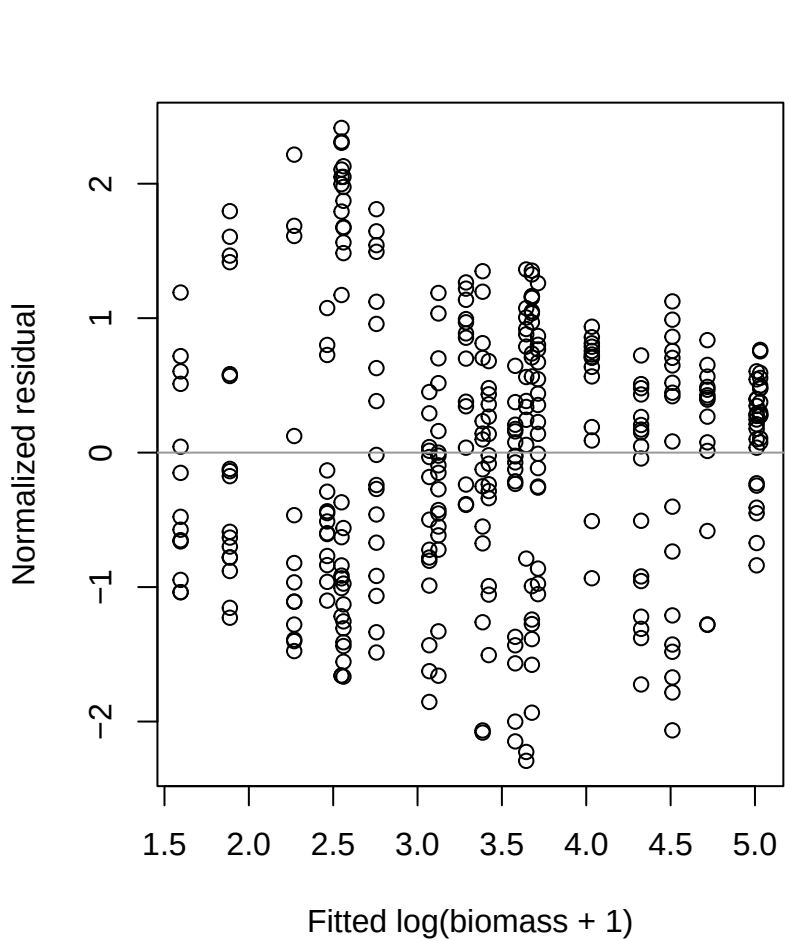


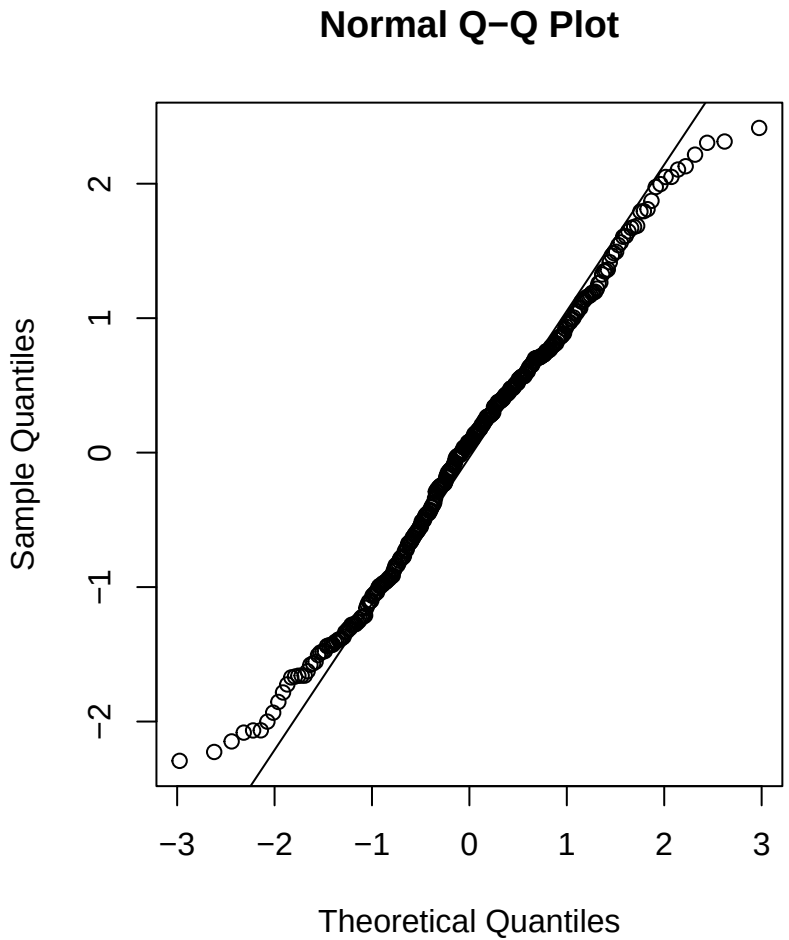



Normalized residual
4
2
0
−2
−4
3.2
3.4
3.6
3.8
4.0
Fitted log(biomass + 1)

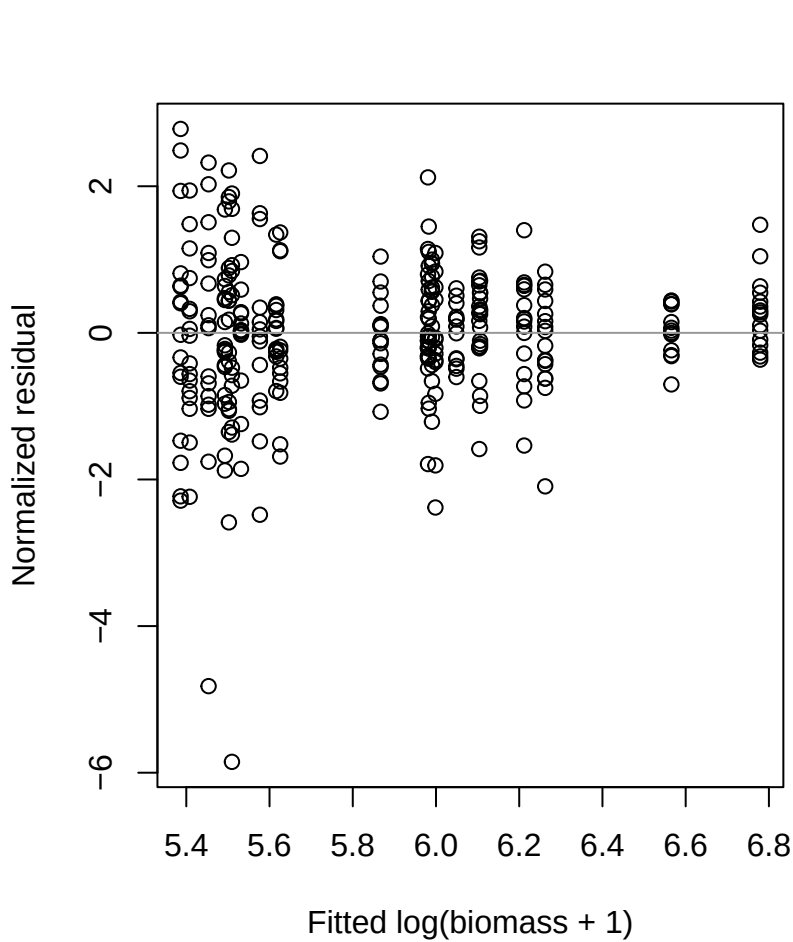

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

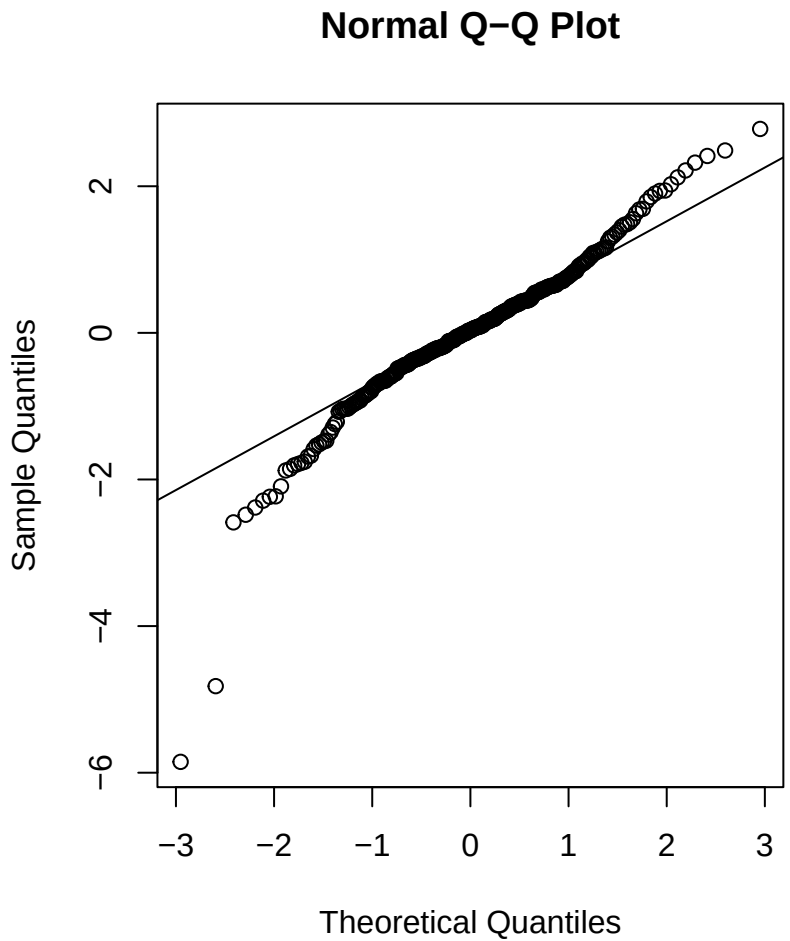
purple_small_mass__2023__2005__site_independent



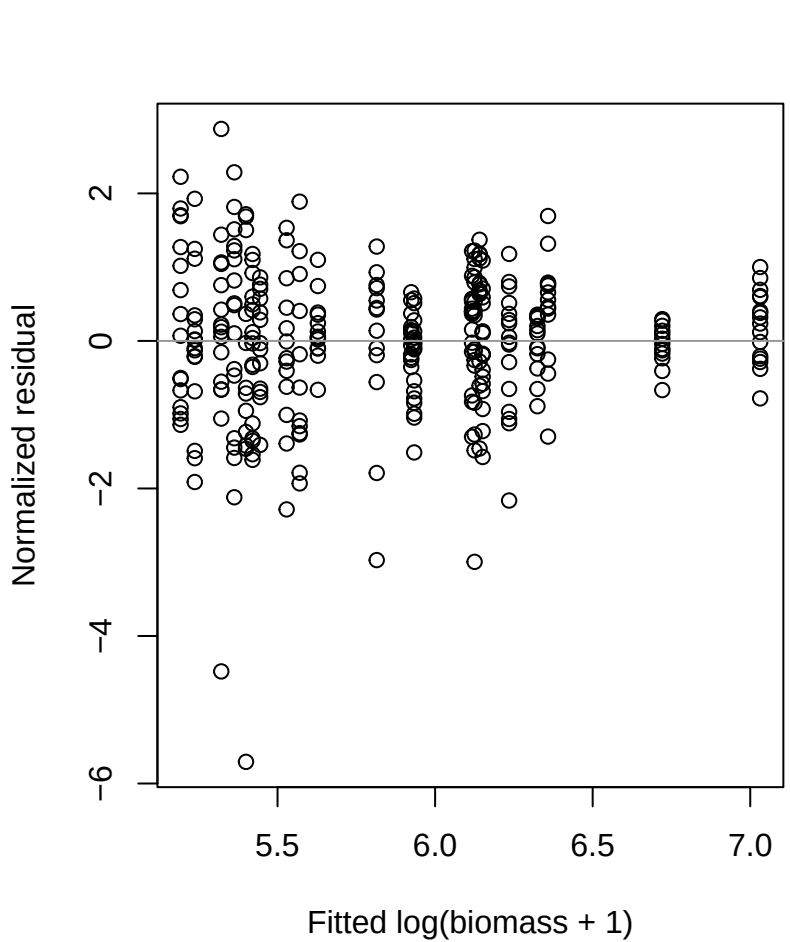


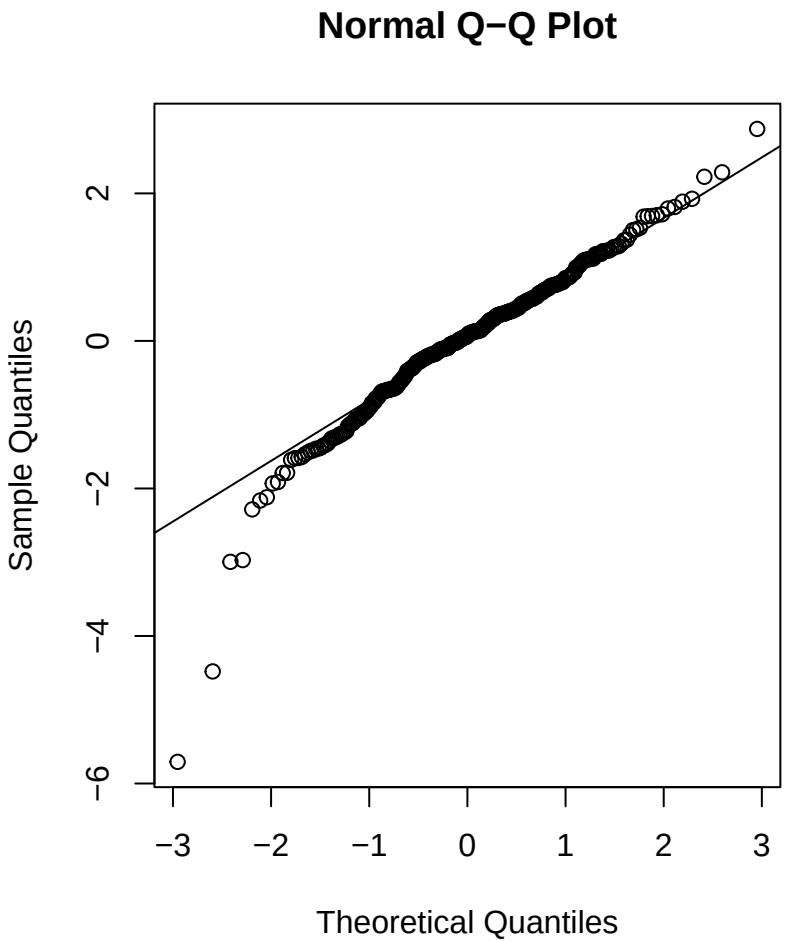
total_biomass__2019__2004__site_calendar_AR1

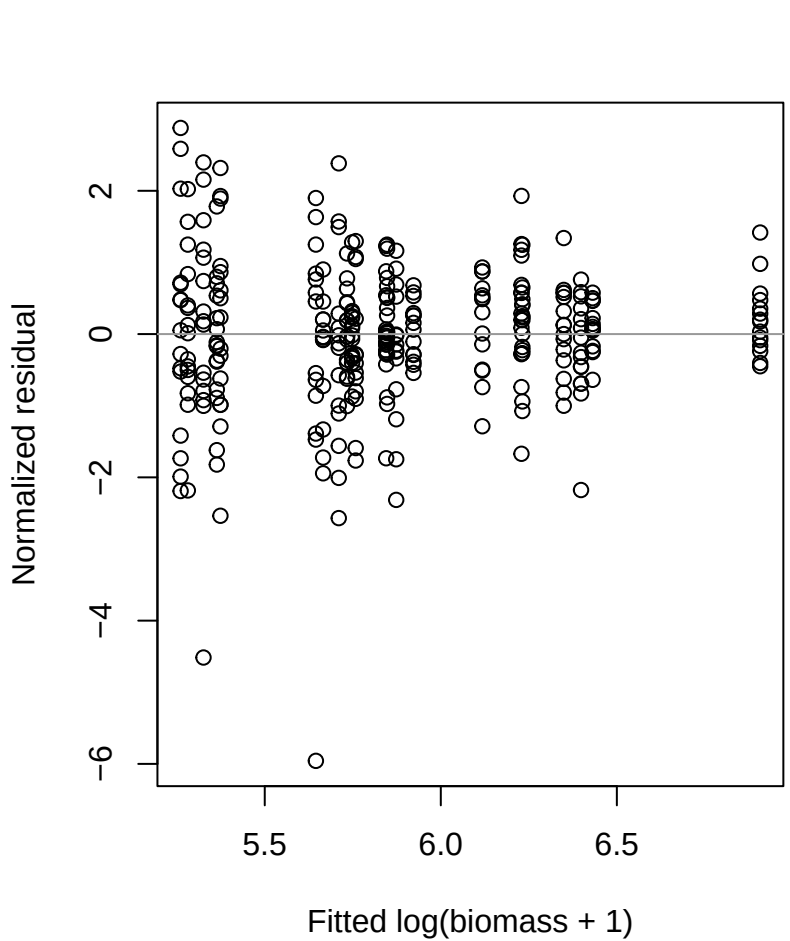


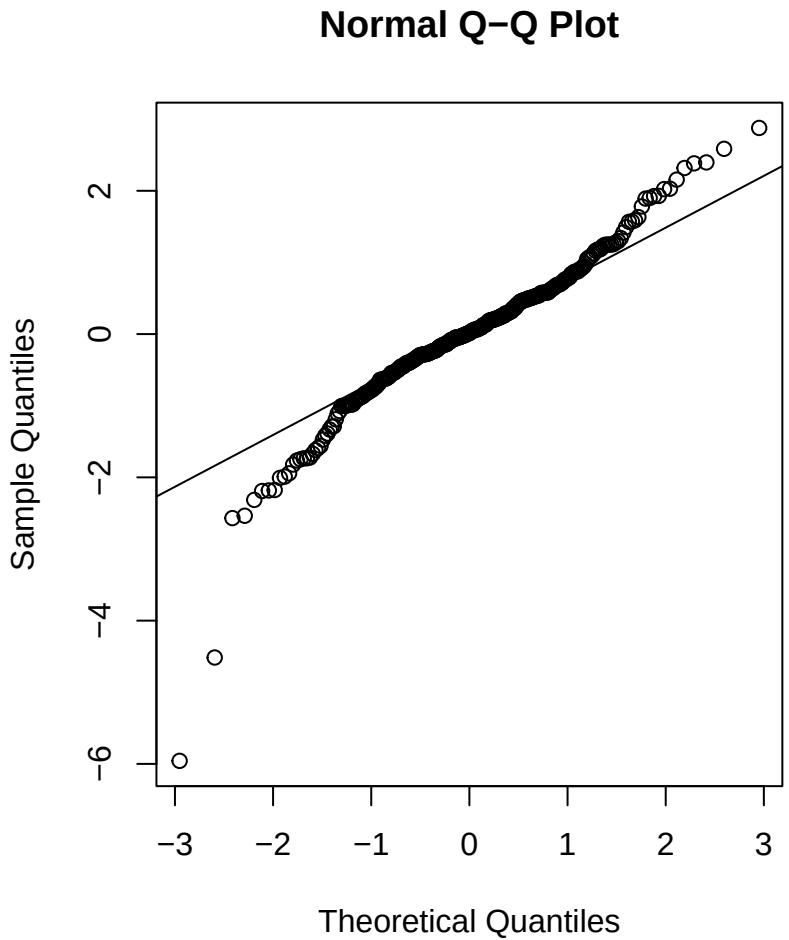


total_biomass__2019__2004__site_independent

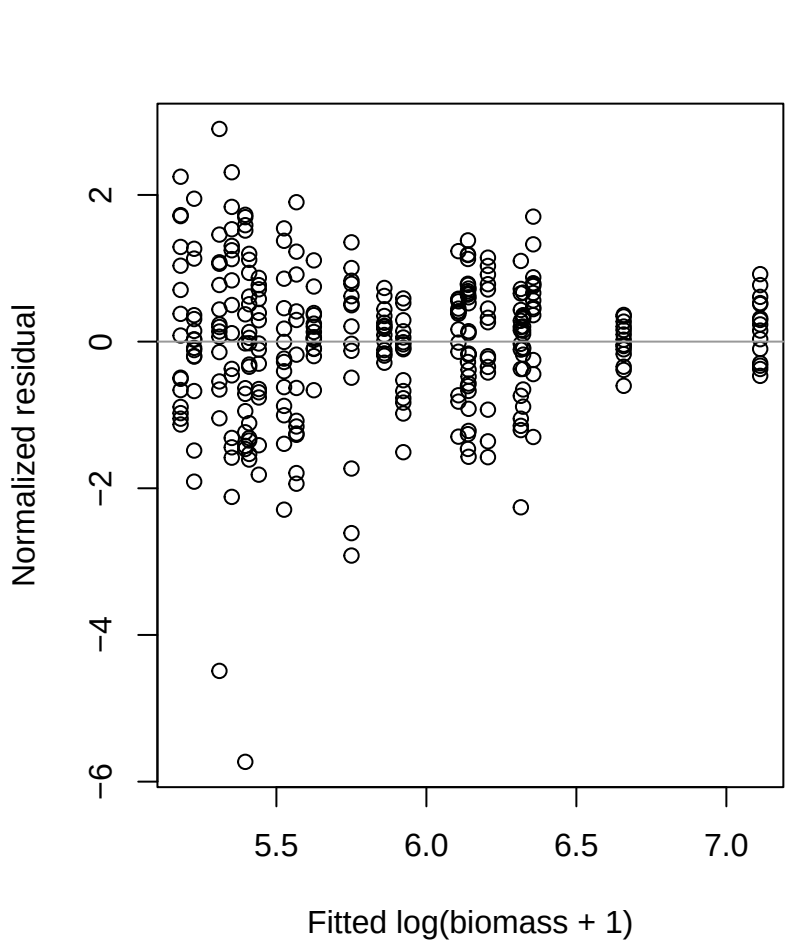


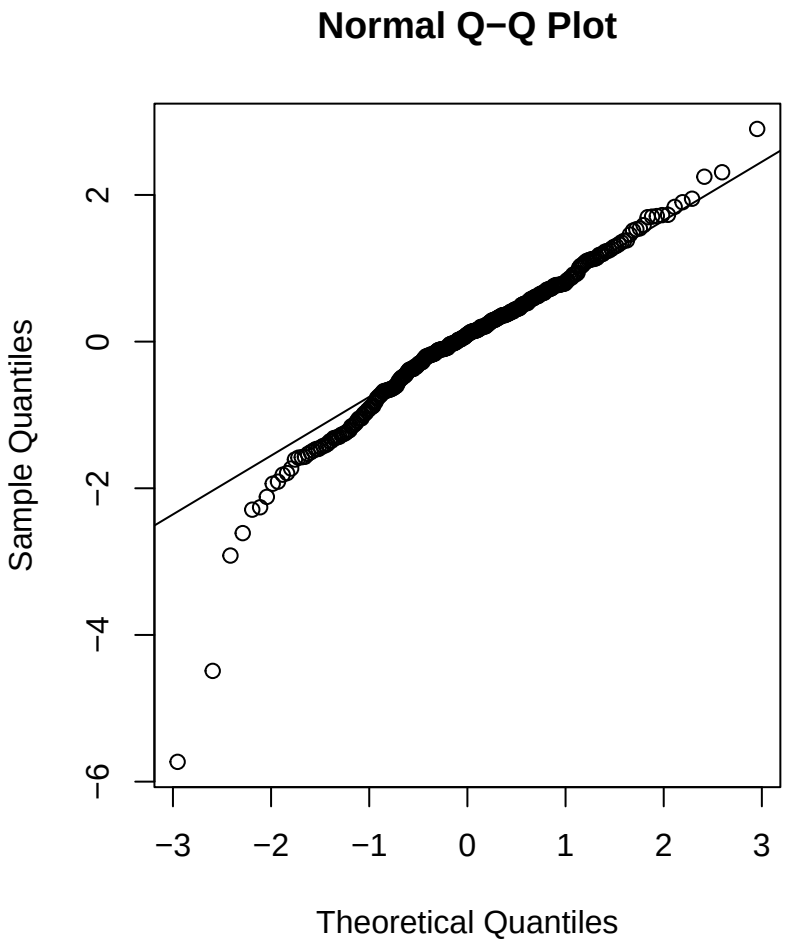


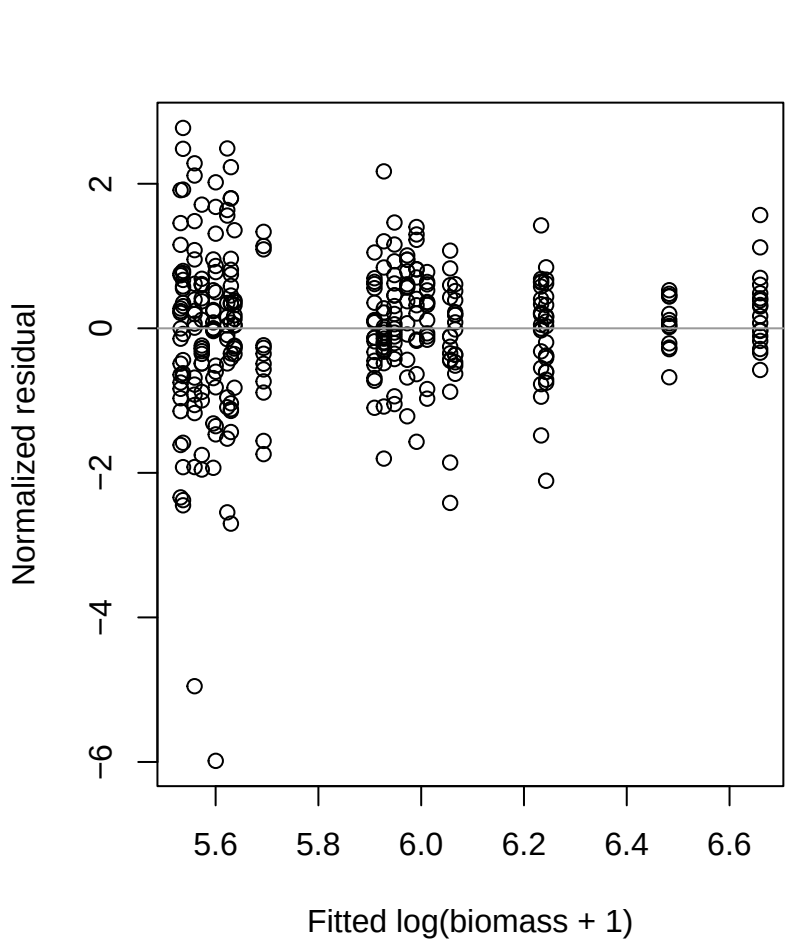

Normalized residual
2
0
−2
−4
−6
5.5
6.0
6.5
Fitted log(biomass + 1)

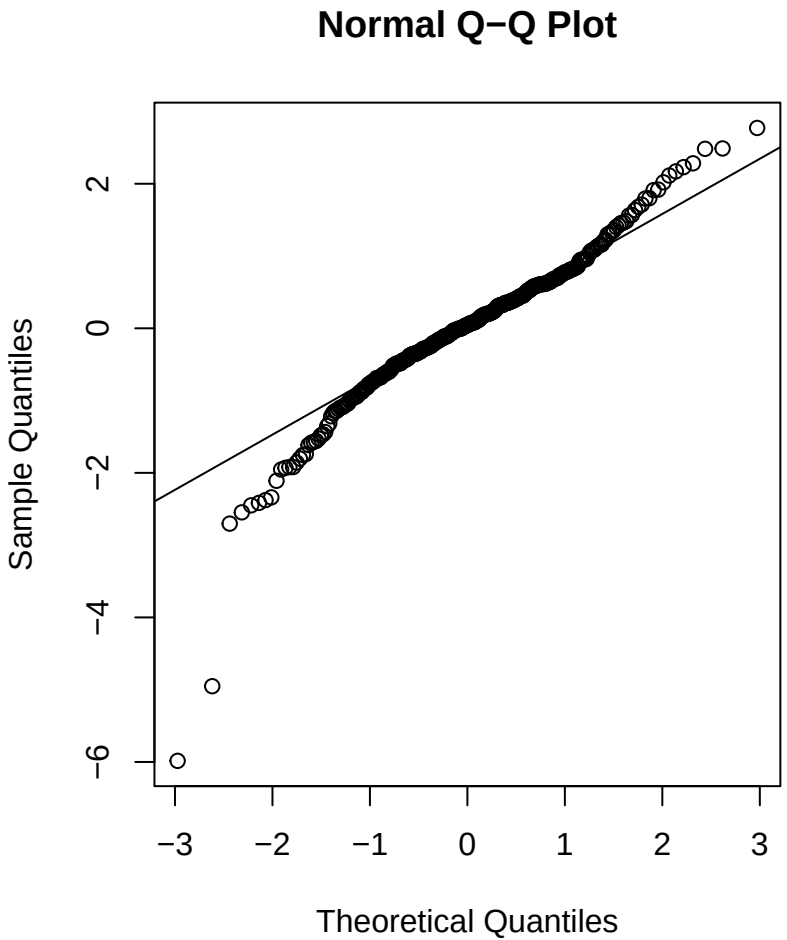

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
Theoretical Quantiles

total_biomass__2019__2005__site_independent

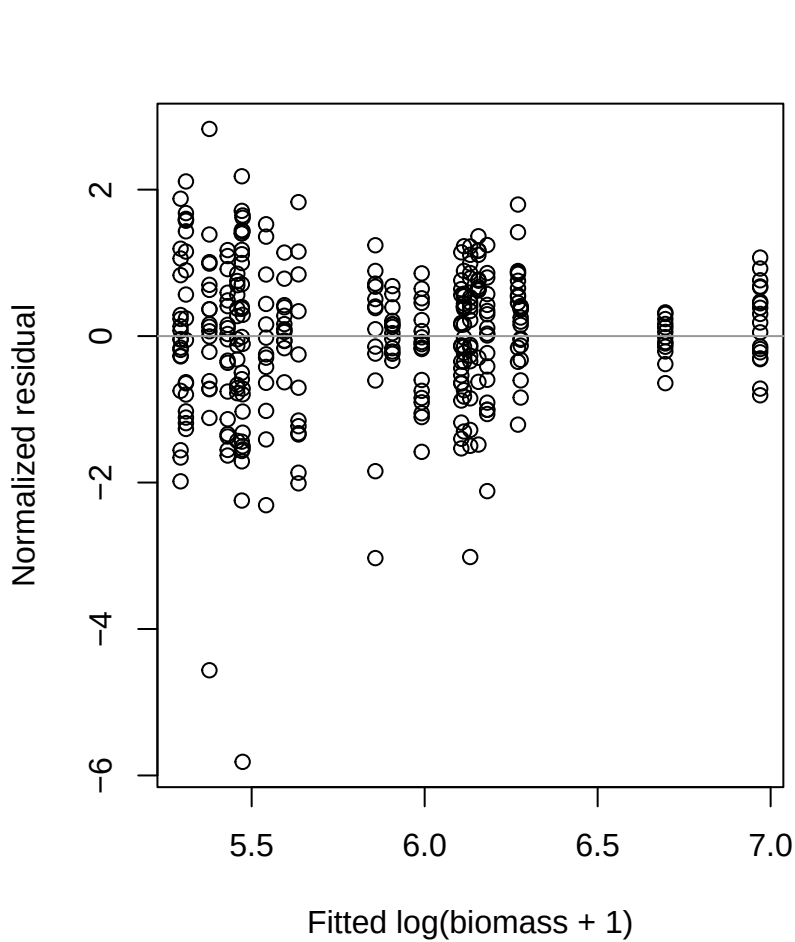


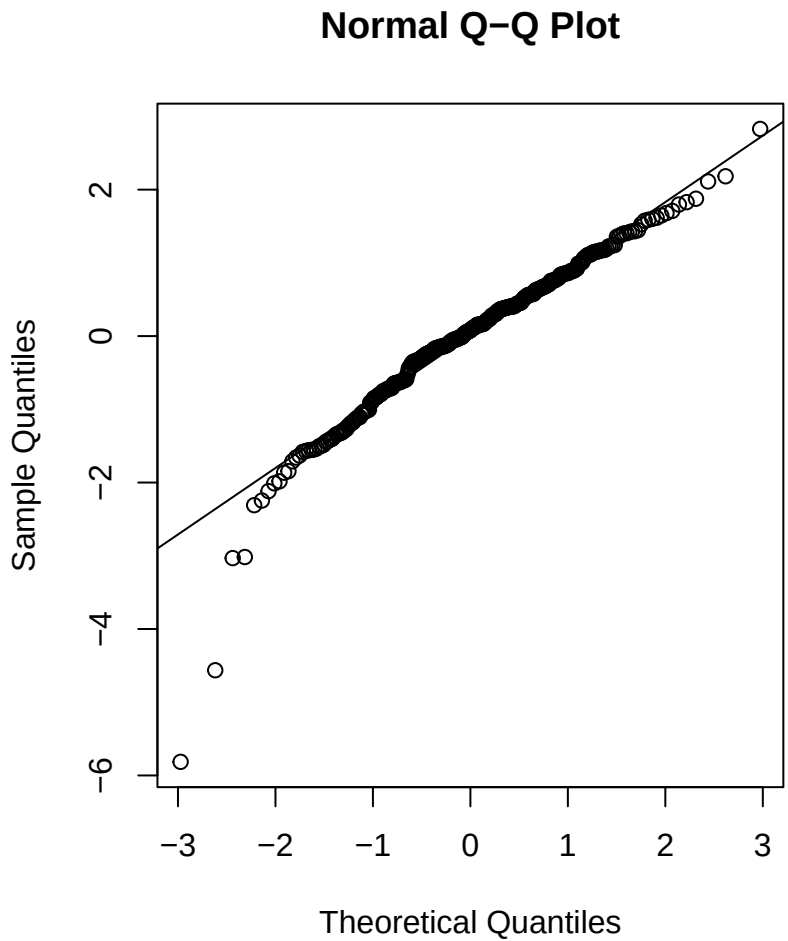



Normalized residual
2
0
−2
−4
−6
5.6
5.8
6.0
6.2
6.4
6.6
Fitted log(biomass + 1)

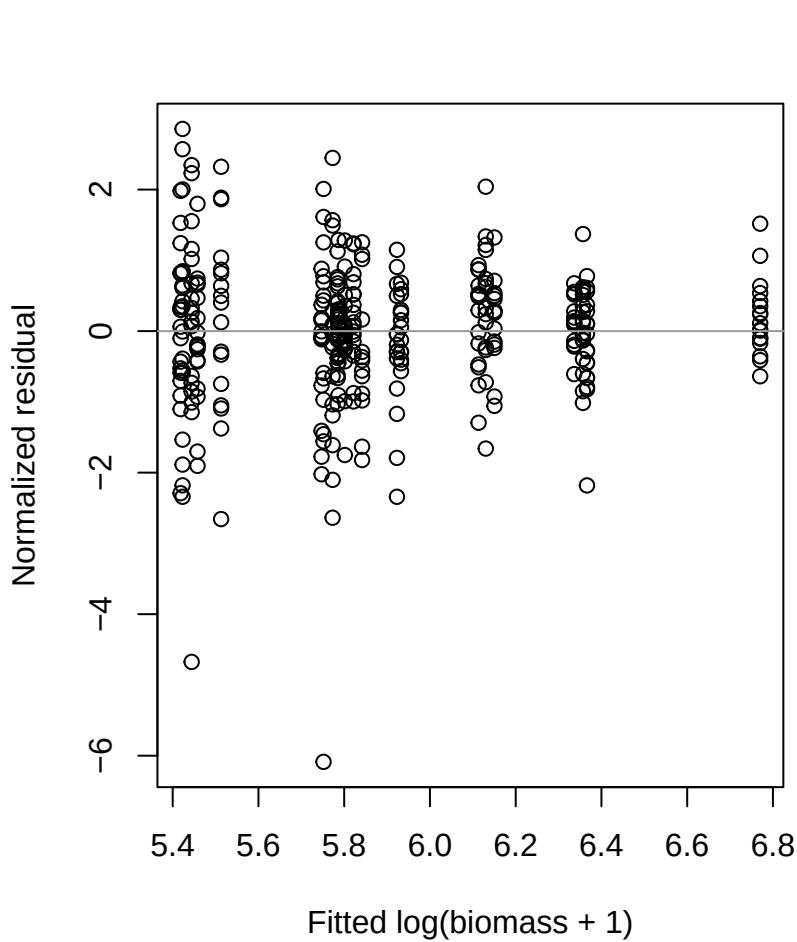

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
Theoretical Quantiles

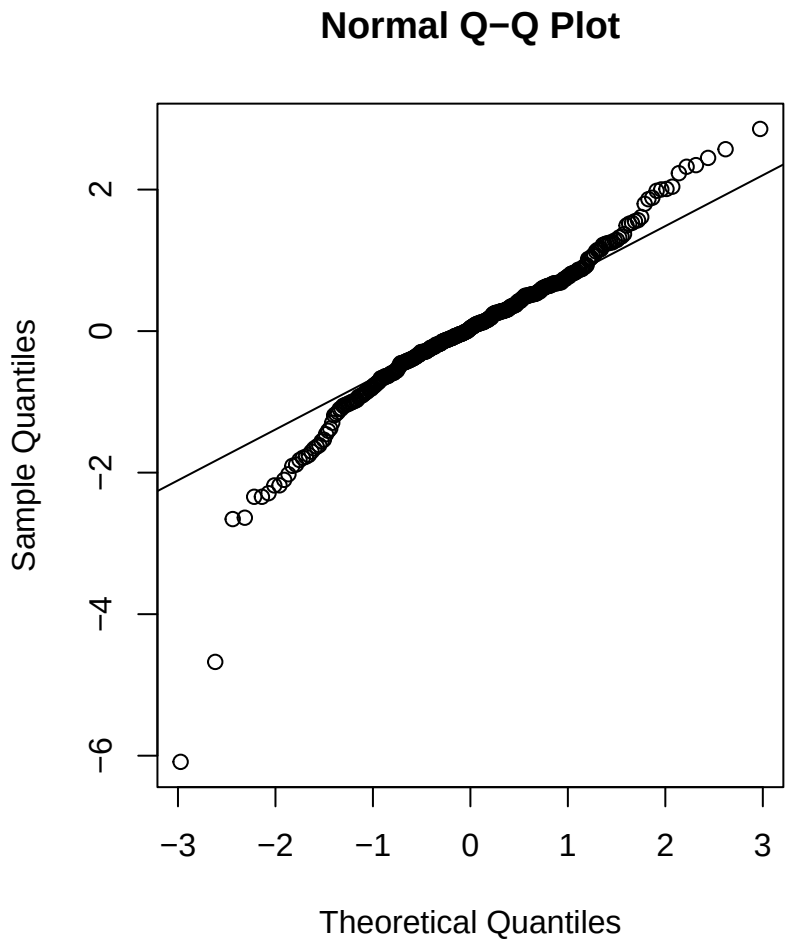
total_biomass__2022__2004__site_independent

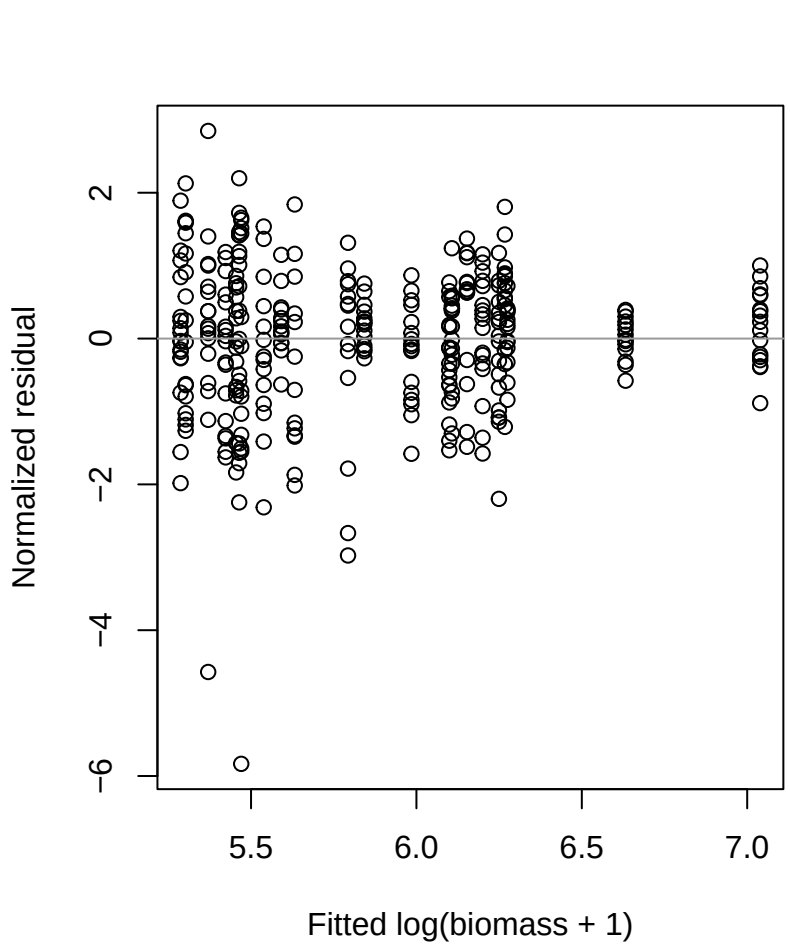


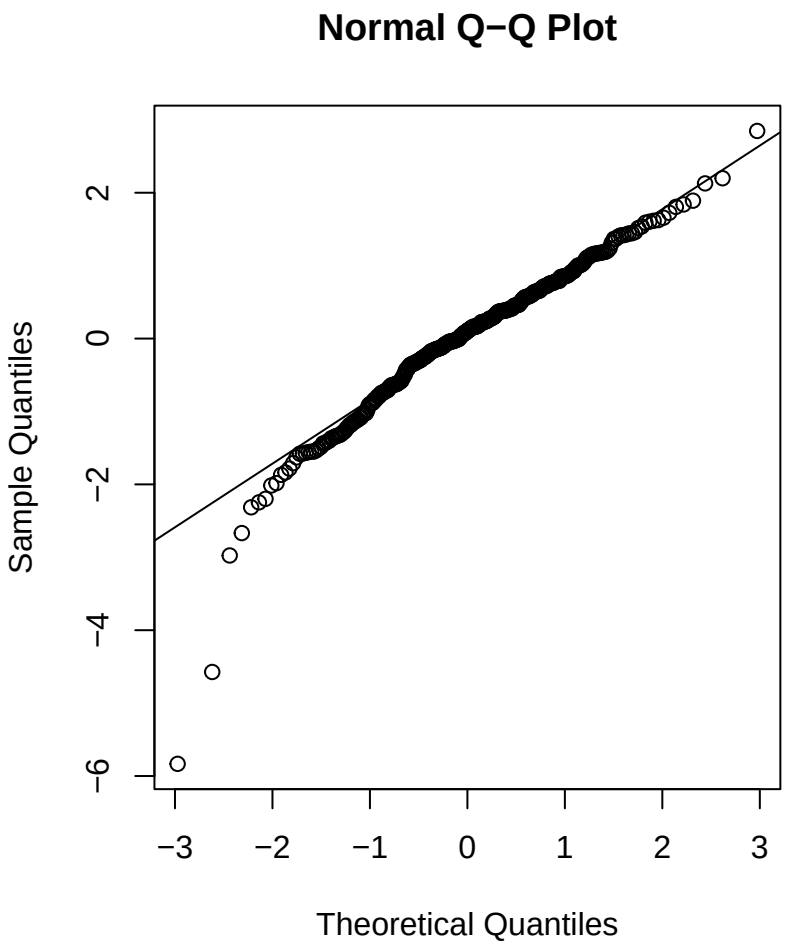


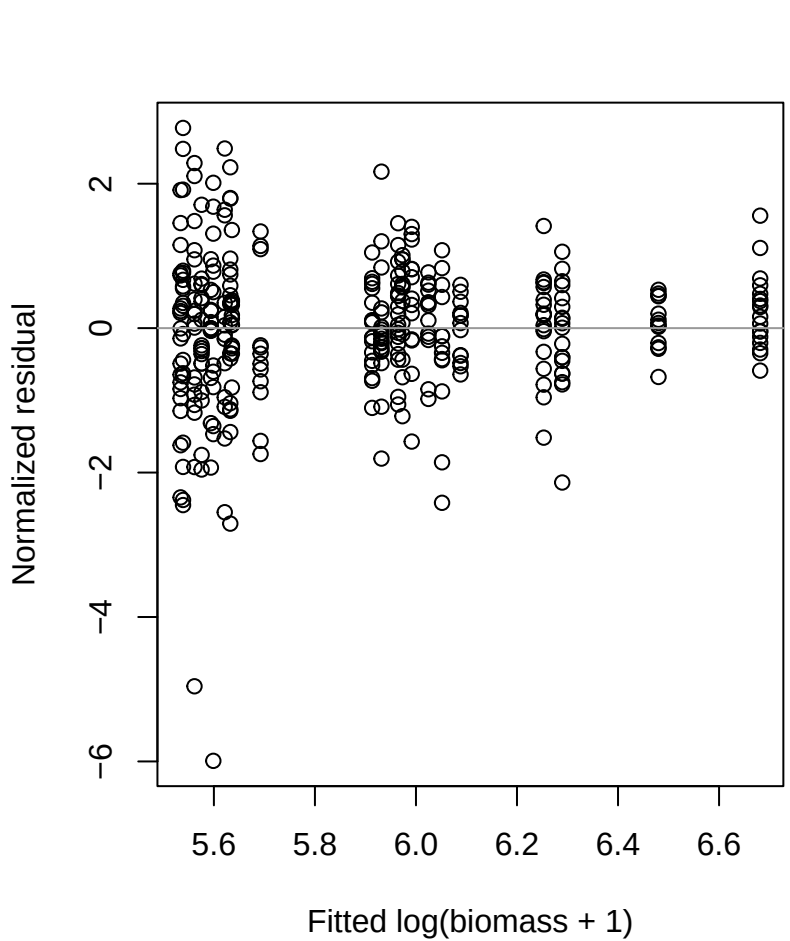
total_biomass__2022__2005__site_calendar_AR1

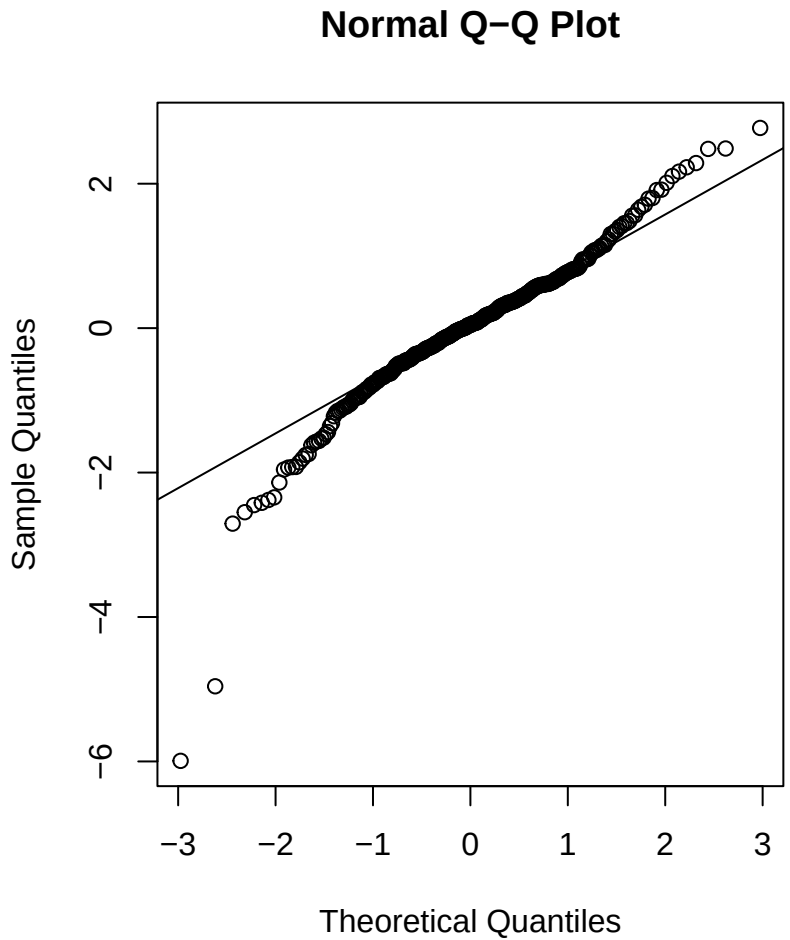


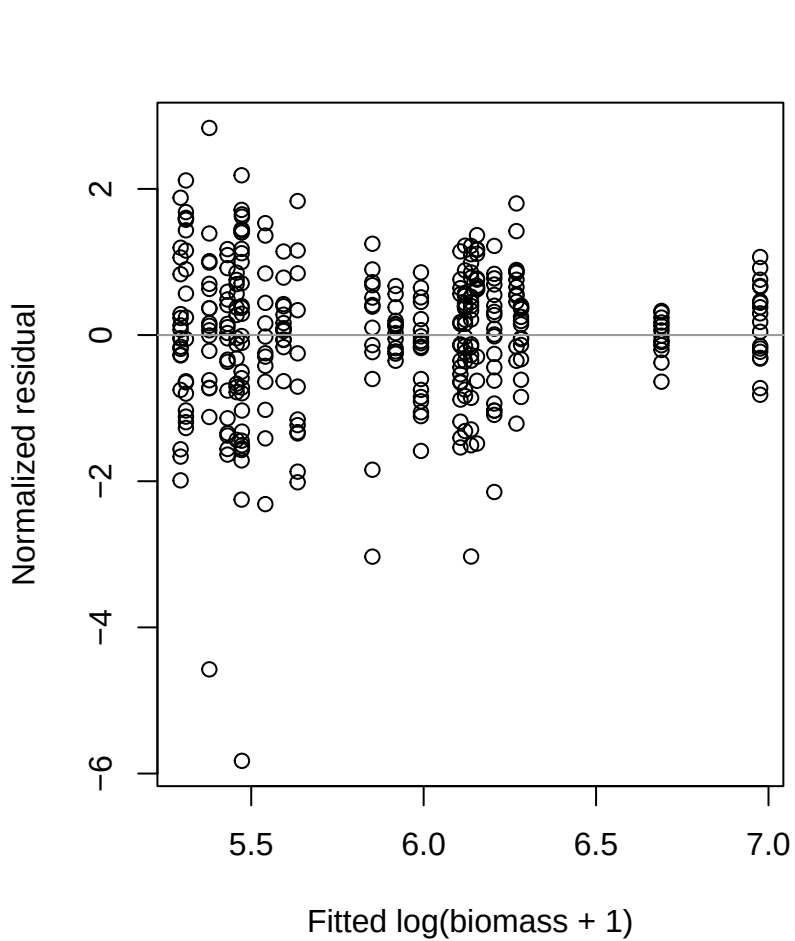


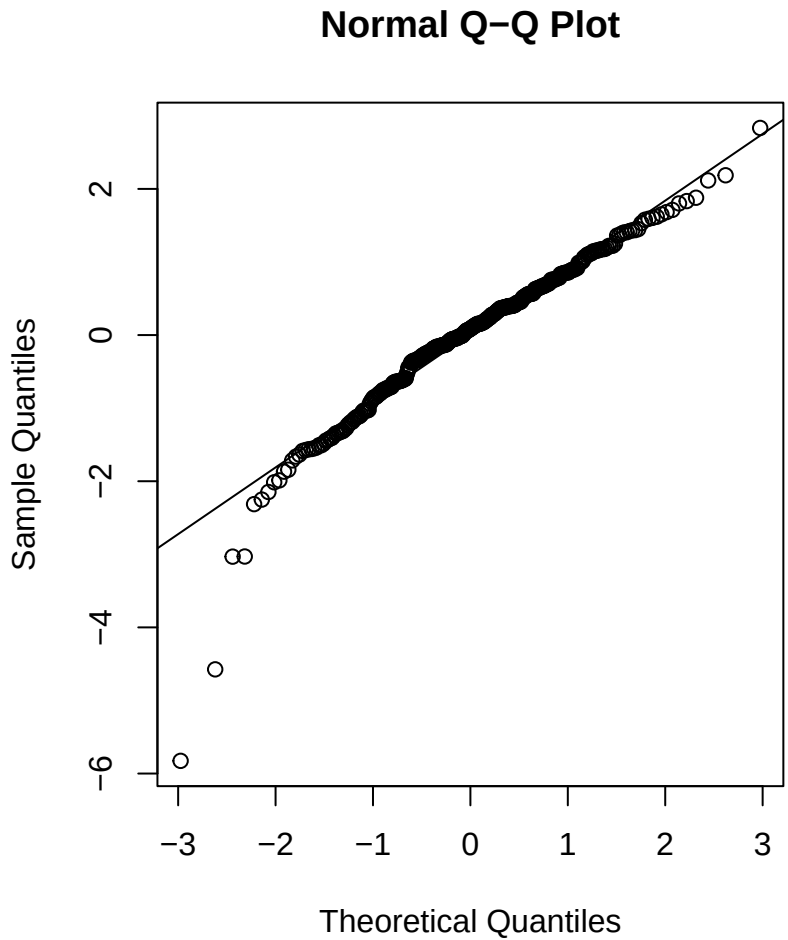

Normalized residual
Fitted log(biomass + 1)
−6
−4
−2
0
2
5.5
6.0
6.5
7.0


Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−6
−4
−2
0
2
−3
−2
−1
0
1
2
3

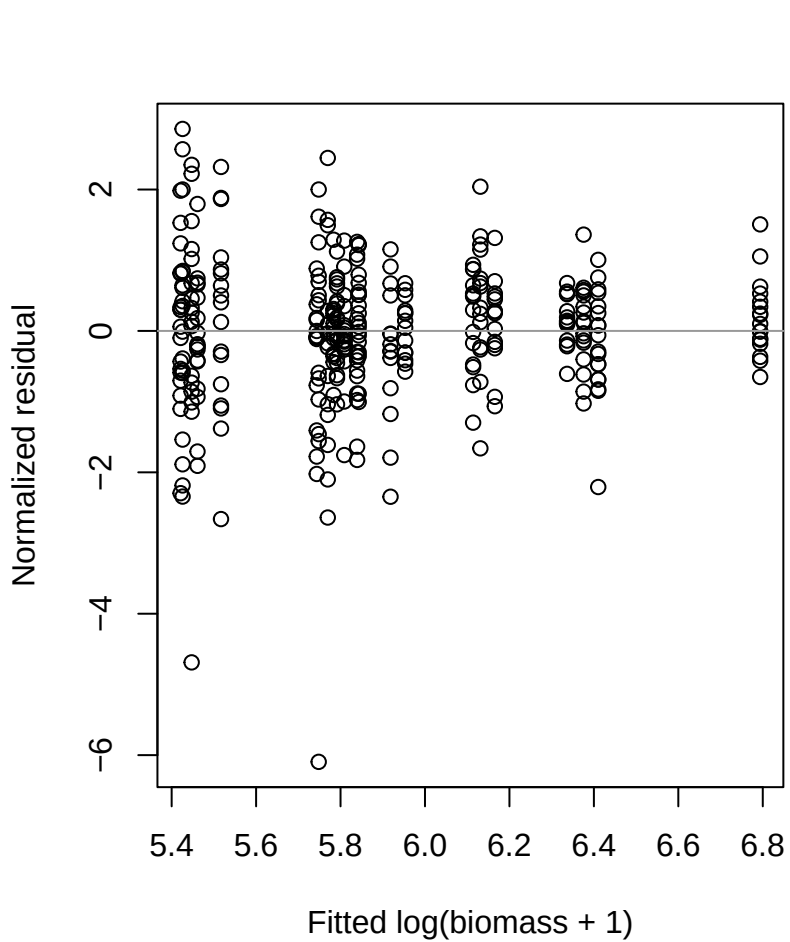

Normalized residual
2
0
−2
−4
−6
5.6
5.8
6.0
6.2
6.4
6.6
Fitted log(biomass + 1)

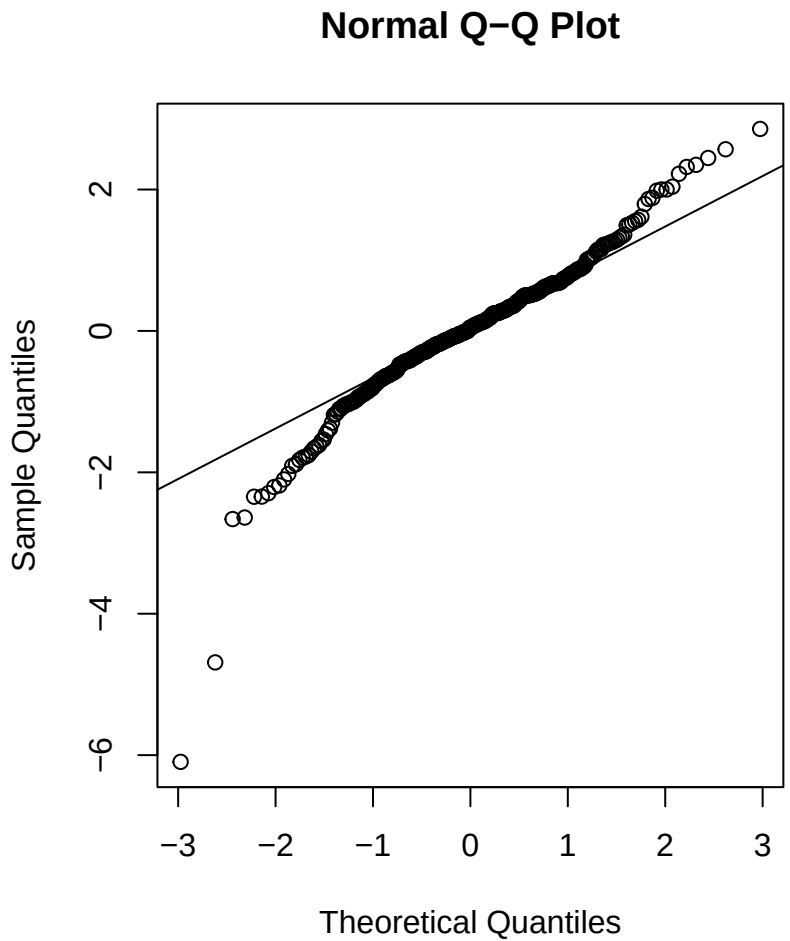

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Normalized residual
Fitted log(biomass + 1)
−6
−4
−2
0
2
5.5
6.0
6.5
7.0

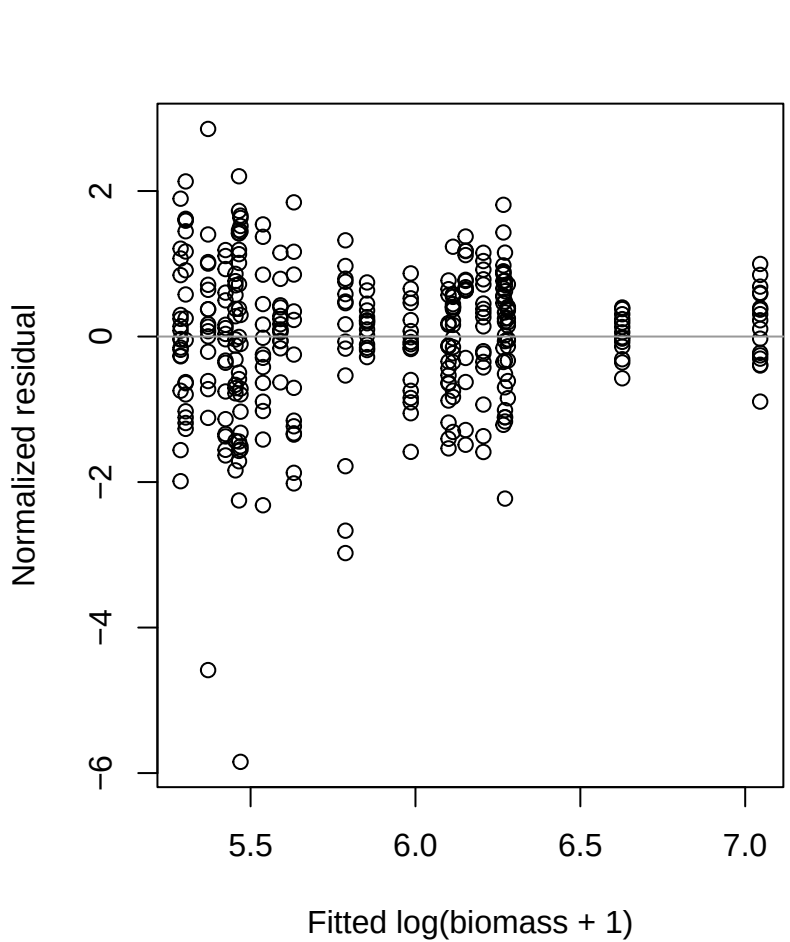

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−6
−4
−2
0
2
−3
−2
−1
0
1
2
3

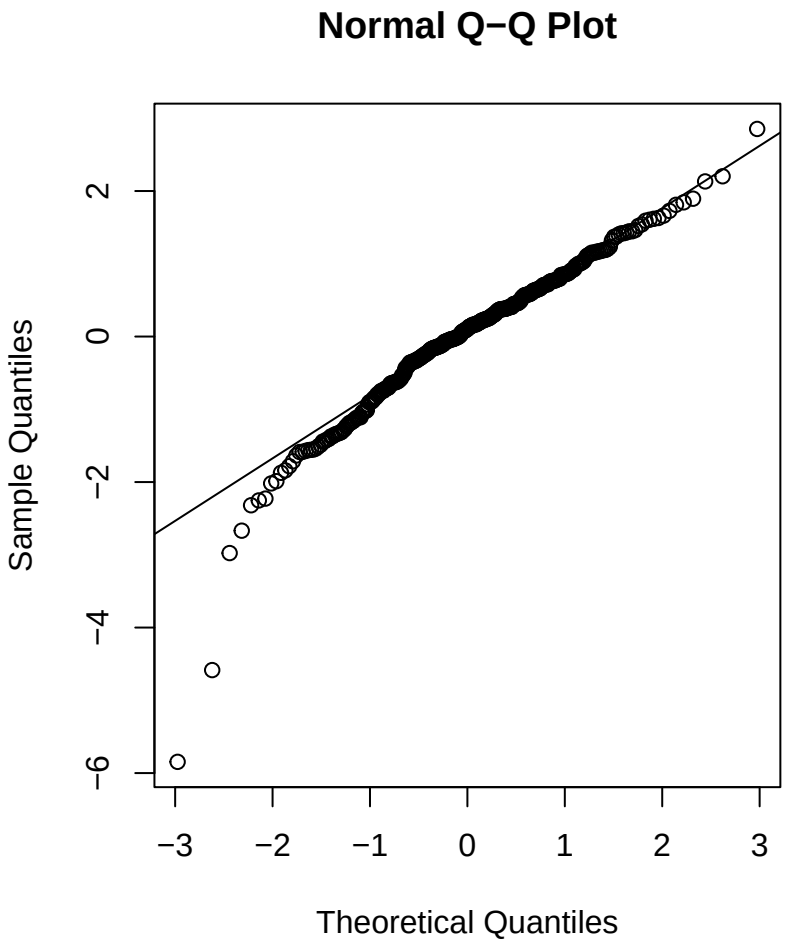
total_biomass__2023__2005__site_calendar_AR1

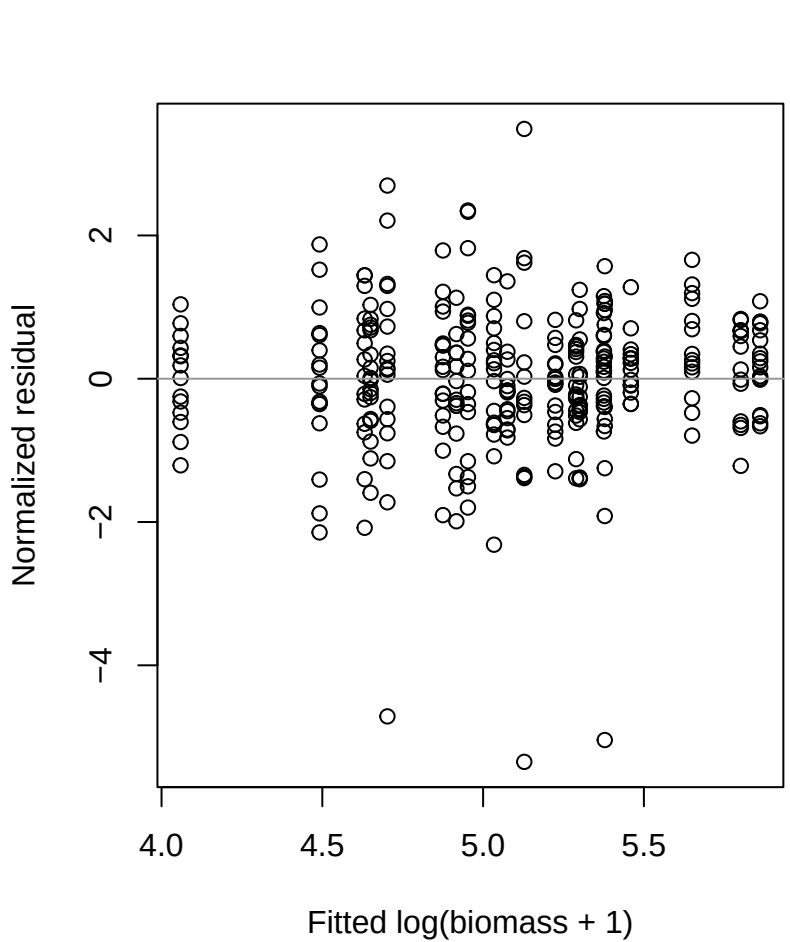


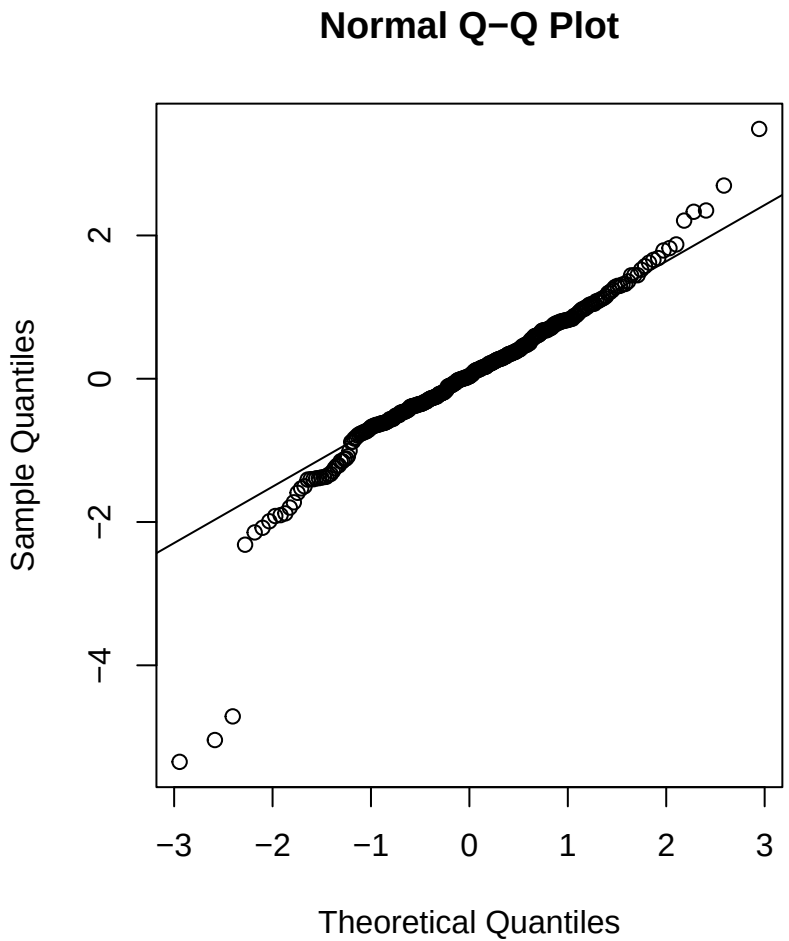


total_biomass__2023__2005__site_independent

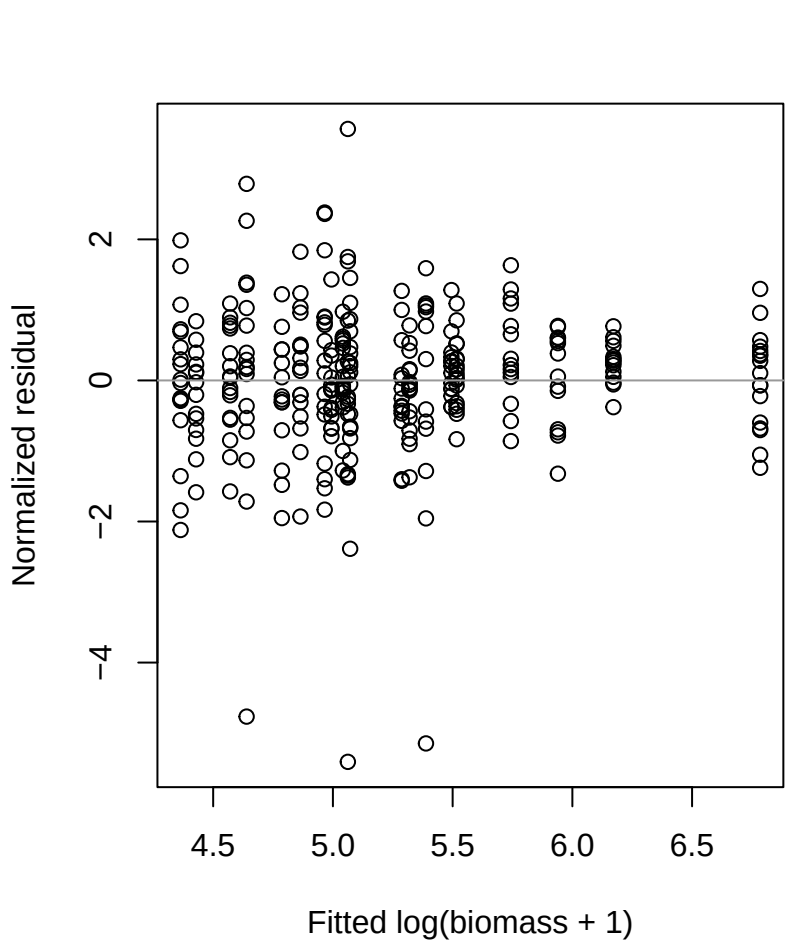


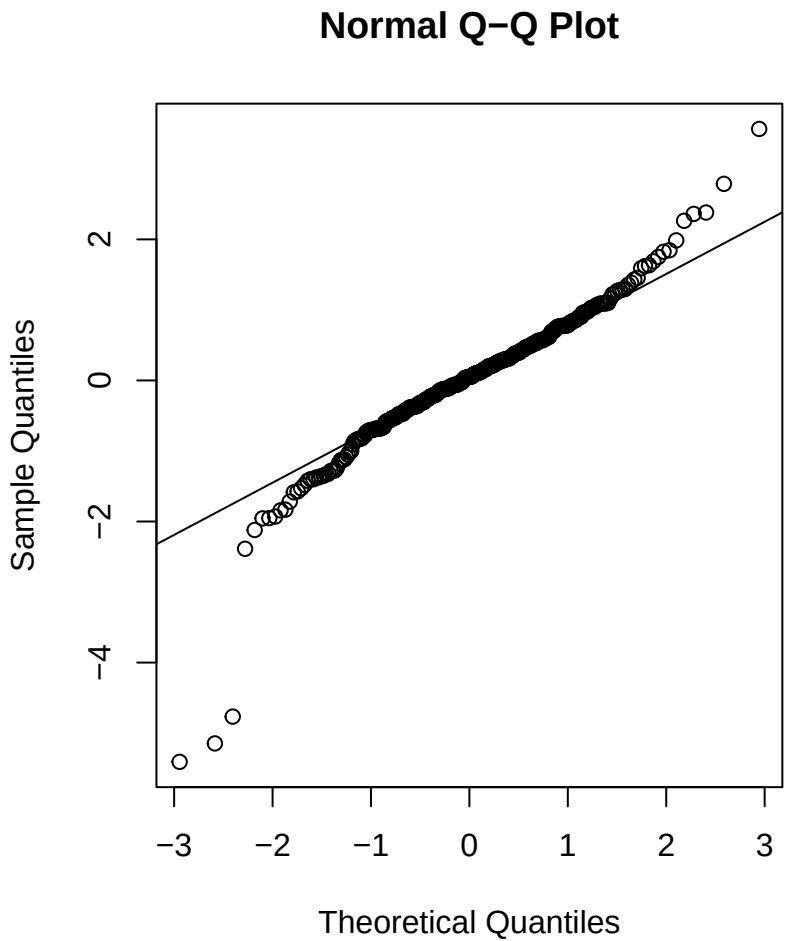



Normalized residual
2
0
−2
−4
4.0
4.5
5.0
5.5
Fitted log(biomass + 1)

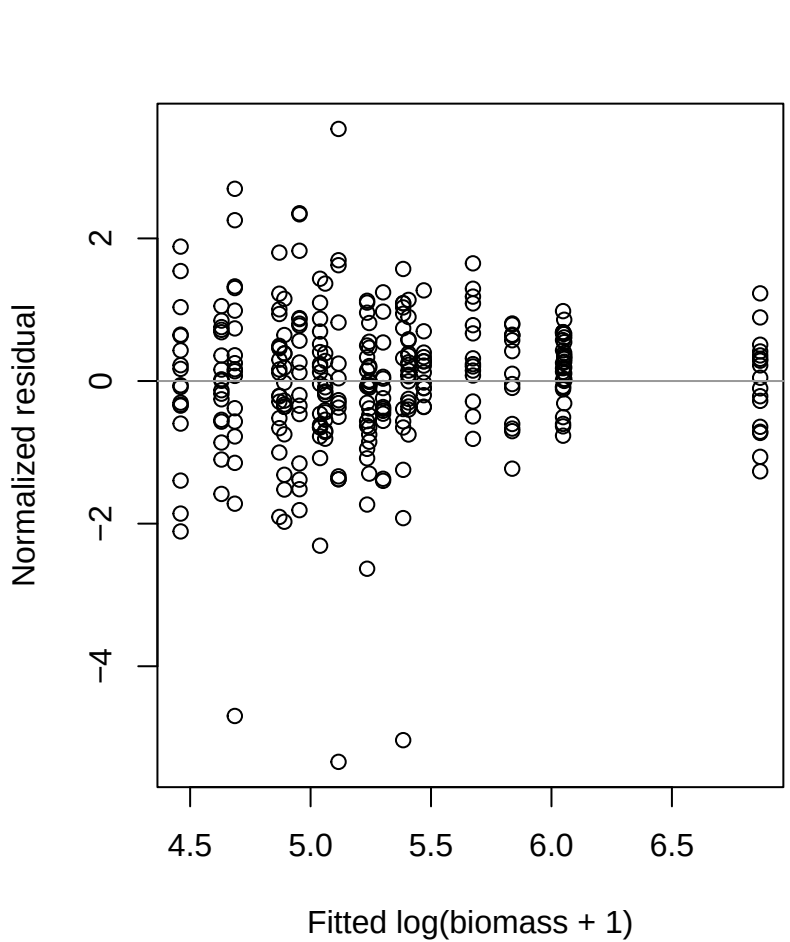

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

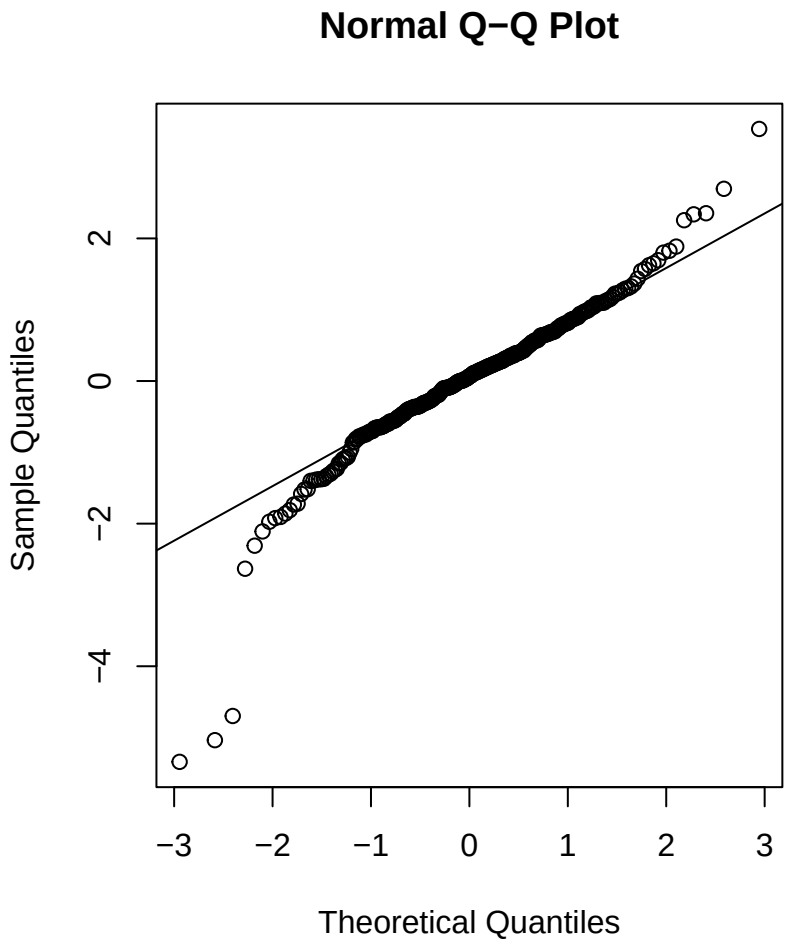
red_biomass__omit__a−k−06

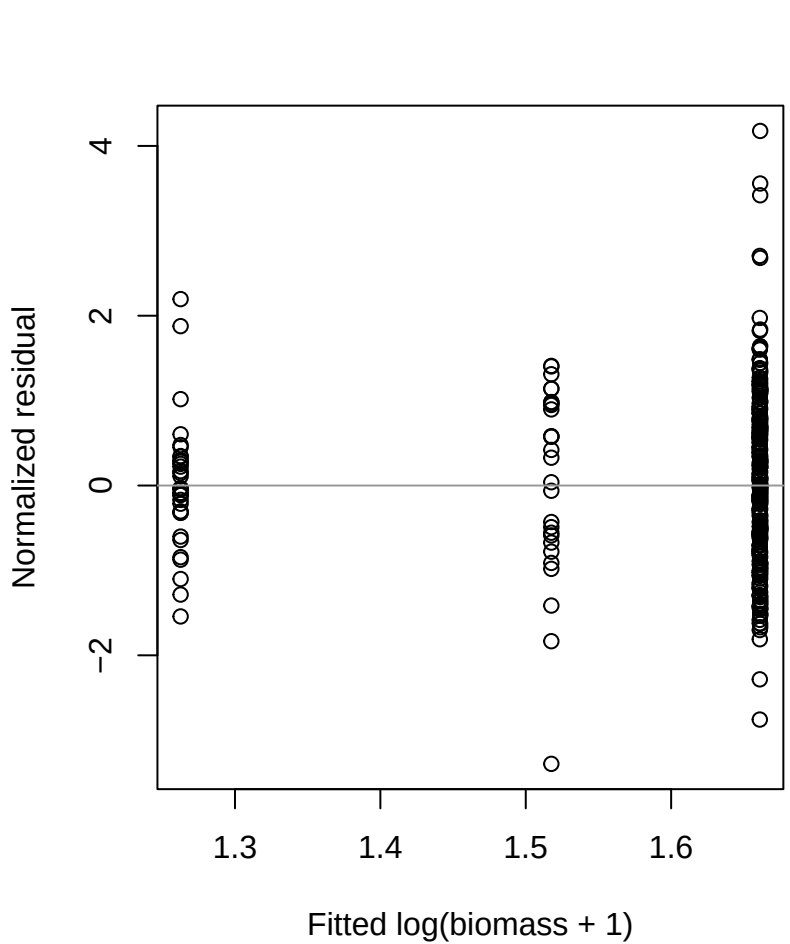


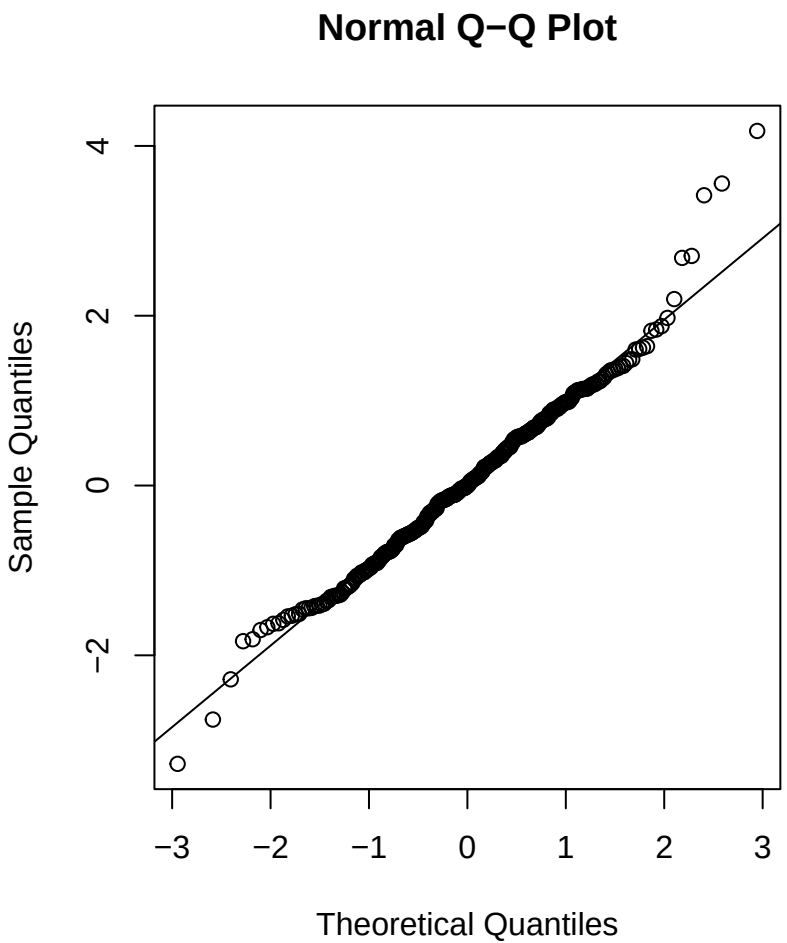


red_biomass__omit__a−k−09

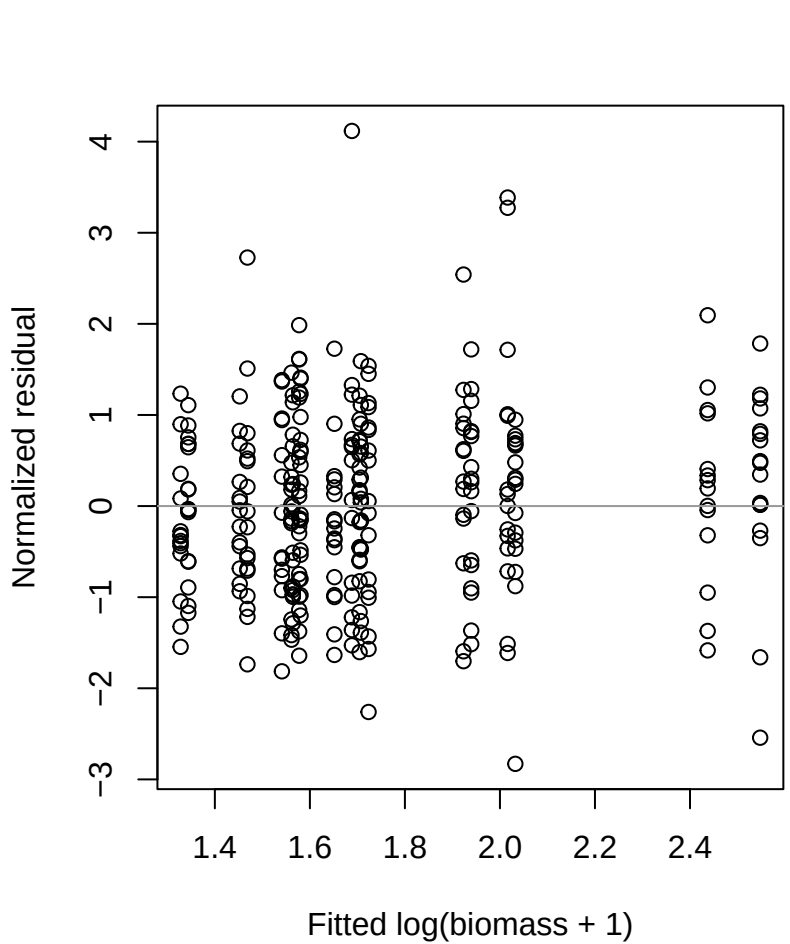


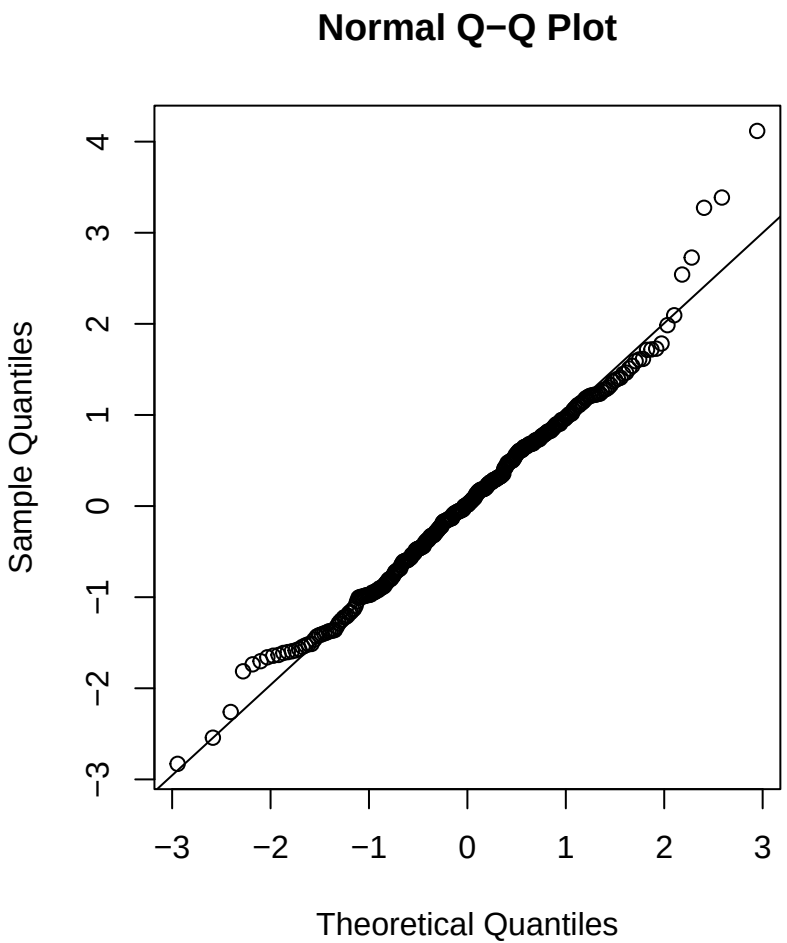


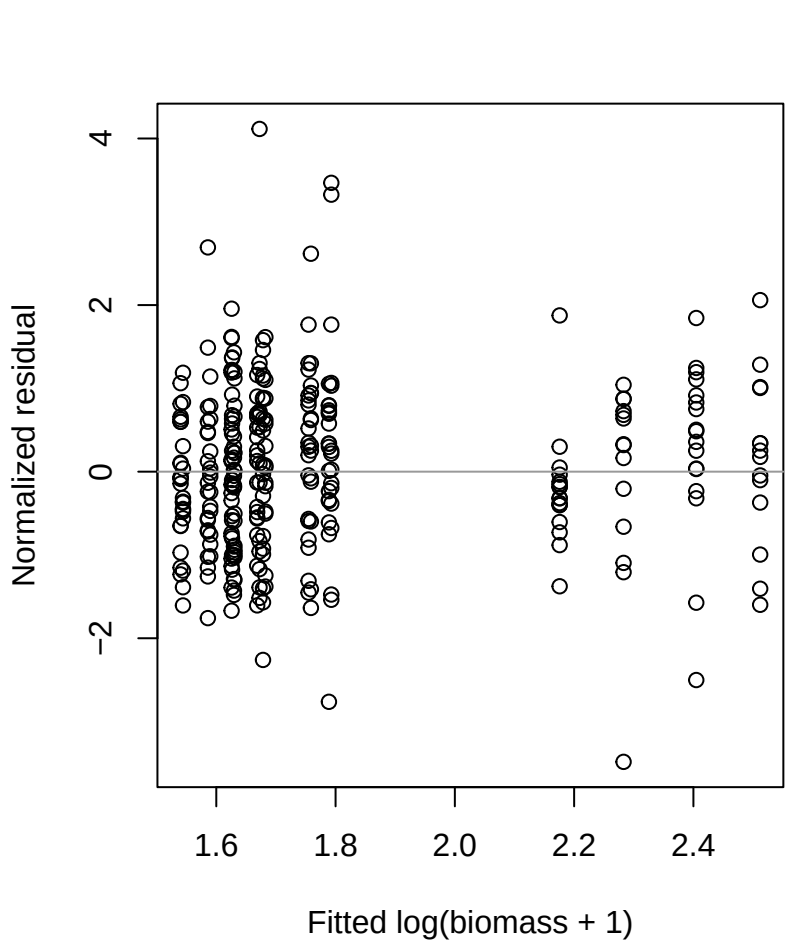

Normalized residual
4
2
0
−2
1.3
1.4
1.5
1.6
Fitted log(biomass + 1)

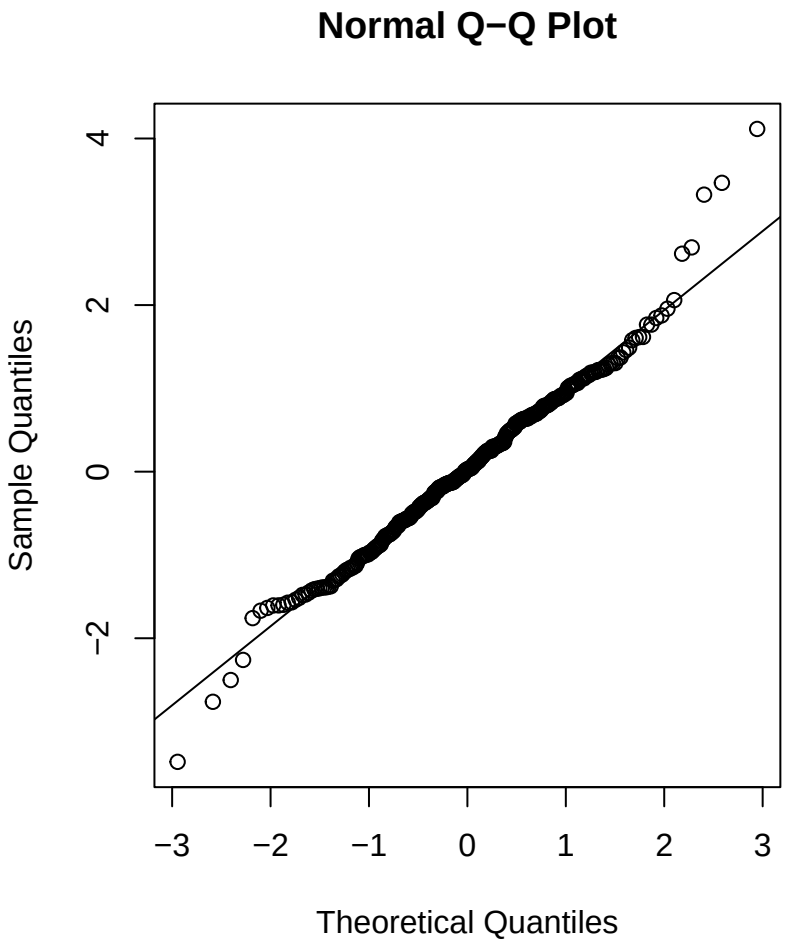

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−3
−2
−1
0
1
2
3
Theoretical Quantiles

red_small_mass__omit__a−k−06

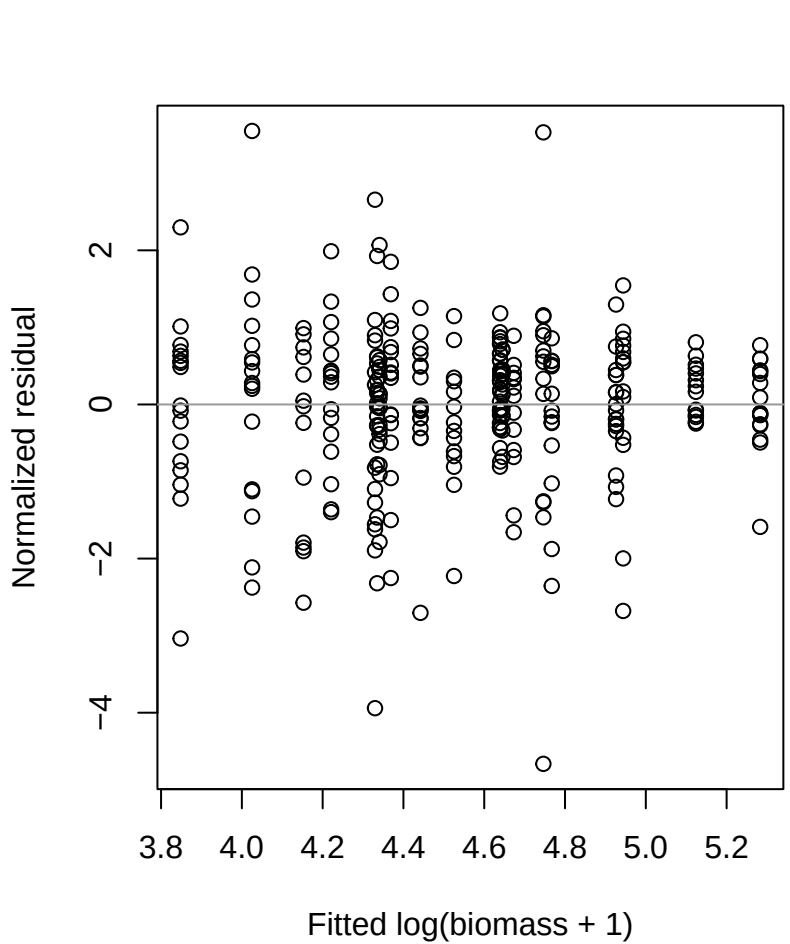


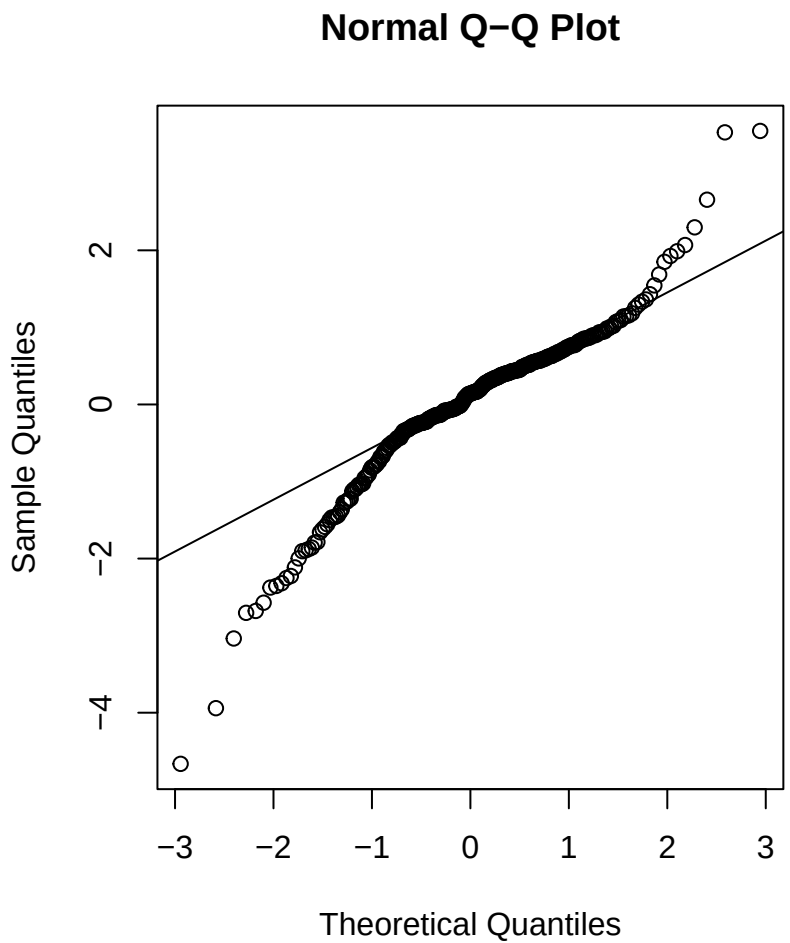



Normalized residual
4
2
0
−2
1.6
1.8
2.0
2.2
2.4
Fitted log(biomass + 1)

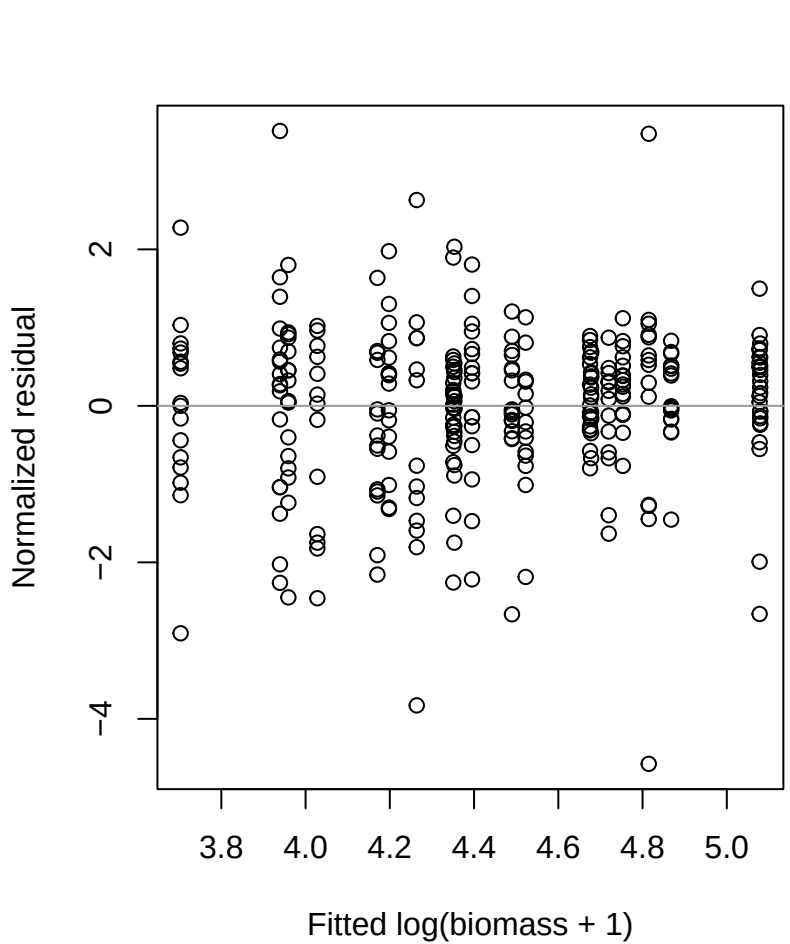

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−3
−2
−1
0
1
2
3
Theoretical Quantiles

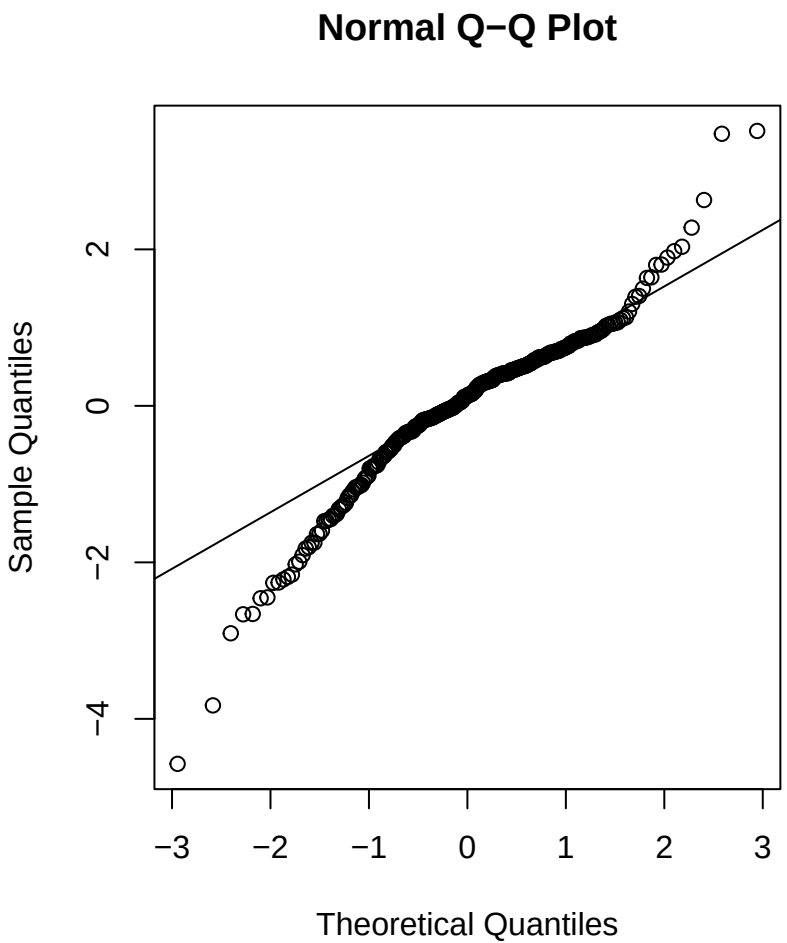
purple_biomass__omit__a−k−02



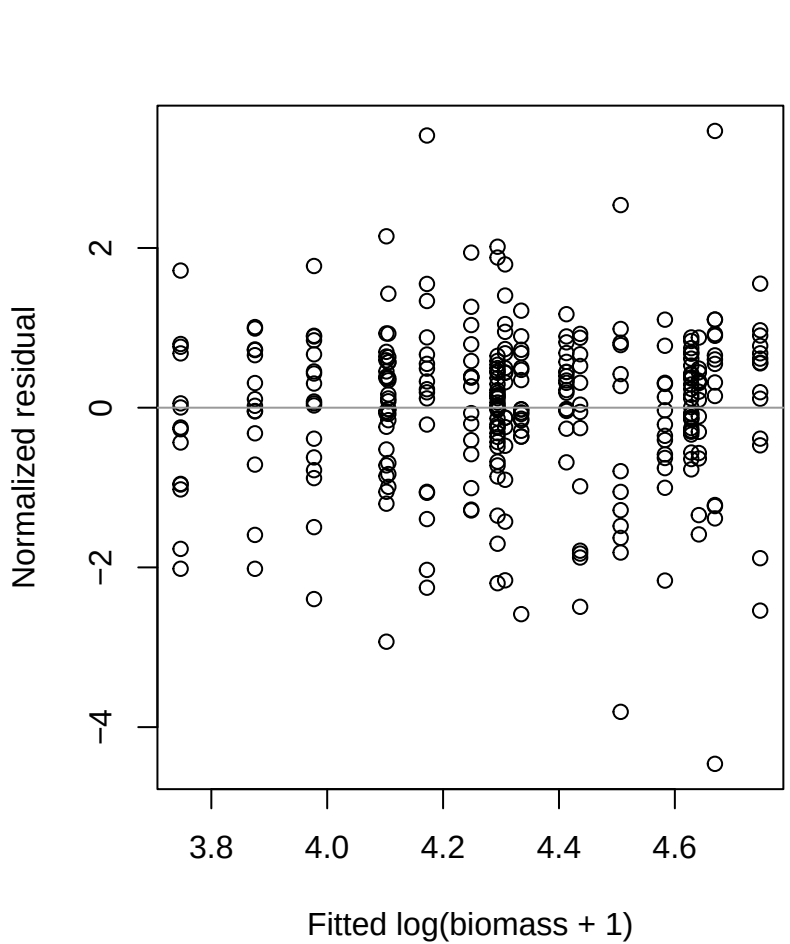


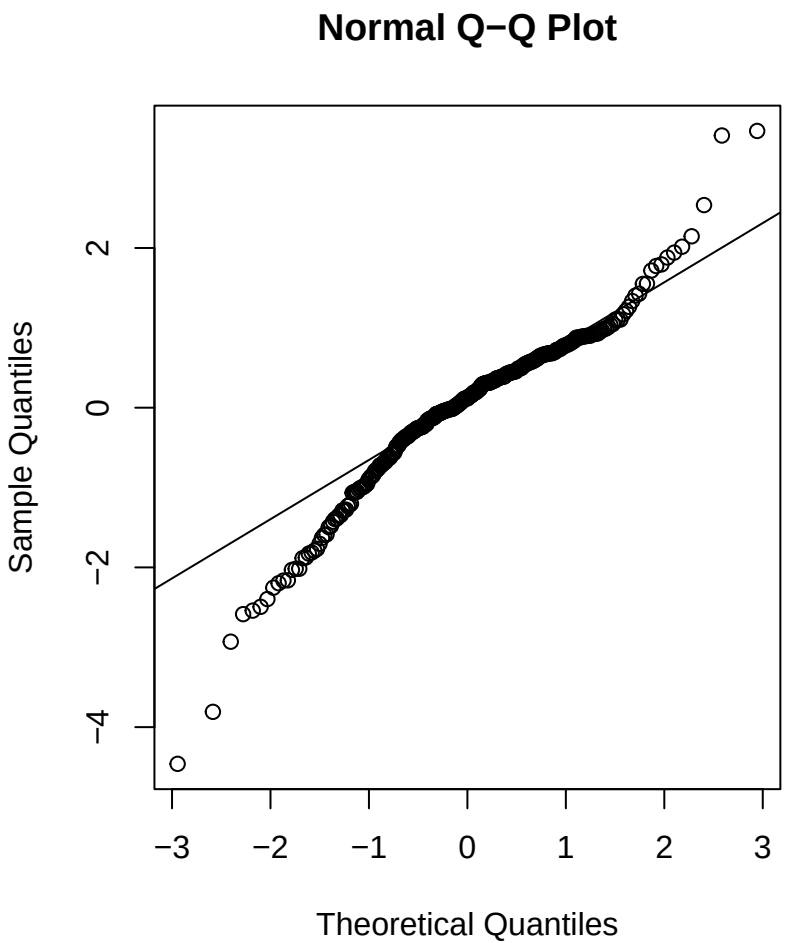
purple_biomass__omit__a−k−06

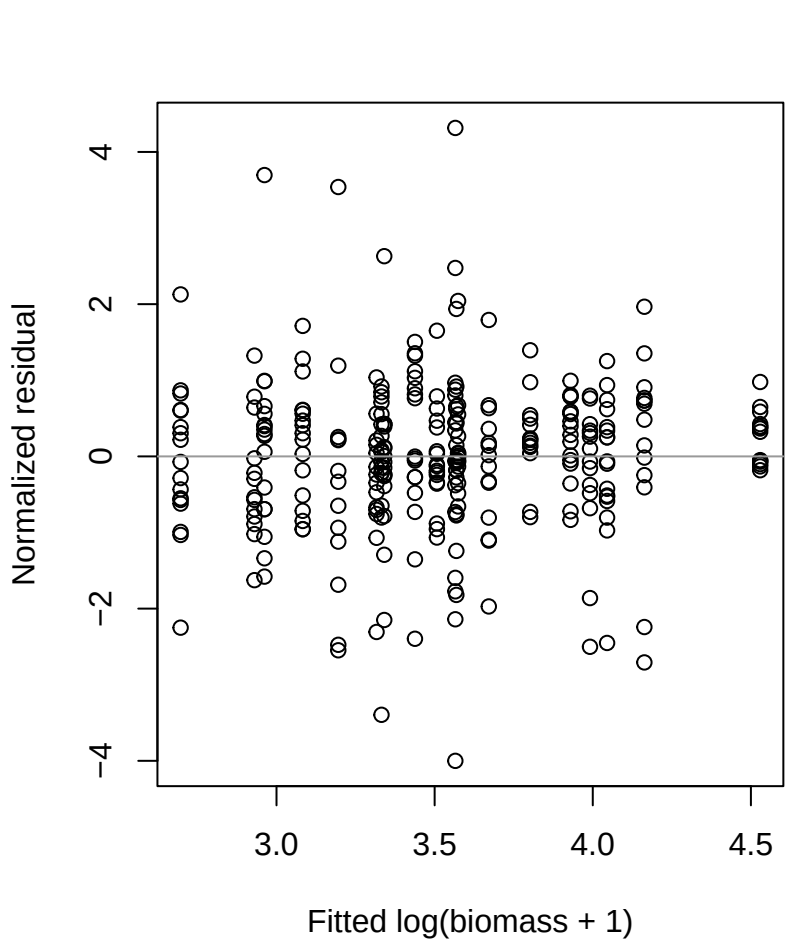


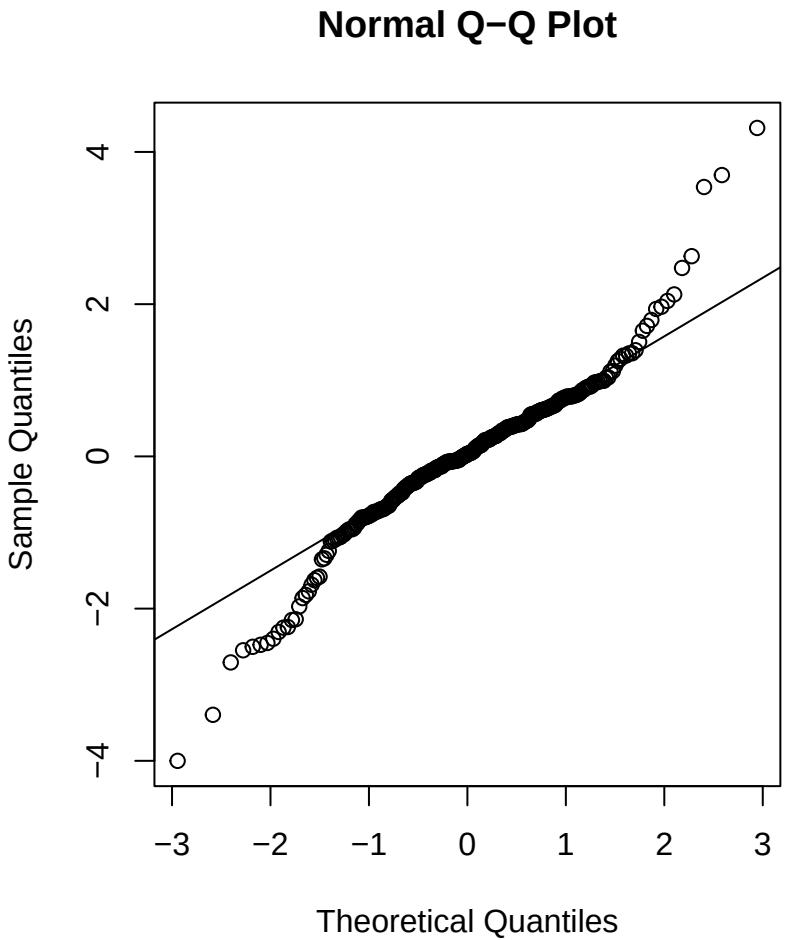


purple_biomass__omit__a−k−09

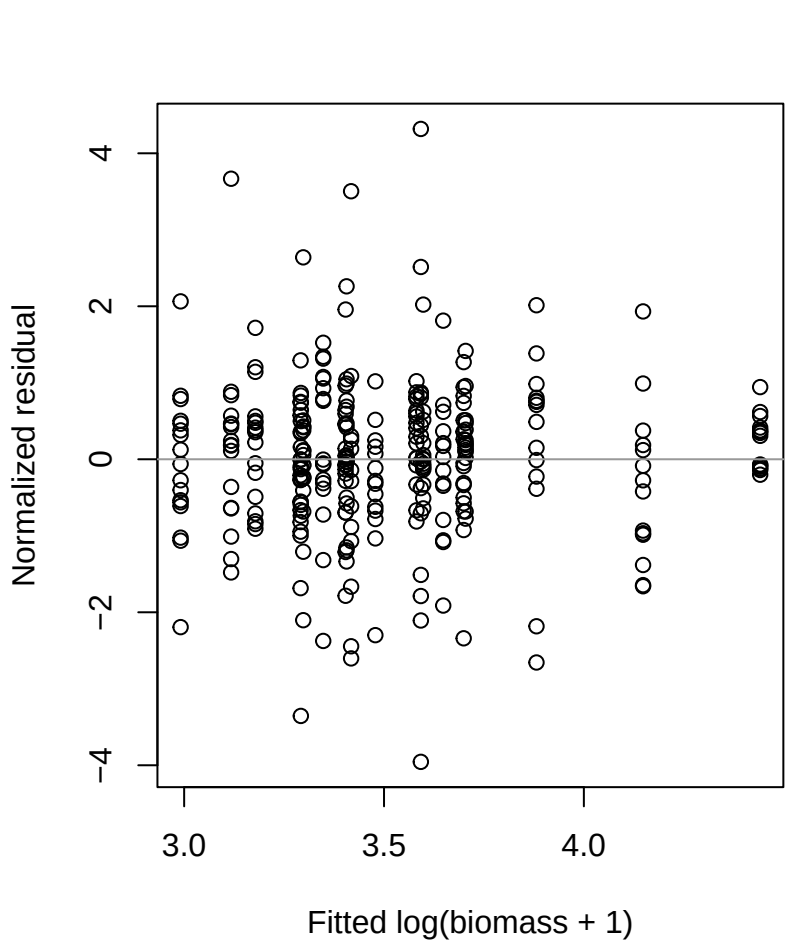


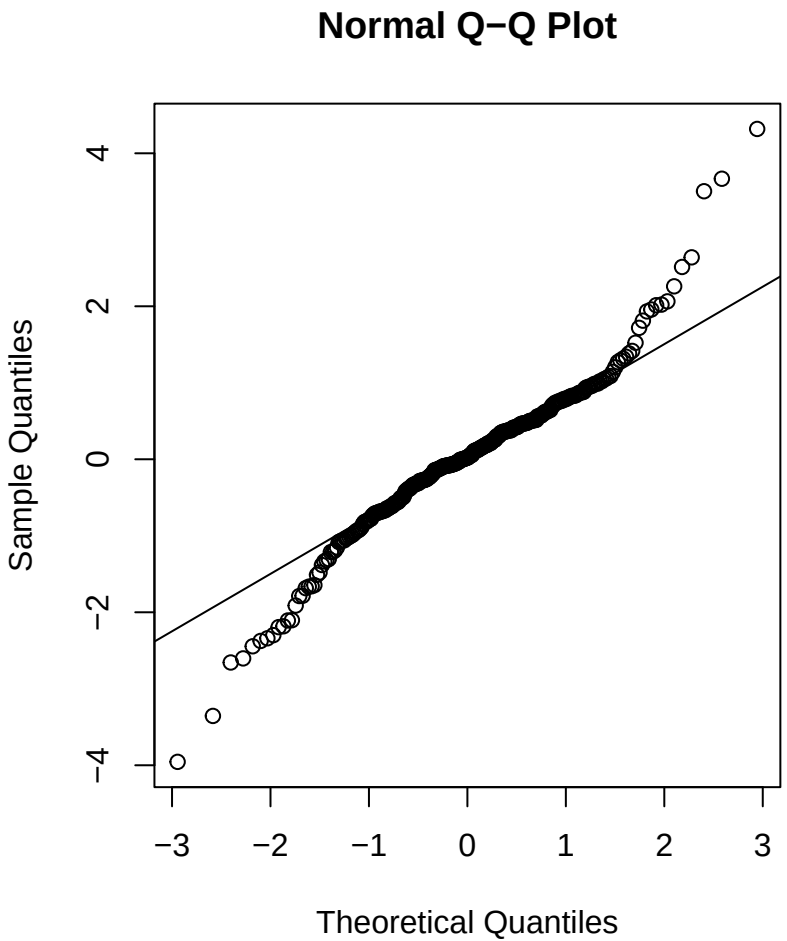


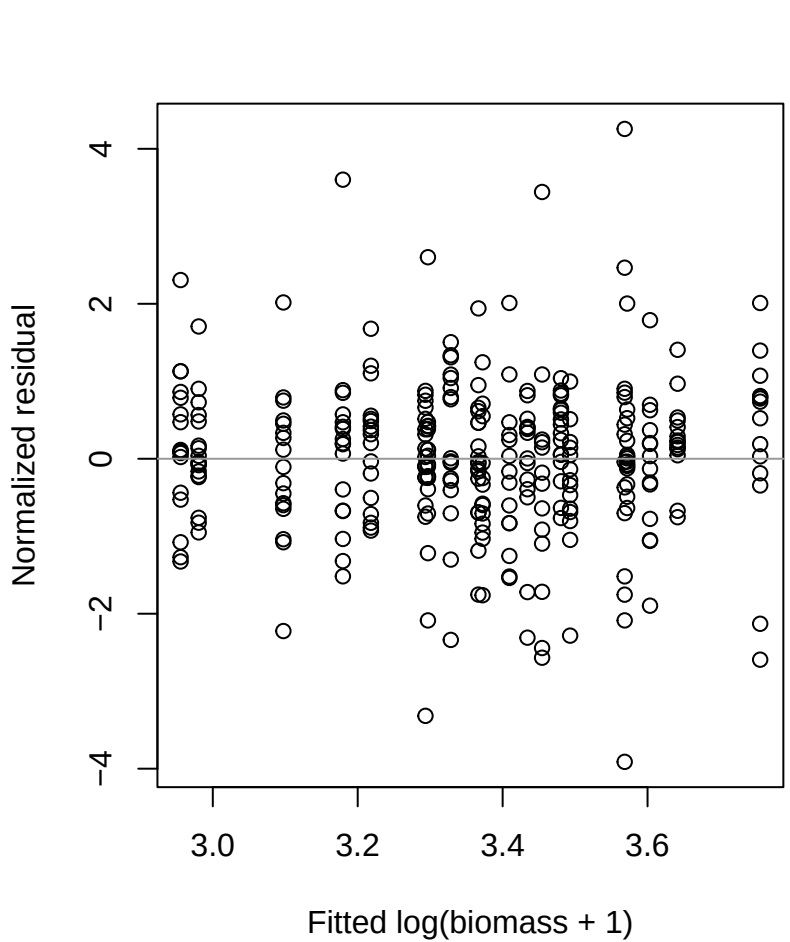

Normalized residual
4
2
0
−2
−4
3.0
3.5
4.0
4.5
Fitted log(biomass + 1)

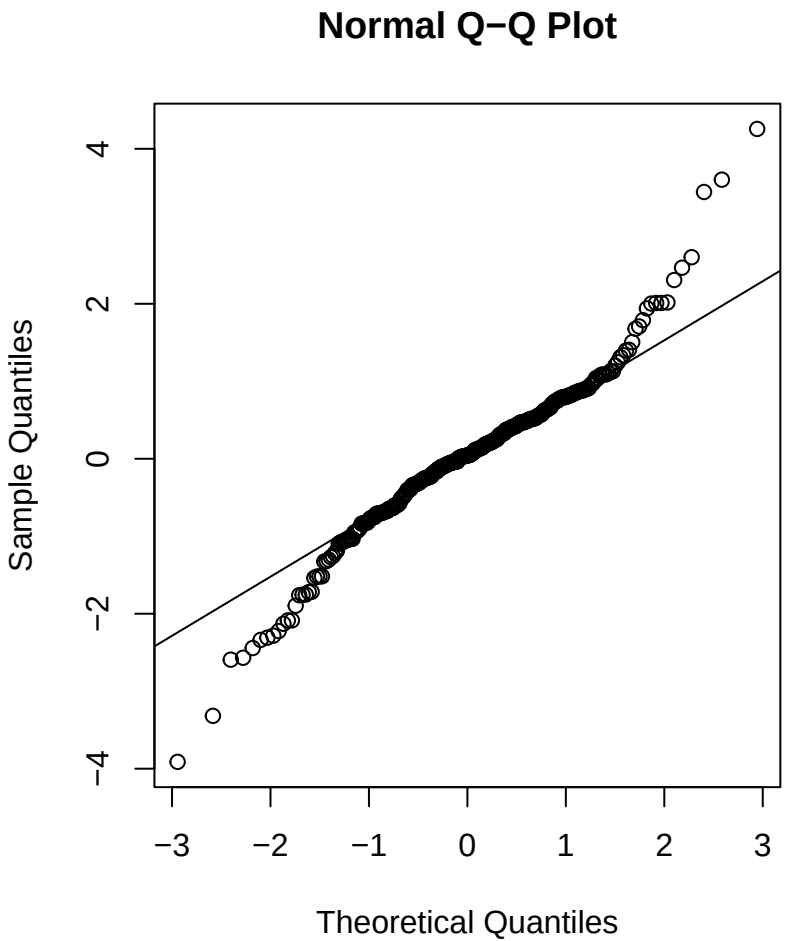

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

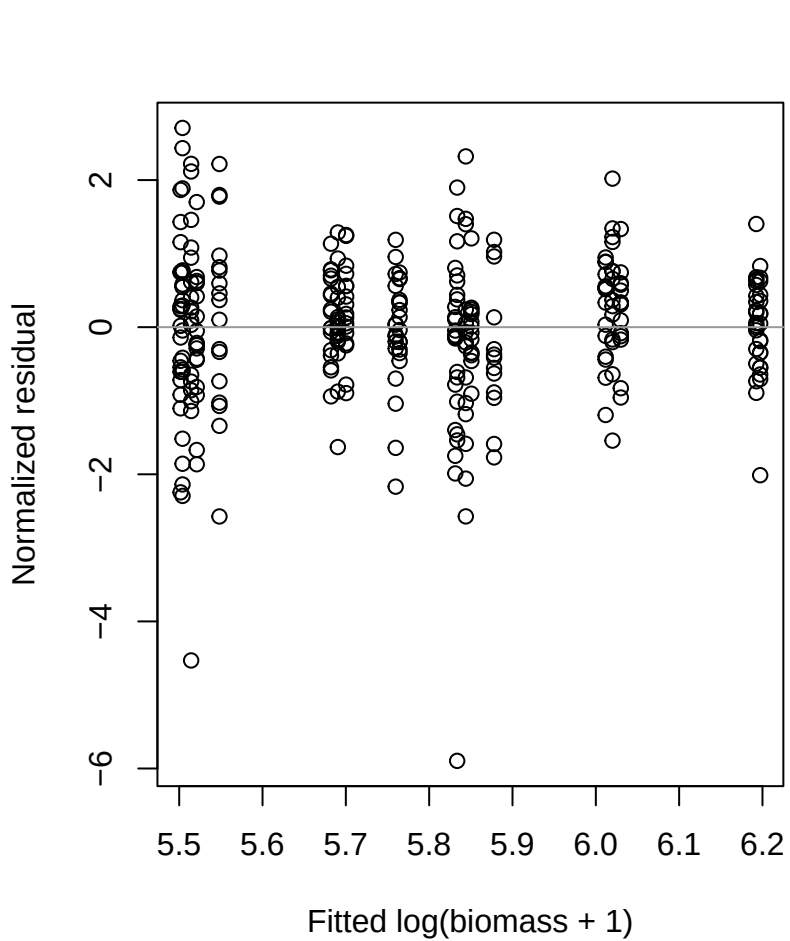
purple_small_mass__omit__a−k−06

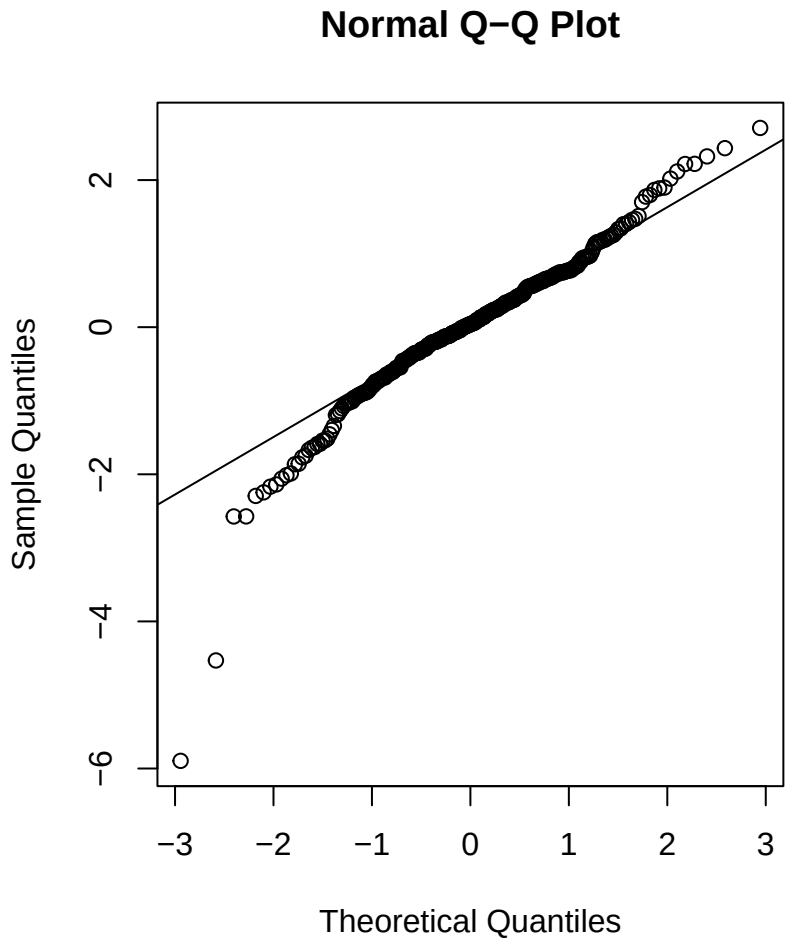




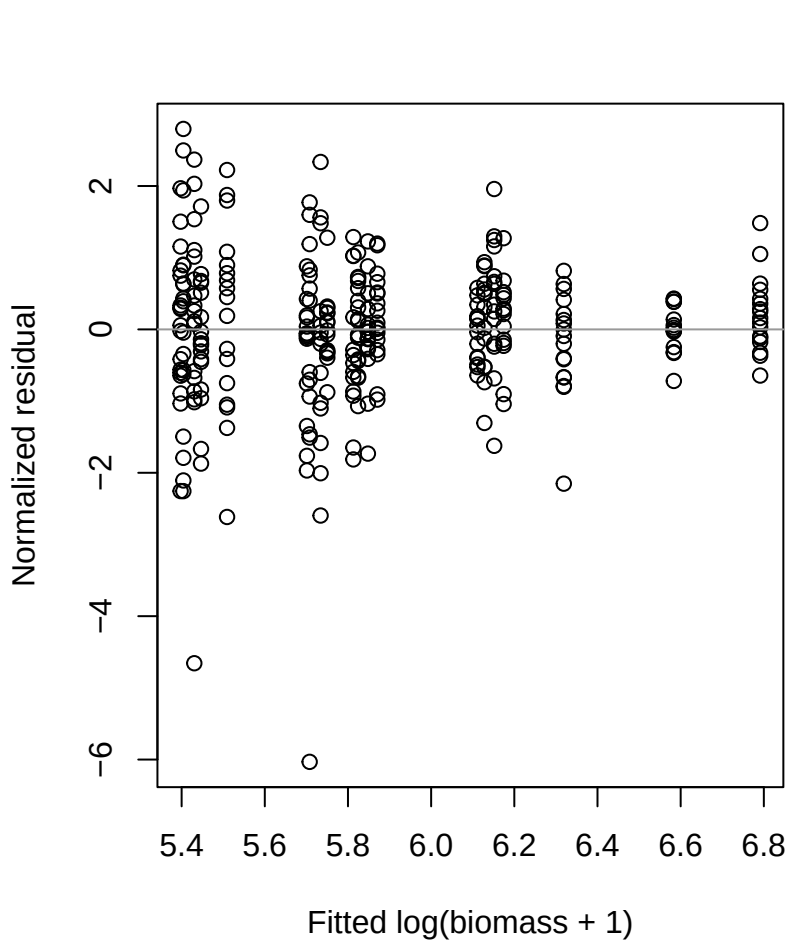

Normalized residual
4
2
0
−2
−4
3.0
3.2
3.4
3.6
Fitted log(biomass + 1)

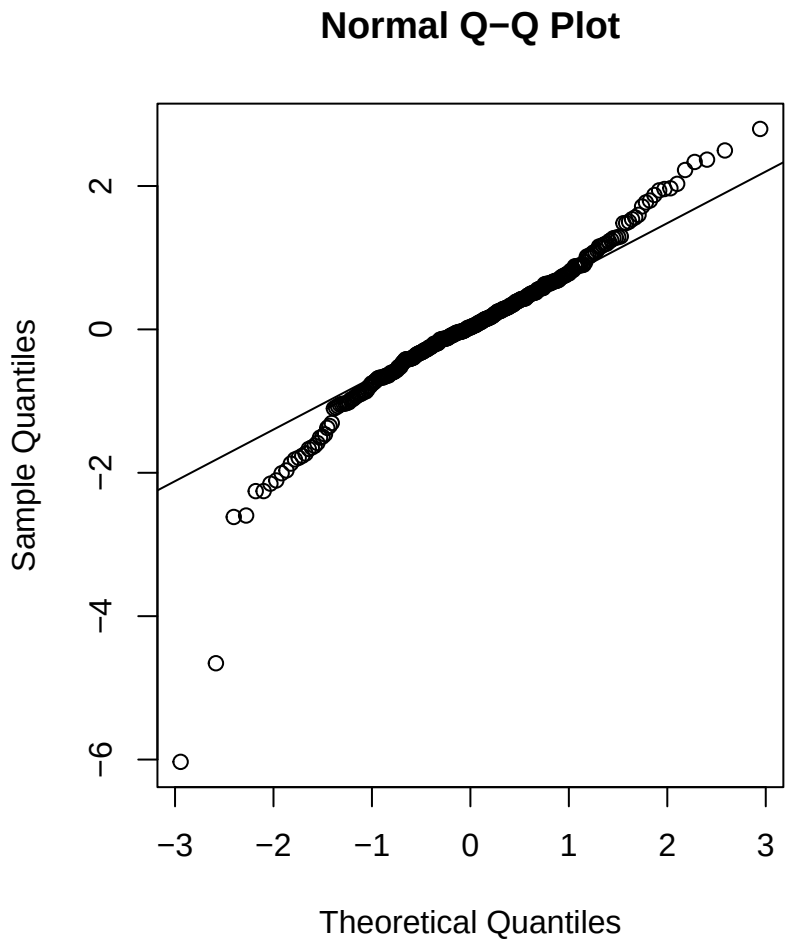

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles

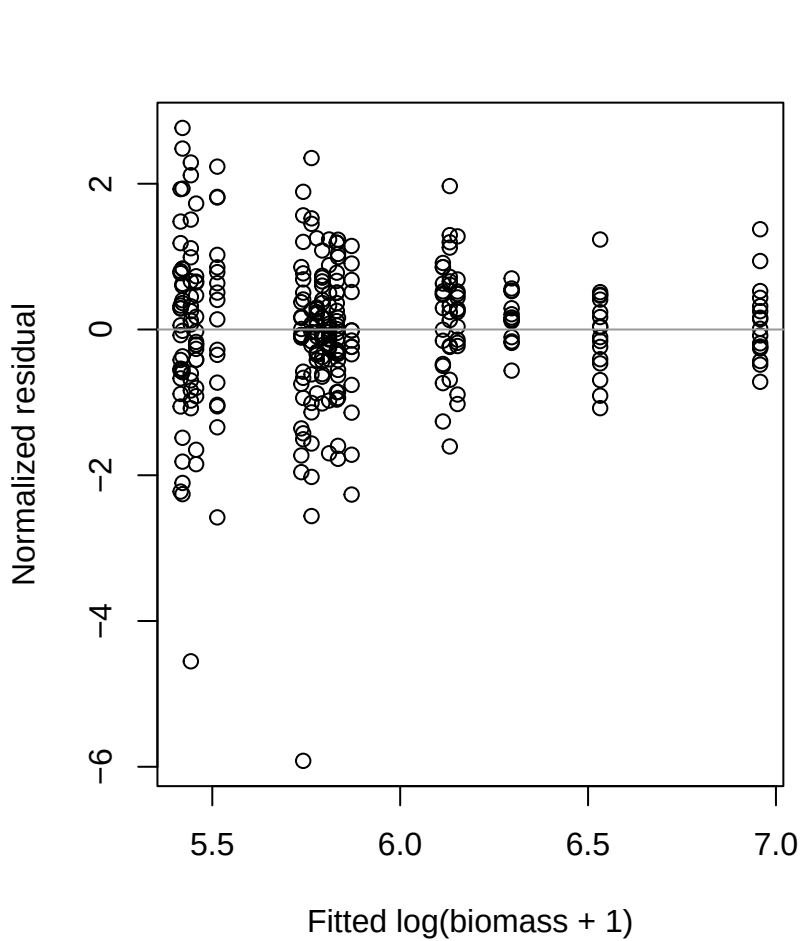

Normalized residual
2
0
−2
−4
−6
5.5
5.6
5.7
5.8
5.9
6.0
6.1
6.2
Fitted log(biomass + 1)

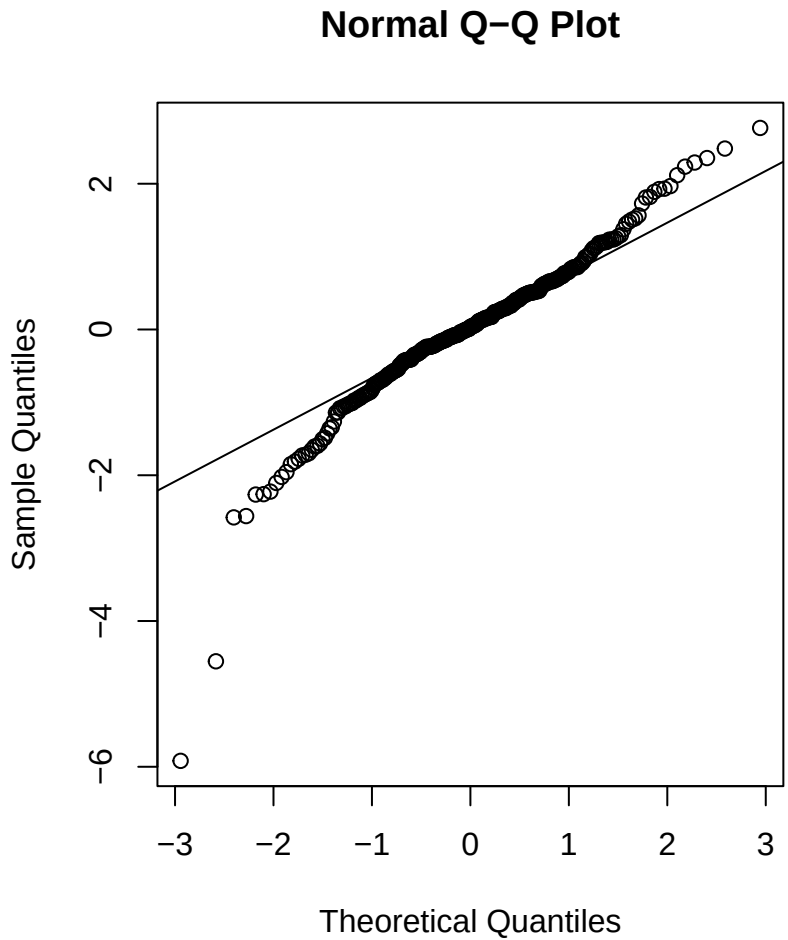

Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
Theoretical Quantiles

total_biomass__omit__a−k−06






Normalized residual
2
0
−2
−4
−6
5.5
6.0
6.5
7.0
Fitted log(biomass + 1)


Normal Q−Q Plot
Sample Quantiles
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
Theoretical Quantiles